



Faculty 1 – Physics and Electrical Engineering

**Organic Electrochemical Transistors (OECT)
combined with enzyme glucose dehydrogenase as
glucose sensors**

Master Project in Control, Microsystems and Microelectronics

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Abstract

Abnormal blood glucose levels, leading to hypoglycemia or hyperglycemia, are key indicators of diabetes mellitus, a chronic metabolic disorder. Continuous and accurate glucose monitoring is vital for effective disease management. Organic Electrochemical Transistors (OECTs) offer a promising platform for enzymatic glucose sensing due to their high sensitivity, low operating voltage, and biocompatibility.

In this work, OECT-based glucose sensors were fabricated using platinum gate electrodes and dual-layer of modified PEDOT:PSS and pure PEDOT:PSS channels for improved stability. Two enzymes, glucose oxidase (GOx) and glucose dehydrogenase (GDH), were examined using two approaches: enzyme immobilization on the channel and floating-enzyme electrolytes. The immobilization method failed due to enzyme wash-off, while the floating-enzyme configuration provided stable and measurable responses.

Both steady-state and real-time measurements confirmed successful glucose detection using GOx and GDH, with comparable qualitative responses. Preliminary observations suggest possible direct electron transfer (DET) in the GDH system, though not conclusively verified. Overall, both enzymes demonstrated potential for OECT-based glucose sensing.

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1 Introduction

Advances in healthcare and living conditions have increased global life expectancy, but with aging populations and lifestyle changes such as reduced physical activity and high-calorie diets, the prevalence of chronic diseases has also risen. Among these, diabetes mellitus is one of the most widespread metabolic disorders, affecting over 500 million people worldwide. [22] Its management relies heavily on maintaining glucose levels within a healthy range, since abnormal concentrations can lead to severe complications including kidney failure, neuropathy, retinopathy, and cardiovascular disease. [28]

Accurate and frequent glucose monitoring is therefore essential for both prevention and treatment. Conventional methods, such as finger-prick blood tests, urine glucose analysis, and continuous glucose monitors (CGMs), are widely used but suffer from limitations: they can be invasive, uncomfortable, or costly, and are not ideal for continuous real-time tracking. [17, 47] These drawbacks highlight the need for alternative sensing technologies that are less intrusive, more convenient, and compatible with wearable formats.

Organic Electrochemical Transistors (OECTs) have recently emerged as promising candidates for glucose sensing. [5, 19] Their low-voltage operation, strong signal amplification, and excellent coupling with biological media enable sensitive detection of biochemical reactions, while their scalability and compatibility with printed electronics make them suitable for next-generation wearable devices. [40] In OECT-based glucose sensors, enzymatic reactions are typically employed to convert glucose into electrochemically detectable signals.

Glucose oxidase (GOx) has been the most common choice, offering high specificity and well-studied kinetics. [19] Several studies have demonstrated that GOx-based OECTs achieve fast and sensitive glucose detection in complex media. However, GOx presents drawbacks such as oxygen dependence and reduced stability under physiological conditions, motivating the search for alternatives. Glucose dehydrogenase (GDH), which operates independently of oxygen, represents one such candidate but has not yet been systematically applied in OECT platforms. [55]

In this work, we investigate GDH-based OECTs for enzymatic glucose detection. Devices are studied under both steady-state and real time conditions using floating enzyme configurations in the electrolyte. While GOx is used as a reference, the focus lies on exploring the potential of GDH to provide stable and sensitive detection across physiologically relevant glucose ranges, thereby opening new opportunities for future wearable glucose monitoring systems.

2 Theory

2.1 Glucose and Its Importance in Biological Systems

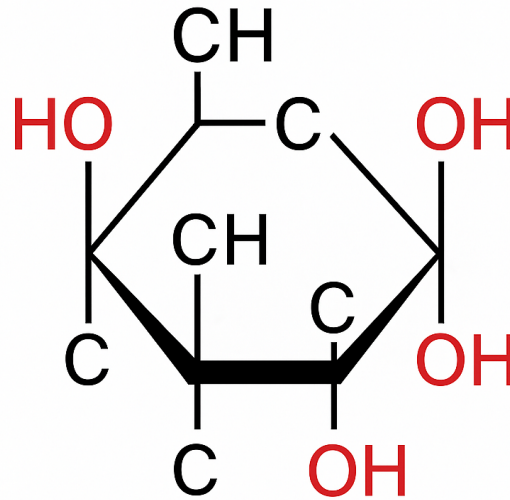


Figure 1: The chemical structure of a Glucose molecule

Glucose is a simple sugar that serves as the main energy source for living organisms. The chemical structure of a typical glucose molecule is depicted in Figure 1. It plays a central role in cellular metabolism, driving pathways such as glycolysis and the citric acid cycle [33]. In humans, glucose is obtained from the digestion of carbohydrates and carried through the bloodstream to body tissues, where it is converted into energy. Maintaining blood glucose levels within a healthy range is vital, as persistent imbalances can disrupt normal body functions. For instance, high blood glucose (hyperglycemia) can damage blood vessels and nerves over time, while low blood glucose (hypoglycemia) may cause dizziness, confusion, or even loss of consciousness [2, 4]. Diabetes mellitus, a chronic condition characterized by elevated glucose levels, remains one of the most pressing global health challenges, affecting millions worldwide [37].

In healthy individuals, the normal glucose concentration in blood typically ranges from 3.9–5.6 mmol/L (70–100 mg/dL) in the fasting state, rising to about 7.8 mmol/L (140 mg/dL) after meals [3, 53]. In contrast, glucose levels in other body fluids such as saliva and sweat are much lower, generally within 0.008–0.2 mmol/L and 0.06–0.2 mmol/L, respectively [26, 60]. These alternative biofluids have attracted interest for non-invasive glucose sensing applications.

Because of its importance, reliable monitoring of glucose levels is essential. Current approaches—such as fingerprick blood tests and continuous glucose monitoring devices—are widely used in clinical and personal health settings. However, these methods also present challenges. Fingerprick tests can be uncomfortable and inconvenient when performed multiple times a day, while continuous monitoring sensors, despite being less invasive, often require frequent calibration, sensor replacement, and can be costly. These drawbacks highlight the need for continued improvement in glucose sensing technologies, with the goal of providing patients with methods that are not only accurate and accessible but also less intrusive and easier to integrate into daily life.

2.2 Currently employed glucose testing techniques

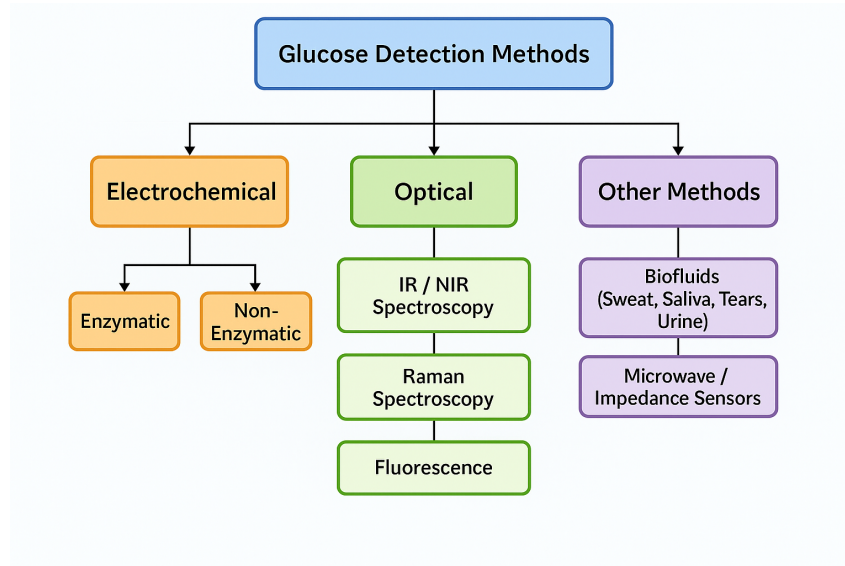


Figure 2: Glucose detection techniques

Glucose detection has progressed through multiple technological routes, with electrochemical methods forming the foundation of today's clinical and personal glucose monitoring systems. Their success was strongly presented by Wang's seminal review on electrochemical glucose biosensors. [51] Alongside these, optical approaches such as Near-Infrared (NIR), Raman, and fluorescence spectroscopy are being advanced as non-invasive techniques, [1] while emerging methods based on alternative biofluids and microwave/impedance sensing aim to support wearable and minimally invasive monitoring platforms. [59] The Figure 2 shows the flowchart of different glucose sensing techniques discussed in further sections.

2.2.1 Electrochemical Glucose Sensors

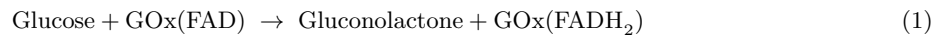
Electrochemical glucose sensors are among the most widely used technologies for glucose monitoring, and they operate by converting glucose oxidation into an electrical signal that can be measured as current, potential, or conductivity. These sensors are typically constructed as three-electrode systems consisting of a working, reference, and counter electrode. The working electrode is functionalized with either a biological recognition element, such as an enzyme, or a catalytic metallic material, enabling efficient glucose oxidation. Based on their signal transduction mechanism, electrochemical biosensors can be broadly categorized as amperometric, potentiometric, or conductometric. [51,57]

Amperometric glucose sensors, [6] which dominate commercial applications, measure the current generated at a constant potential as glucose undergoes enzymatic or direct oxidation at the electrode surface. The steady-state current is proportional to glucose concentration, making this method the foundation for finger-prick glucometers and continuous glucose monitoring systems. [51] Potentiometric glucose sensors measure the potential difference between a working and a reference electrode, which varies as a function of glucose concentration in the sample. [18] Conductometric glucose sensors instead monitor changes in electrical conductivity caused by enzymatic or catalytic reactions that produce or consume ionic species, thereby altering the solution's conductivity in proportion to glucose content. [15] Together, these classifications form the basis of electrochemical glucose sensing technology employed in clinical and personal monitoring devices.

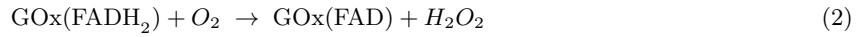
Depending on the catalytic element used at the electrode surface, electrochemical glucose sensors are broadly divided into two main categories: enzymatic sensors, which rely on biological catalysts, and non-enzymatic sensors, which utilize metallic or nanostructured materials for direct glucose oxidation. [45]

2.2.1.1 Enzymatic Electrochemical Glucose Sensors

Enzymatic electrochemical glucose sensors utilize enzymes such as glucose oxidase (GOx) immobilized on the electrode surface to selectively catalyze glucose oxidation. GOx is the most commonly applied enzyme because of its strong specificity toward glucose and proven reliability in clinical diagnostics. The sensing process involves a cascade of biochemical and electrochemical reactions. First, glucose is oxidized to gluconolactone, while the FAD cofactor of GOx is reduced:



The reduced enzyme is then regenerated by molecular oxygen, producing hydrogen peroxide:



Finally, the hydrogen peroxide undergoes electrochemical oxidation at the electrode, generating electrons that are measured as current:



In this cycle, the FADH₂ cofactor produced in Equation 1 is continuously reoxidized back to FAD by oxygen (or by artificial redox mediators in some sensor designs), enabling sustained enzyme activity. The gluconolactone formed is rapidly hydrolyzed to gluconic acid in aqueous solution, serving as the stable end product of glucose oxidation but not directly involved in electron transfer. The measurable signal therefore arises either from the electrochemical oxidation of H₂O₂ at the electrode (Equation 3) or from direct electron transfer through mediators. The resulting current is directly proportional to glucose concentration, providing the basis for modern glucometers that use disposable enzyme-coated strips as well as continuous glucose monitoring (CGM) devices that detect glucose in interstitial fluid. [18,57]

GOx has long been the gold standard in glucose biosensing due to its high specificity for glucose, well-established catalytic performance, and successful commercialization in numerous sensor platforms. Nevertheless, GOx-based sensors also face important limitations, including their dependence on oxygen as an electron acceptor, the accumulation of hydrogen peroxide that can foul electrodes, and sensitivity to pH and temperature that reduces long-term stability. [11] To overcome these drawbacks, glucose dehydrogenase (GDH) has emerged as a promising alternative. Unlike GOx, GDH relies on artificial redox mediators rather than oxygen, eliminating oxygen-related constraints and avoiding hydrogen peroxide formation. The catalytic mechanism of PQQ-dependent glucose dehydrogenase (PQQ-GDH) involves the oxidation of glucose to gluconolactone, coupled with the reduction of the enzyme-bound cofactor pyrroloquinoline quinone (PQQ) to its reduced form, PQQH₂, as shown in Figure 3. In this reaction, glucose donates two electrons and two protons (-2e^- , -2H^+) to PQQ, forming gluconolactone and PQQH₂. The reduced cofactor then transfers its electrons and protons to an external electron acceptor, such as phenazine methosulfate (PMS), which is reduced to PMSH₂. This two-step electron transfer ensures regeneration of the oxidized PQQ cofactor, enabling continuous enzymatic turnover. Importantly, this mechanism bypasses

the need for oxygen as the terminal electron acceptor, in contrast to GOx, making PQQ-GDH highly suitable for glucose biosensing applications in oxygen-limited environments. [55]

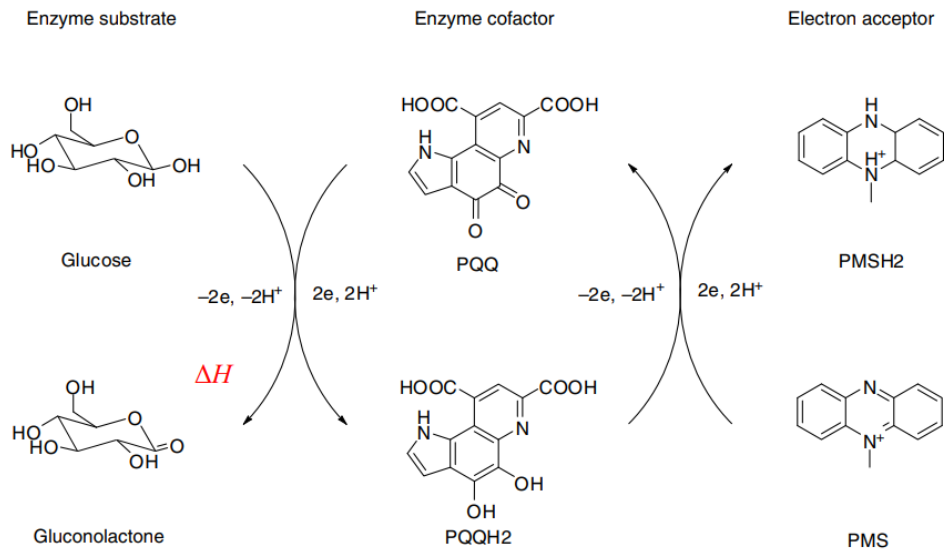


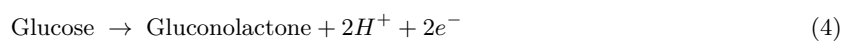
Figure 3: Principle of PQQ-GDH catalyzed glucose oxidation, adapted from Xie et al. [55]. Glucose is oxidized to gluconolactone while the enzyme cofactor PQQ is reduced to PQQH₂. The reduced cofactor subsequently transfers electrons to an external mediator such as PMS, regenerating PQQ and allowing continuous enzymatic turnover.

Certain GDH variants, such as FAD-GDH and PQQ-GDH, demonstrate improved stability and faster electron transfer kinetics compared to GOx, making them attractive candidates for next-generation wearable and implantable continuous glucose monitoring platforms [23]. Enzymatic glucose sensors are thus widely deployed in both clinical and personal healthcare, with GOx remaining the most established enzyme while GDH is increasingly being explored for overcoming the limitations of traditional designs. [45]

2.2.1.2 Non-Enzymatic Electrochemical Glucose Sensors

Non-enzymatic electrochemical glucose sensors eliminate the need for biological catalysts by relying on metallic or nanostructured electrodes that directly catalyze glucose oxidation. Materials such as nickel, copper, cobalt, platinum, and gold, along with their oxides and nanocomposites, have been extensively studied due to their high electrocatalytic activity and large effective surface areas. These properties enable direct electron transfer from glucose at the electrode surface without enzymatic mediation. [13, 27]

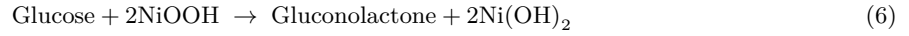
The overall oxidation of glucose can be summarized as:



In nickel-based electrodes, which are among the most investigated systems, catalysis proceeds through the reversible redox transition of nickel hydroxide (Ni(OH)₂) to nickel oxyhydroxide (NiOOH) under alkaline conditions. The first step is the electrochemical oxidation of Ni(OH)₂:



The NiOOH formed in this step then acts as an active catalytic species that oxidizes glucose to gluconolactone while being reduced back to Ni(OH)₂, thereby completing the catalytic cycle:



This continuous regeneration of NiOOH ensures sustained catalytic activity and a current output proportional to glucose concentration. [13, 27]

Non-enzymatic glucose sensors are currently being actively developed for next-generation wearable and implantable monitoring systems, and they are also being studied for use in alternative biofluids such as sweat and saliva. [8] Their main advantages include long-term operational stability, independence from oxygen concentration, reduced environmental sensitivity, and lower production costs compared to enzymatic systems. Nonetheless, they face challenges such as poor selectivity due to interference from other electroactive species and the requirement of strongly alkaline media for optimal performance, which limits direct clinical translation at present. [13, 27]

2.2.2 Optical Glucose Detection Methods

Optical glucose detection methods, [1] rely on the interaction of light with glucose molecules and include techniques such as near-infrared (NIR) spectroscopy, [7] Raman spectroscopy, [29] and fluorescence sensing. [39] IR and NIR spectroscopy measure the absorption of light at specific wavelengths corresponding to vibrational modes of glucose, which can be modeled by the Beer–Lambert law:

$$A = \epsilon c l \quad (7)$$

where A is absorbance, ϵ is the molar absorptivity, c is the concentration of glucose, and l is the optical path length. Raman spectroscopy, [29] by contrast, detects inelastic scattering of monochromatic light to provide a molecular fingerprint of glucose, offering high specificity but often requiring advanced signal enhancement to overcome interference from biological tissues. Fluorescence-based sensing employs glucose-sensitive fluorophores or nanoparticles that emit light upon glucose binding, producing an intensity or lifetime change proportional to concentration. These optical methods are attractive for their potential in non-invasive, continuous monitoring without the need for blood sampling. However, challenges such as weak signal intensity, interference from water and other biomolecules, and complex calibration procedures remain significant barriers to routine clinical use. [7]

2.2.3 Other Glucose Detection Methods

Other glucose detection methods aim to provide alternatives to electrochemical and optical sensing, with a focus on wearable, non-invasive platforms. Biofluid-based sensing detects glucose in sweat, saliva, tears, or urine, offering non-invasive sampling that can be integrated into devices like smartwatches, skin patches, or contact lenses. [8, 59] These approaches are particularly promising for continuous monitoring, though the correlation between glucose concentrations in alternative biofluids and blood glucose levels remains a major limitation. Beyond biofluids, microwave and impedance-based sensors exploit changes in dielectric properties or electrical impedance of tissues and fluids due to glucose variations. [30] Such methods avoid enzyme-related instability and reagent consumption, making them cost-effective and potentially long-lasting. However, they are still in the early stages of development, and issues of sensitivity, miniaturization, and robustness must be resolved before clinical deployment. [59]

2.3 Organic Electrochemical Transistor(OECT)

In 1984, White first proposed a variant of the transistor composed of three electrodes, an organic semiconducting channel, and an electrolyte medium, which laid the foundation for what would later be known as the Organic Electrochemical Transistor (OECT) which is depicted in Figure 4. [52] The operating principle of the OECT relies on the electrochemical coupling between the gate, the electrolyte, and the channel. When a voltage is applied at the gate electrode, ions from the electrolyte migrate into or out of the organic channel. This ion injection induces doping or dedoping of the channel material, thereby modulating its charge carrier density. As a result, the conductivity of the channel changes, and the output drain current (I_D) is directly regulated by this doping state. [5, 40] In other words, the channel acts as the active medium where the ionic–electronic interaction takes place, while the gate simply provides the driving force.

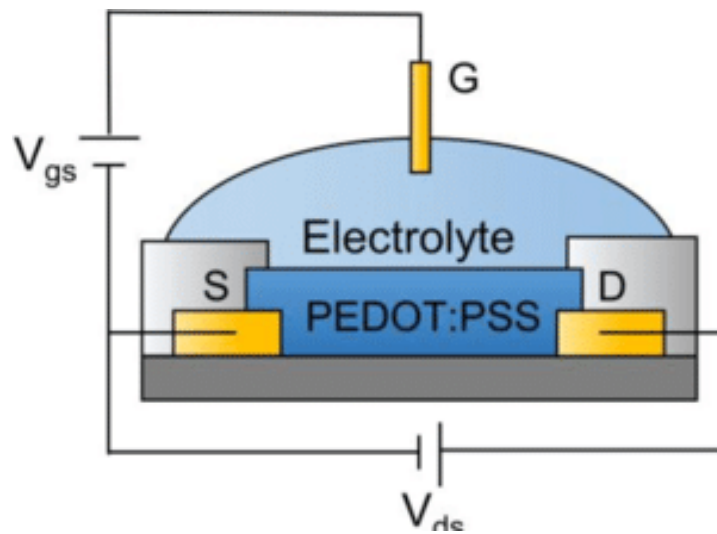


Figure 4: Schematic representation of a typical Organic Electrochemical Transistor (OECT), showing the source (S), drain (D), gate (G), and electrolyte-channel interaction.

Compared with conventional field-effect transistors (FETs), OECTs possess several distinctive advantages. They operate at low voltages, exhibit compatibility with aqueous electrolytes, and in many cases are intrinsically biocompatible. [21] In addition, they provide high signal amplification and are capable of transducing ionic fluxes into electronic signals, which is a particularly important feature since biological systems communicate primarily through ions, whereas conventional electronics rely on electrons. These characteristics make OECTs especially appealing for applications in biosensing, neuromorphic computing, and bioelectronic interfaces, where direct interaction with biological environments is required. [40]

Since the heart of OECT operation lies in the doping and dedoping dynamics of the channel, the choice of the channel material is crucial. Among the various organic semiconductors studied, poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) has emerged as the most widely employed material due to its high conductivity, mixed ionic–electronic transport capability, and electrochemical stability in aqueous media. [40, 58] Its unique core–shell morphology allows efficient charge transport while maintaining compatibility with a broad range of electrolytes. PEDOT:PSS has therefore become the benchmark channel material for OECT research and development. See Figure 5

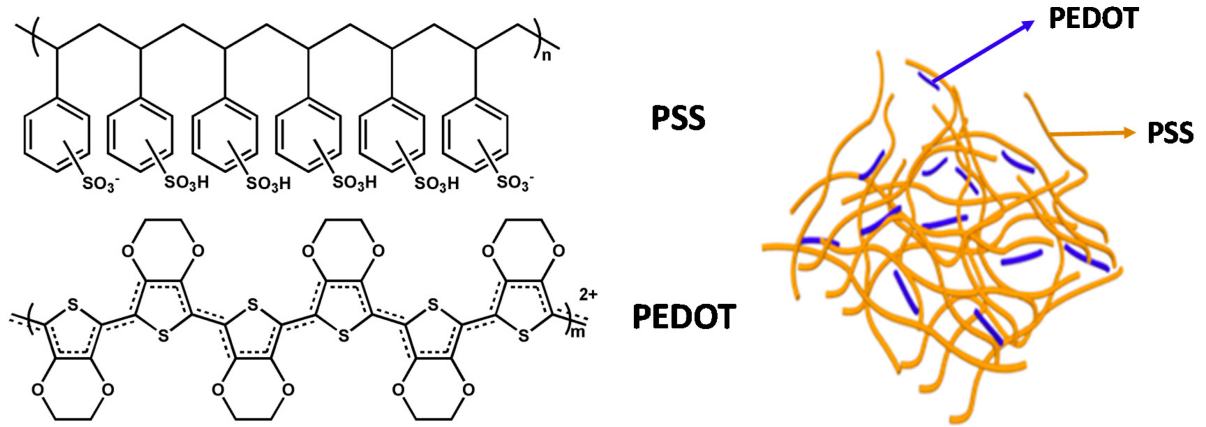


Figure 5: Chemical structure and schematic core-shell morphology of poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS). [20]

Thanks to these combined advantages, OEECTs are regarded as promising candidates for next-generation technologies such as biosensors, bioelectronic computing platforms, and flexible health monitoring systems. The next section will focus on the detailed operation of OEECTs, with an emphasis on how PEDOT:PSS channels enable efficient signal transduction.

2.3.1 p-type Depletion and Accumulation Mode in OEECTs

In OEECTs, the performance and stability of the device are strongly influenced by the nature of the channel material. P-type materials, particularly poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS), are predominantly employed due to their favorable electrochemical properties. PEDOT:PSS exhibits high electrical conductivity in its doped state and can be efficiently dedoped by cation injection from the electrolyte upon application of a gate voltage, enabling large modulation of channel current and high transconductance. [5,41] This volumetric doping process, where ions penetrate the bulk of the channel, enhances the sensitivity and dynamic range of OEECTs. [41] Moreover, p-type materials such as PEDOT:PSS are chemically stable in ambient and aqueous environments, which is a significant advantage over n-type materials. The development of n-type OEECTs has historically been limited by challenges such as poor air and water stability, lower electron mobility, and the scarcity of stable, high-performance n-type organic semiconductors. [16,44] In contrast, p-type OEECTs also offer inherent biocompatibility, softness, and processability, making them well-suited for bioelectronic applications, including neural interfaces and wearable sensors. [54]

The drain current in an OEECT when no gate voltage is applied is constant and is shown in Figure6. In the accumulation mode, the application of a negative gate voltage to a p-type OEECT pushes anions (negative charge carriers) to the organic semiconductor, resulting in the accumulation of holes (positive charge carriers) in the channel. This results in an increase in the density of charge carriers in the channel, which improves the conductivity and enables a greater drain current to travel from the source to the drain electrode. This is the accumulation mode, in which the transistor varies its current from low magnitude("off") to high magnitude("on") which is shown in Figure7.

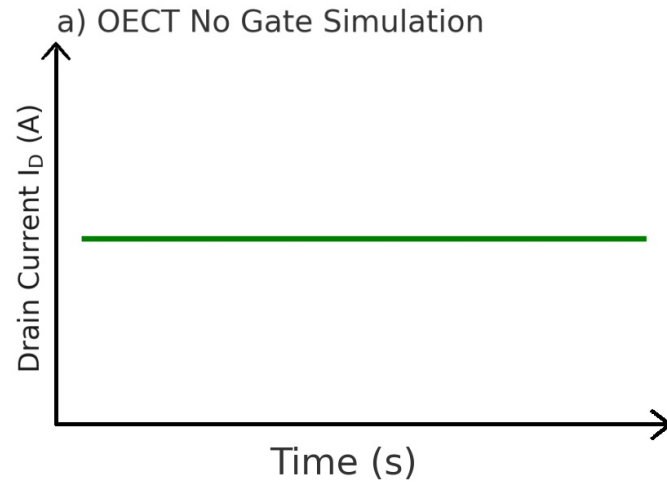


Figure 6: OEET with no gate simulation.

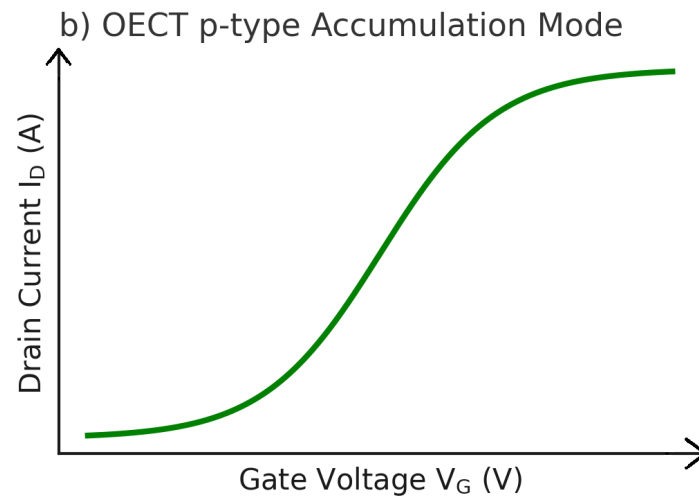


Figure 7: p-type OEET in accumulation mode.

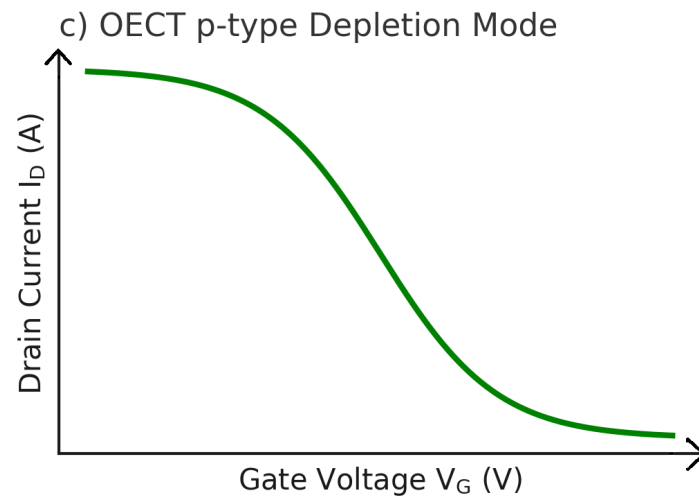


Figure 8: p-type OEET in depletion mode.

When a positive gate voltage is applied to a p-type OECT in the depletion mode, it attracts cations to the semiconductor, resulting in a depletion of holes in the channel. The discharge current is reduced as a result of the decrease in hole density, which in turn decreases the conductivity of the organic semiconductor. In this depletion mode, the transistor becomes “off” i.e. the current between the source and drain becomes either minimal or nonexistent and is shown in Figure 8. The OECT may be completely disabled, thereby obstructing current flow, if the gate voltage is sufficiently positive.

2.3.2 Electronic and Ionic Circuit

To model the device’s behaviour the OECT is divided into an electronic and ionic circuit. The electronic circuit in an OECT shown in Figure 9, refers to the pathway through which electronic charge carriers, typically holes in p-type materials like PEDOT:PSS flow from the source to the drain electrode through the organic semiconductor channel. The magnitude of this drain current (I_D) depends on the conductivity of the channel, which is modulated by the concentration of mobile charge carriers. [5] Unlike traditional field-effect transistors, where conductivity is controlled via interfacial electric fields, OECTs rely on bulk modulation: ions from the electrolyte penetrate the channel and change its doping level, thereby altering its conductivity. [50] This interaction effectively transforms the channel into a variable resistor whose resistance is controlled by the ionic environment set by the gate voltage. Thus, the electronic circuit in an OECT operates by transporting electronic charges under an applied drain-source voltage (V_{DS}), while its resistance is tuned by ion-induced doping, making it fundamentally different from purely electronic transistors. [5]

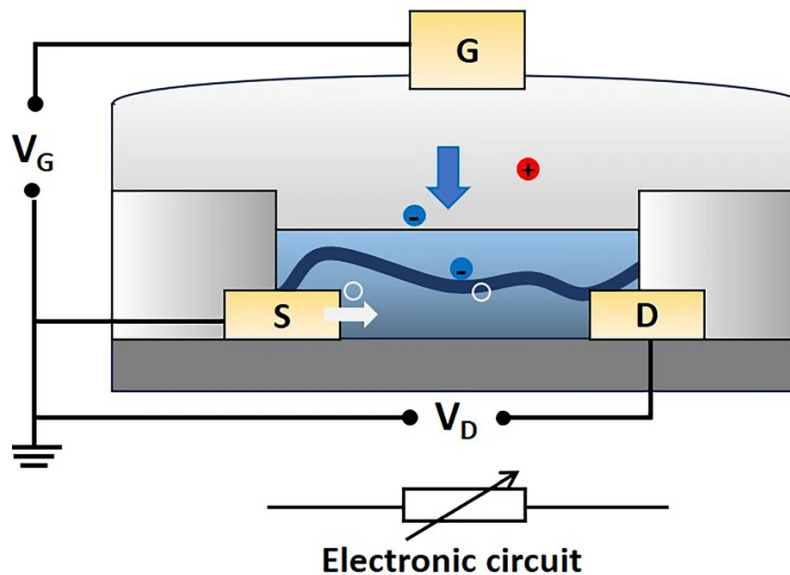


Figure 9: Diagram illustrating the electronic circuit of a p-type OECT structure. Upon application of V_G (<0 V), anions are injected into the channel, and are counterbalanced by the holes injected from source electrode, leading to the electrochemical doping of the channel. [61]

The ionic circuit in an OECT plays a key role in modulating the channel conductivity by governing the movement of ions from the electrolyte in response to an applied gate voltage. When a voltage is applied at the gate, ions in the electrolyte migrate toward either the gate or the channel, depending on the polarity, and redistribute to form electric double layers at the respective interfaces. [5] The electrolyte behaves like a resistive medium due to ionic

transport, while both the gate–electrolyte and electrolyte–channel interfaces behave as capacitive elements. [5,50] In the case of a platinum (Pt) gate electrode, the gate–electrolyte interface is non-faradaic, forming a stable electric double layer that acts as a pure capacitor rather than undergoing redox reactions. [61] However, as noted by Bernards and Malliaras, [5] this large double-layer capacitance causes most of the applied gate voltage to drop across the gate–electrolyte interface, resulting in weaker voltage coupling to the electrolyte–channel interface and thus less efficient modulation of the channel conductivity. In contrast, redox-active gates such as Ag/AgCl can transfer the applied potential more effectively to the channel by undergoing Faradaic reactions, thereby enabling stronger channel modulation. Hence, the ionic circuit—with its combination of resistive and capacitive elements—can be modeled as a series resistor and capacitors that define both the transient and steady-state behavior of the device, as depicted in Figure.10.

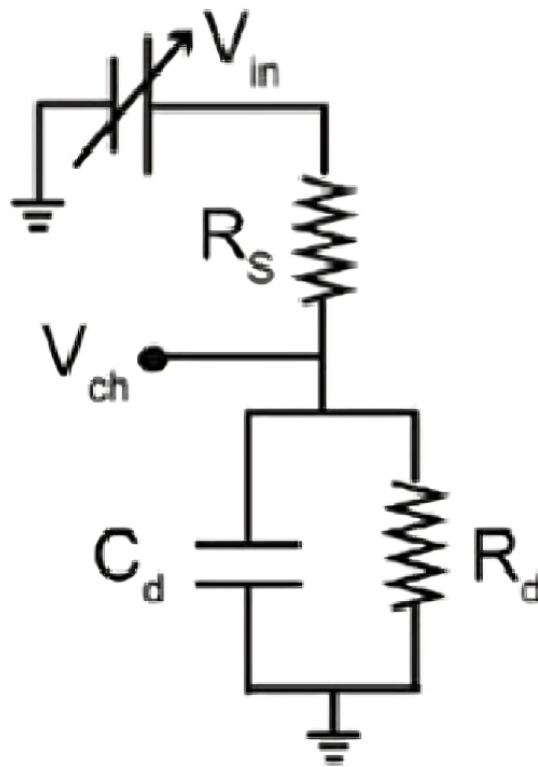


Figure 10: Equivalent ionic circuit model of an OEET, where R_s represents the ionic resistance of the electrolyte, C_d and R_d represents the channel capacitance. [10]

2.3.3 Transient-State Behaviour in Organic Electrochemical Transistors

The transient response of OEETs describes how the drain current evolves over time when a gate voltage is applied (refer to Figure 12). This behavior was first modeled by Bernards and Malliaras using a distributed RC transmission-line framework, where the channel is represented as a ladder of resistors and capacitors that capture the coupled ionic and electronic dynamics. [5] Figure11 illustrates this equivalent circuit for a differential channel element.

The time constant, τ , characterizes the dynamic response of an OEET when the gate voltage is switched on or off. It reflects how quickly the drain current evolves from its initial value toward a new steady state as ions move into

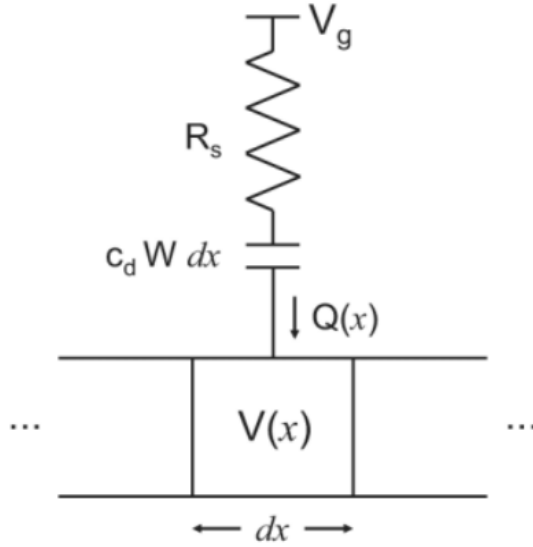


Figure 11: Equivalent circuit of a differential channel element dx in the transient OECT model. [5] The resistor R_s represents ionic resistance, while the capacitor C_d accumulates ionic charge that modulates the local channel potential.

or out of the channel. Physically, τ can be described by an effective RC constant, $\tau = RC$, where R represents the ionic resistance of the electrolyte and C the double-layer capacitance at the channel/electrolyte interface. Conventionally, τ is defined as the time required for the drain current to reach 63% of its final value after a step change in gate voltage. A smaller τ indicates faster charging and discharging kinetics, which is desirable for applications such as biosensing and neuromorphic computing where rapid signal processing is required. Thus, τ provides a direct measure of the response speed of the OECT.

The change in drain current, ΔI_D , quantifies the modulation depth of the OECT channel under the influence of a gate voltage. It is defined as the difference between the steady-state current during the gate pulse ($I_{D,on}$) and the baseline current before the pulse ($I_{D,off}$). A larger ΔI_D corresponds to stronger modulation of channel conductivity, which is directly linked to the sensitivity of the device. In biosensing applications, ΔI_D reflects how effectively biochemical interactions at the channel/electrolyte interface are transduced into an electrical signal. While τ determines how quickly this change occurs, ΔI_D determines how strongly the device responds. Together, they provide complementary insight into OECT performance, combining both sensitivity and temporal resolution.

2.3.4 Steady-State Behavior in Organic Electrochemical Transistors

When a gate voltage is applied to an OECT, ions from the electrolyte migrate into or out of the organic channel and thereby dope or dedope the semiconductor. Once the ionic distribution and electronic carrier density stop evolving in time, the device reaches the steady state. In this regime, the channel conductivity is fixed by the local ionic compensation of electronic charges along the channel length, and the drain current is fully determined by the spatial electrostatic potential profile. [5,40] Rather than deriving the full transport equations, device behavior is usually described by standard characterization curves.

In the linear (ohmic) regime ($|V_D| < |V_G - V_{th}|$), the transfer characteristics (drain current versus gate bias at

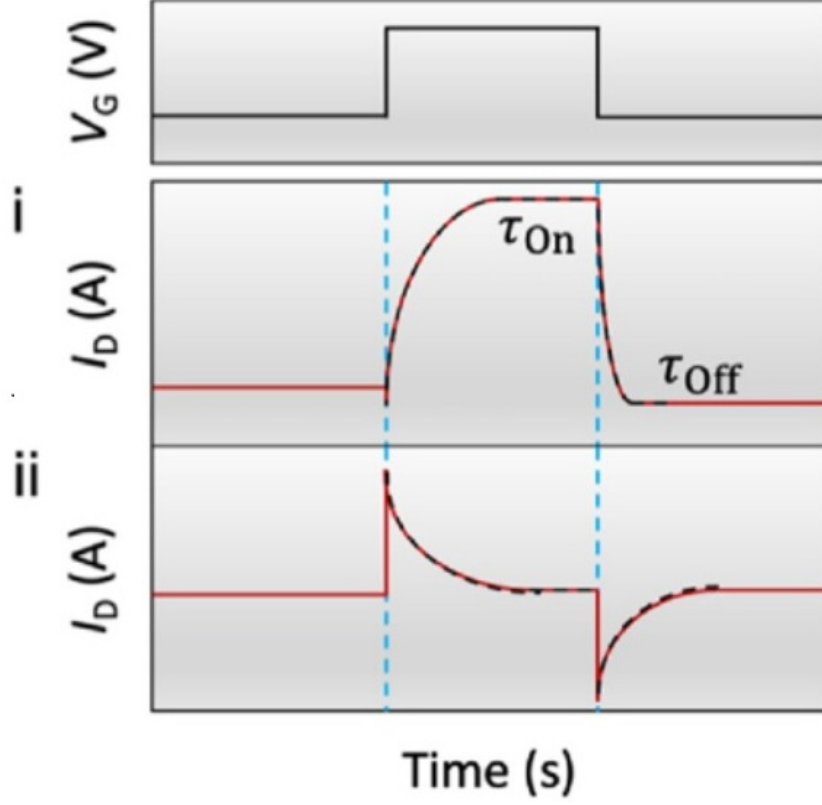


Figure 12: The transient characteristics of an OEET where I_D in response to a square V_G pulse is measured while V_D is maintained constant. Two main types of behaviours are observed: a monotonic relaxation of I_D towards the steady-state value (i) or a "spike and recovery" behavior (ii). The fit is an exponential decay function (black dotted lines), from which the characteristic time is extracted. [36]

fixed V_D) follow the widely used relation

$$I_D \approx \frac{\mu C^* W T}{L} \left[(V_G - V_{th}) V_D - \frac{1}{2} V_D^2 \right], \quad (8)$$

where μ is the electronic mobility, C^* the volumetric capacitance of the channel (F m^{-3}), W, T, L are the channel width, thickness, and length, and V_{th} is the threshold potential for redox-driven modulation. [5, 40] The corresponding small-signal transconductance at fixed V_D is

$$g_m = \frac{\partial I_D}{\partial V_G} \approx \frac{\mu C^* W T}{L} V_D, \quad (9)$$

i.e., proportional to V_D and independent of V_G in this regime. [5]

At larger drain bias, when $|V_D| \geq |V_G - V_{th}|$, the channel pinches off near the drain and the device enters saturation, for which

$$I_{D,\text{sat}} \approx \frac{\mu C^* W T}{2L} (V_G - V_{th})^2, \quad (10)$$

$$g_{m,\text{sat}} = \frac{\partial I_{D,\text{sat}}}{\partial V_G} \approx \frac{\mu C^* W T}{L} (V_G - V_{th}), \quad (11)$$

making the threshold dependence explicit in the saturated transconductance. [5, 40]

The output characteristics (drain current versus drain bias at fixed gate voltage) reveal the transition from the ohmic region to current saturation as the channel becomes fully dedoped near the drain, consistent with depletion-mode operation for PEDOT:PSS. [5, 40] In addition to DC characterization, electrochemical impedance

spectroscopy (EIS) is commonly used to extract C^* and ionic resistances by fitting equivalent-circuit models that include the gate–electrolyte capacitance, electrolyte resistance, and a parallel $R_{\text{ch}} \parallel C_{\text{ch}}$ branch for the channel to capture the mixed ionic–electronic response. [34, 61] These models generalize earlier purely resistive channel descriptions and account for the frequency-dependent behavior observed in modern PEDOT:PSS OECTs.

Moreover the dependence of the drain current (I_d) on the drain voltage (V_d) at a constant gate bias (V_g), illustrate the dual nature of conduction in p-type OECTs (refer to Figure 13). At low drain voltages, the device operates in the linear regime, where holes in the polymer channel sustain conduction and I_d increases proportionally with V_d . As V_d increases further, ions from the electrolyte penetrate the channel and locally compensate the hole carriers near the drain. This dedoping process progressively reduces the hole density, eventually creating a region with very low conductivity. The device then enters the saturation regime, where the drain current reaches a plateau. [5, 14] Unlike in conventional MOSFETs, where pinch-off arises purely from an electric field effect, in OECTs the transition to saturation is governed by volumetric ion penetration and electrochemical modulation of the channel. The exact saturation behavior in p-type OECTs is strongly influenced by parameters such as the volumetric capacitance of the polymer, ion mobility in the electrolyte, channel thickness, and gating efficiency at the electrolyte-channel interface. [40] This mixed ionic-electronic mechanism explains why the saturation regime in OECTs differs fundamentally from that in classical FETs.

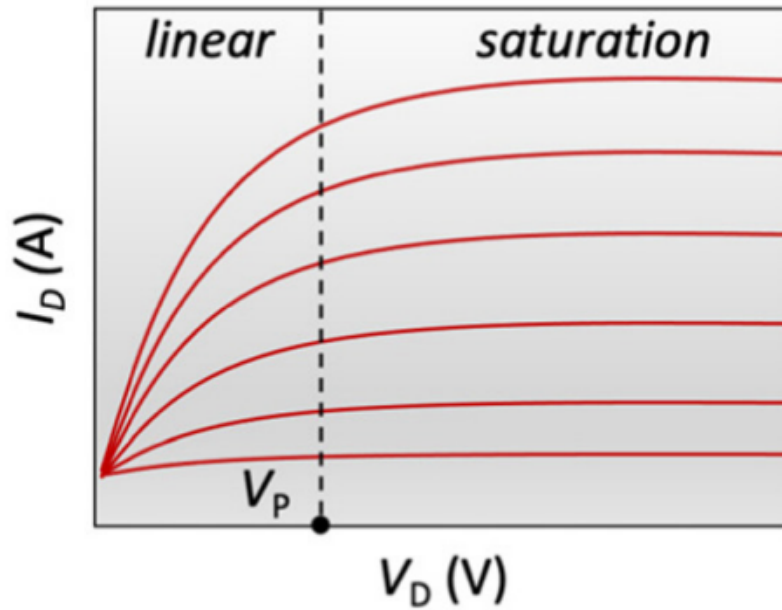


Figure 13: Output characteristics of an OECT showing drain current (I_D) versus drain voltage (V_D) at different fixed gate voltages (V_G). The onset of the saturation regime and the pinch-off voltage (V_P) are indicated. [36]

The transfer characteristics of an OECT, which plot the drain current I_d as a function of gate voltage V_g at a constant drain voltage V_d , provide insight into how effectively the gate regulates channel conductivity (refer to Figure 14). The on current, I_{on} , represents the maximum current obtained when the channel is fully doped and thus highly conductive. In a typical PEDOT:PSS-based OECT, this condition occurs at $V_g = 0$ V, where the polymer remains in its pristine, p-doped state.

The off current, I_{off} , is the minimum current measured when V_g becomes sufficiently high in magnitude to dedope the channel. In this state, the injection of ions compensates the mobile holes, significantly reducing the channel's

conductivity. [40]

These two current values define the on-off current ratio, which is expressed as:

$$\text{On-Off Ratio} = \frac{I_{\text{on}}}{I_{\text{off}}} \quad (12)$$

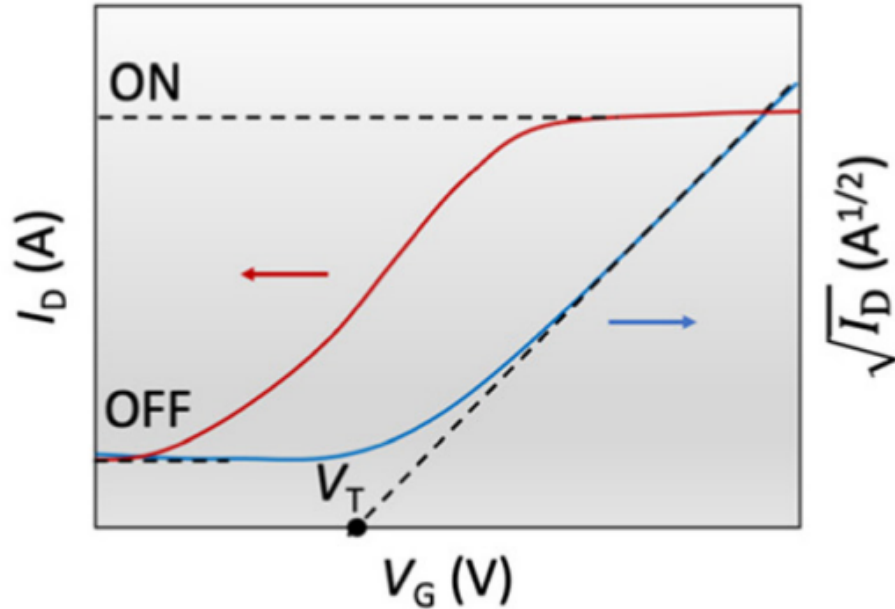


Figure 14: Transfer characteristics showing I_D versus V_G (red curve) of an accumulation mode OEET. The ON/OFF ratio is determined by the ratio of the maximum saturation current (I_{ON}) to the minimum saturation current in the OFF state (I_{OFF}). The square root of I_D versus V_G (blue curve) is used to extract the threshold voltage (V_T) from the x-axis intercept. [36]

This ratio serves as a critical performance indicator, especially for digital switching and biosensing applications, where the ability to clearly distinguish between "high" and "low" current states is vital. [16] On-off ratios in OEETs vary widely depending on the device design and materials used. Devices with narrow channels, high charge-carrier mobility, and low-leakage electrolytes may achieve values exceeding 10^5 , whereas devices with thicker channels or limited ionic mobility may show lower ratios in the range of 10^2 to 10^3 . [25]

The threshold voltage, V_{th} , marks the gate voltage at which the drain current begins to deviate significantly from its baseline, indicating the onset of electrochemical modulation in the channel. A common method for extracting V_{th} involves plotting $\sqrt{I_d}$ versus V_g and extrapolating the linear region back to the voltage axis, based on the relation:

$$I_d \propto (V_g - V_{\text{th}})^2 \quad (13)$$

Although this quadratic approximation originates from traditional FET theory, it provides a consistent framework for comparative analysis across different OEET materials and architectures. In OEETs, V_{th} is influenced by a range of factors, including the redox potential of the polymer, ion diffusion rates, the pH and ionic strength of the electrolyte, and even device aging. [21] Shifts in V_{th} over time may reflect changes in stability or the presence of bias-stress effects, which are especially important in long-term sensing applications.

Beyond material properties, the on-off ratio is also influenced by electrolyte composition, channel geometry, and the electrochemical potential window of the gate electrode.

Transconductance, g_m , defined as

$$g_m = \frac{dI_d}{dV_g} \quad (14)$$

is a central figure of merit in OECTs, as it quantifies how sensitively the drain current responds to changes in gate voltage. Unlike conventional FETs, where gating modulates only the surface charge at the semiconductor–insulator interface, OECTs operate through volumetric doping, meaning that the entire bulk of the channel is electrochemically coupled to the gate potential. This mechanism allows OECTs to achieve transconductance values orders of magnitude higher than those of similarly sized FETs. [40]

Following the Bernards–Malliaras model, the steady-state expression for the transconductance of a depletion-mode OECT is given by:

$$g_m = \frac{WT}{L} \mu C^* \left(V_g - V_{th} - \frac{V_d}{2} \right) \quad (15)$$

where W , T , and L are the channel width, thickness, and length, respectively; μ is the electronic mobility; C^* is the volumetric capacitance of the channel; V_g is the applied gate voltage; V_{th} is the threshold voltage; and V_d is the drain voltage. [5]

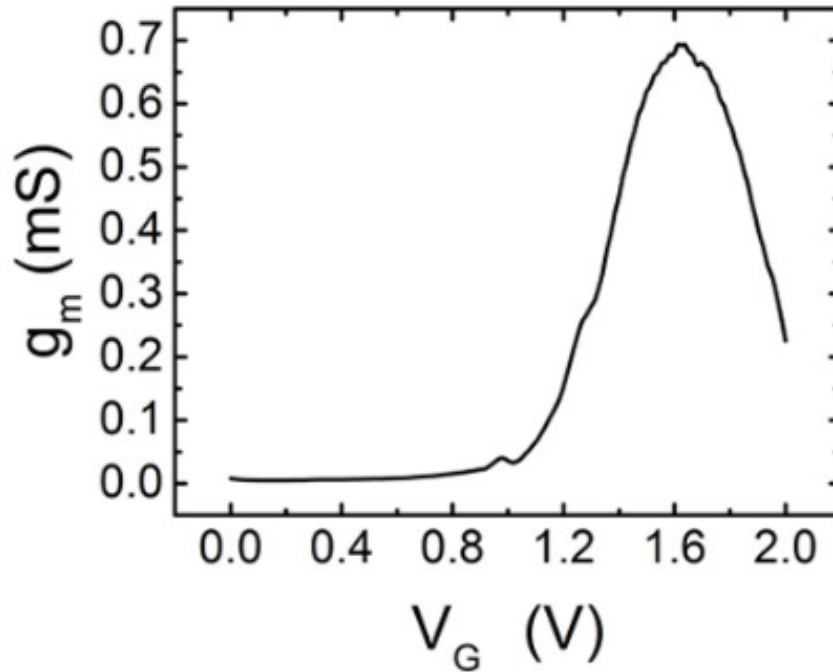


Figure 15: Transconductance characteristics of an OECT with fixed $V_D = -0.4$ V, showing the bias point where maximum amplification is achieved. [42]

This relation highlights that g_m scales linearly with the product μC^* , making these intrinsic material properties key performance drivers. Importantly, g_m is not constant with gate bias but reaches a maximum at an intermediate V_g . This bias point corresponds to the operational voltage at which ionic injection and electronic conduction

are optimally balanced, delivering the highest amplification. [40] Such a condition defines the practical working window of OECTs for sensing and analog signal processing (refer to Figure 15).

From a practical perspective, each electrical figure of merit plays a complementary role. A high I_{on} ensures a strong readout signal, while a low I_{off} minimizes background leakage. A large on-off ratio improves binary switching performance, and a high transconductance amplifies weak inputs to enhance sensitivity. Finally, the threshold voltage, V_{th} , defines the operational window and provides insight into device stability. Together, these parameters form the foundation for evaluating and optimizing OECT performance.

2.3.5 OECT in Enzymatic Sensing

OECTs functionalized with GOx have proven to be a highly effective platform for enzymatic glucose sensing. Their ability to operate in aqueous environments at low voltages, along with inherent signal amplification, makes them particularly attractive for biosensing applications. In this configuration, GOx catalyzes the oxidation of β -D-glucose to gluconolactone, producing hydrogen peroxide (H_2O_2) as a byproduct in the presence of oxygen. The electrochemical oxidation of H_2O_2 at the platinum gate electrode generates electrons that modulate the gate potential, introducing a redox-mediated mechanism of gate modulation that distinguishes OECTs from purely capacitive field-effect transistors. [5]

As glucose is consumed, the resulting H_2O_2 oxidation and electron flow at the gate effectively increase the gate potential. Bernards and Malliaras demonstrated that the impact of glucose on the transistor can be captured phenomenologically by a concentration-dependent voltage offset, V_{offset} , such that:

$$V_{g,\text{eff}} = V_g + V_{\text{offset}}. \quad (16)$$

This empirical approach collapses transfer curves obtained at different glucose concentrations onto a single universal curve. The observation indicates that the primary influence of glucose is to shift the apparent gate voltage axis, consistent with the Faradaic modulation mechanism, without altering the intrinsic transistor characteristics.

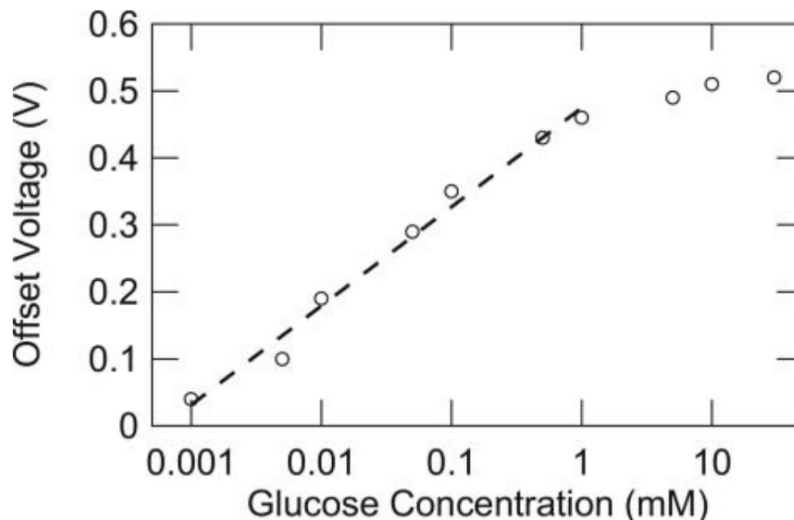


Figure 16: Offset voltage as a function of glucose concentration, fitted up to 1 mM, indicating Nernstian behavior. [5]

The modulation of gate potential by enzymatic reactions in OECTs follows a Nernstian trend, where the redox

potential shifts logarithmically with the ratio of oxidized to reduced species:

$$E = E^0 + \frac{RT}{nF} \ln \left(\frac{[\text{Ox}]}{[\text{Red}]} \right) \quad (17)$$

In glucose sensing, H_2O_2 acts as the oxidized species, with its concentration directly linked to glucose levels. Bernards et al. experimentally demonstrated a logarithmic increase in gate offset voltage with increasing glucose concentration up to 1 mM, consistent with Nernstian behavior (refer to Figure 16. [5]).

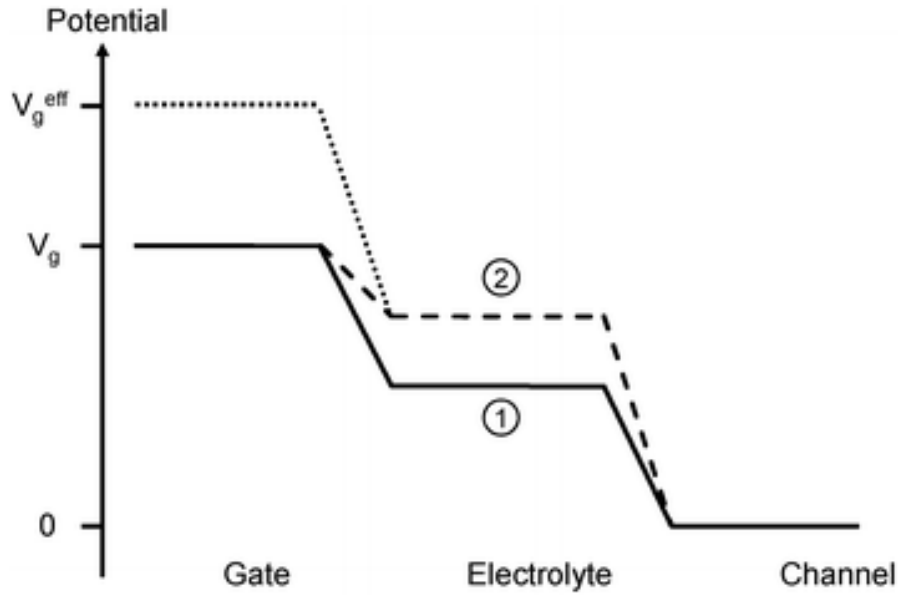


Figure 17: Potential diagram showing changes in electrolyte and effective gate voltage before and after glucose introduction. [5]

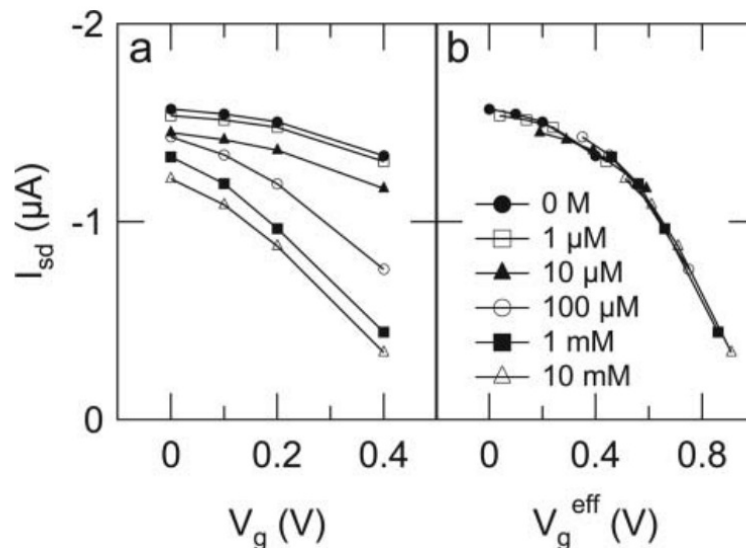


Figure 18: (a) Transfer curves shifting with glucose concentration. (b) Same data plotted against $V_{g,\text{eff}}$, collapsing onto a universal curve. [5]

The potential profile across the gate-electrolyte-channel system shifts depending on the presence of glucose. In the absence of glucose, the system behaves primarily as a capacitive element. When glucose is introduced, enzymatic reactions at the gate elevate the electrolyte potential, resulting in an increase in the effective gate voltage $V_{g,\text{eff}}$.

This shift modulates the transistor's operating point without changing the externally applied gate bias (refer to Figure17).

Further validation of this mechanism is shown in the transfer characteristics (refer to Figure18). Figure18a presents the source-drain current (I_d) versus gate voltage (V_g) curves, which shift with increasing glucose concentration due to changes in $V_{g,\text{eff}}$. When the same data are replotted against $V_{g,\text{eff}}$, the curves collapse onto a single universal curve, demonstrating that the observed changes in transistor behavior are primarily driven by Faradaic modulation of the gate (refer to Figure18 b). [5]

Beyond the foundational work of Bernards et al., the GOx-based OECT sensing architecture has been further developed by several research groups. Prof. Feng et al. demonstrated stable real-time glucose detection in saliva using GOx-functionalized OECTs, [56] while Mohamed et al. reported innovative cylindrical shape OECTs for glucose detection in sweat. [32]. Other researchers have integrated these systems into flexible and microfluidic platforms for minimally invasive monitoring in biofluids such as saliva. [38]

3 Materials and Methods

3.1 Mask Design and Writing Procedure

The mask design process begins with the creation of electrode and path geometries using AutoCAD, a widely adopted tool in semiconductor layout design. The layout must be composed of closed polylines or polygons to ensure that the shapes are properly recognized during subsequent processing. In this specific case, the chip has dimensions of 21 mm $\times$ 21 mm and incorporates three square channels of 500 μm $\times$ 500 μm each, together with three in-plane platinum gate electrodes (GT1, GT2, and GT3) of varying sizes. Although these in-plane gates are included in the chip layout, they are not employed in the present experiments. Instead, a platinum-coated glass substrate electrode is used as the gate, positioned in an overhanging configuration above the channel and dipped into the electrolyte during testing. An image of the chip layout is provided in Figure 19, illustrating the spatial arrangement of the channels and gate electrodes within the chip area.

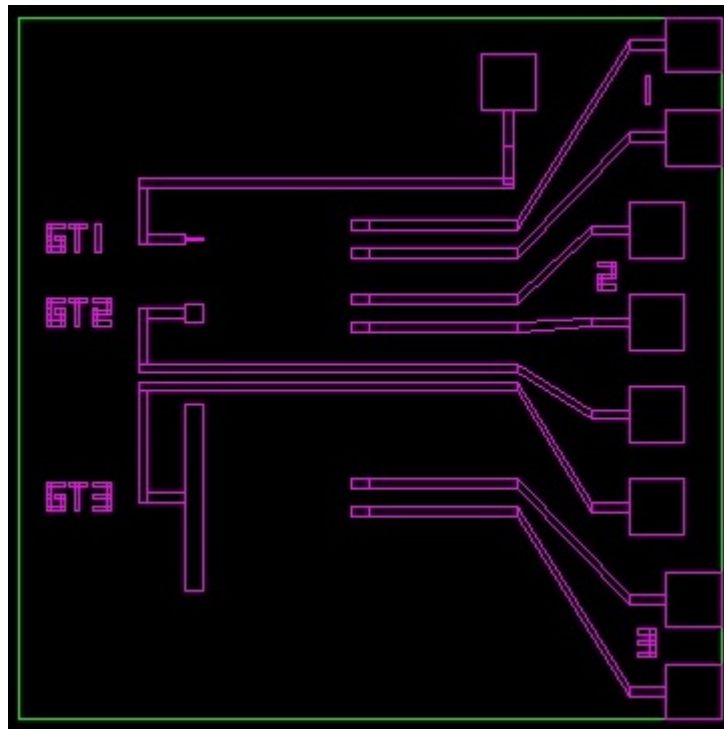


Figure 19: CAD design of the fabricated chip, showing gates GT1, GT2, and GT3 on the left, and the corresponding transistor channels on the right.

Once the AutoCAD layout is complete, users must move and align each chip design by snapping the bottom-left corner to the top-right corner of a positioning square within the pre-defined wafer template. These squares ensure uniform placement of chips across the wafer. Any extraneous structures outside the wafer boundary must be removed. The file is then saved in .dxf format using a recent AutoCAD version (2018 or later).

3.2 Sample Fabrication

3.2.1 Deposition and Patterning of Electrodes

The process starts from a thermally oxidized silicon wafer, which provides a SiO_2 insulating base of about 500 nm thick for electrical isolation and mechanical support. Before any polymer is applied, the surface is silanized

with APTES to promote adhesion of the subsequent polyimide (PI) coating. The first PI layer is spin-coated and cured, serving primarily as an adhesion/buffer layer for the metal. A thin Au film is then sputtered on top and patterned into source, drain, and gate interconnects using a first photoresist (PR1): after UV exposure and development, the exposed Au is etched and PR1 is stripped, completing the first photolithography.

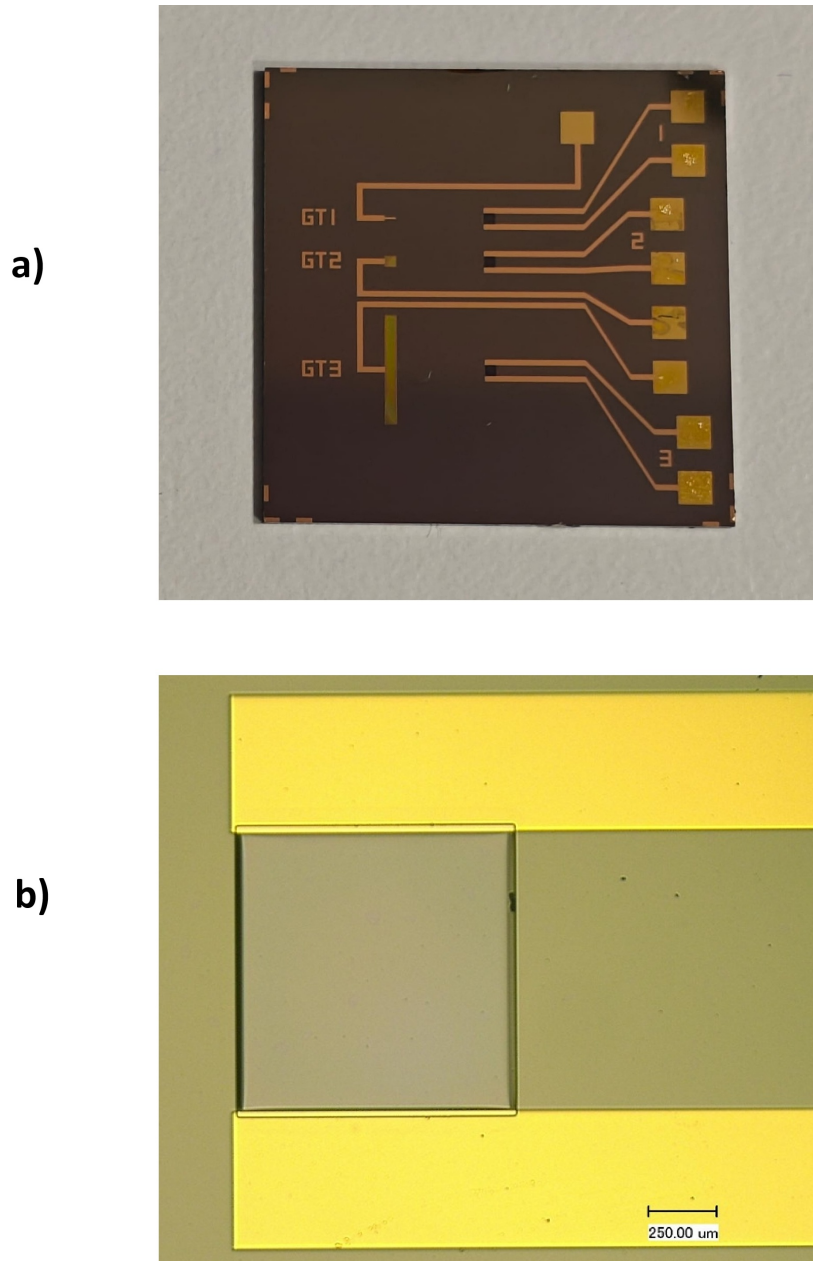


Figure 20: a) Optical image of the fabricated OECT chip. b) The zoomed image of the channel which separates the gold source and drain electrodes appearing in yellow by a $500\text{ }\mu\text{m} \times 500\text{ }\mu\text{m}$ square window. The scale bar represents $250\text{ }\mu\text{m}$.

Next a third PI layer, comparable in thickness to the first one, is deposited on top to act as a sacrificial layer for later pattern definition. Only after this sacrificial PI is in place, a thicker second photoresist (PR2) is applied to withstand the subsequent long reactive-ion etch (RIE) step (about 12 min) used to open the contact vias and define the channel regions. After exposure and development of PR2, the RIE defines the openings through the PI, and PR2 is subsequently removed. Finally, PEDOT:PSS is deposited and patterned. The sacrificial third PI layer is lifted off only after PEDOT:PSS processing, releasing the final channel geometry without damaging the

polymer film.

This multistep process enables the realization of OECT devices with high precision, multilayer insulation, and well-defined electrode and channel structures suitable for flexible electronics, biosensing, and neuromorphic applications. A microscopic image of the fabricated OECT channel is shown in Figure 20, where the gold source and drain electrodes appear yellow, separated by a $500\text{ }\mu\text{m} \times 500\text{ }\mu\text{m}$ square window that defines the channel area. The deposition and patterning of the active channel material, PEDOT:PSS, which completes the device structure, will be described in the following section.

3.2.2 Deposition of PEDOT:PSS

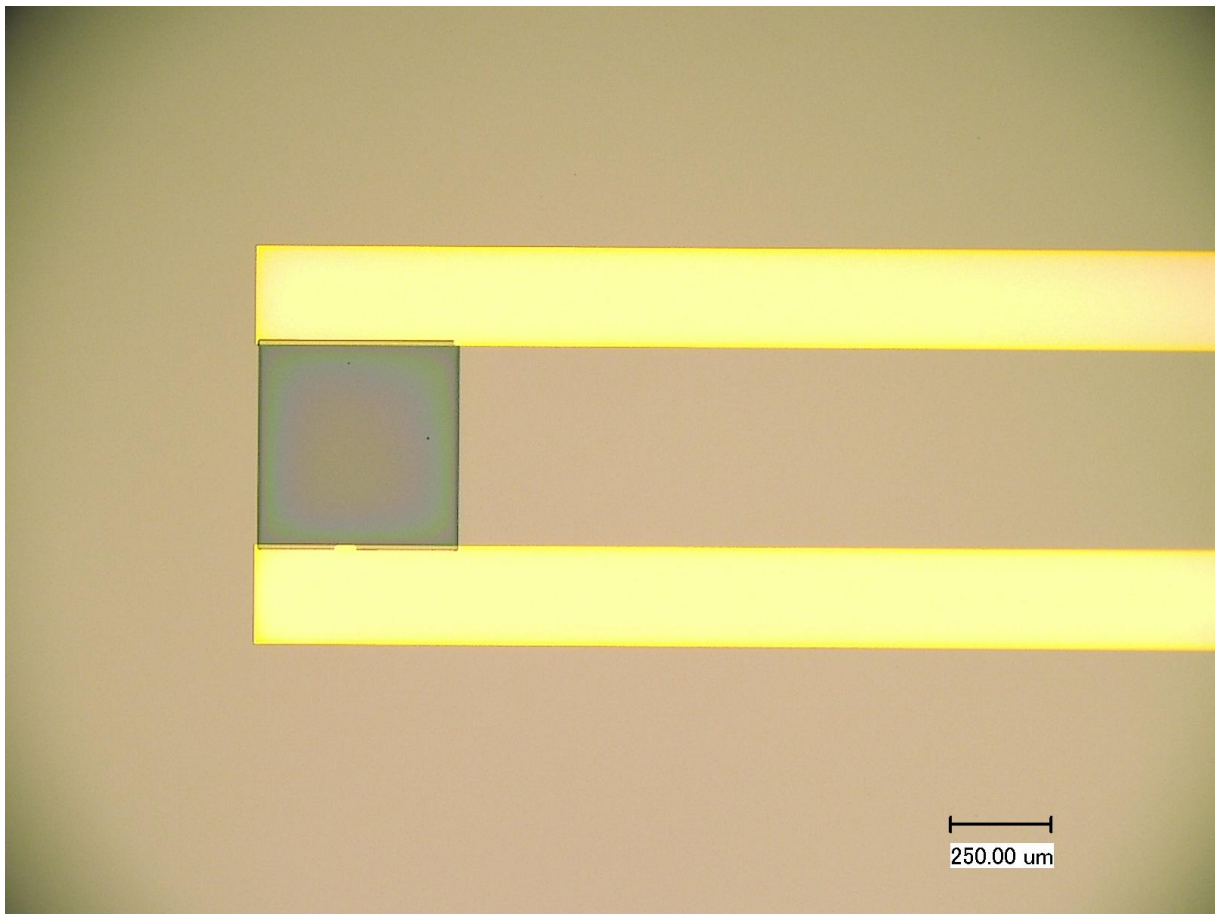


Figure 21: a) Optical image of the fabricated chip after deposition of the mPEDOT:PSS layer. b) Microscopic image of a single OECT channel after sacrificial layer removal. The gold source and drain electrodes are visible in yellow, and the active polymer film uniformly fills the $500\text{ }\mu\text{m} \times 500\text{ }\mu\text{m}$ channel window. The scale bar represents $250\text{ }\mu\text{m}$.

To prepare the active channel layer, a modified PEDOT:PSS (mPEDOT:PSS) solution was formulated to enhance electrical performance and film quality. The composition of the solution was PEDOT:PSS (Clevios PH1000) with 5% ethylene glycol (EG), 0.1% dodecylbenzene sulfonic acid (DBSA, 70% wt. in water), and 1% (3-glycidyloxypropyl)trimethoxysilane (GOPS). All components were mixed thoroughly and stirred for 15 minutes to ensure homogeneity. Approximately $200\text{ }\mu\text{L}$ of the prepared solution was then spin coated onto the chip at 1000rpm for 1 minute, specifically targeting the exposed channel region between the source and drain electrodes. The substrate was baked on a hot plate at 110°C for 10 minutes to facilitate solvent evaporation and promote uniform polymer film formation. This single-step deposition of the modified formulation provided a balance between

adhesion, film stability, and electrical conductivity, eliminating the need for a secondary coating step. The final fabricated OECT with mPEDOT:PSS-coated channel after sacrificial layer removal is shown in Figure 21. While initial tests were also carried out using pure PEDOT:PSS without additives, the modified formulation described here was adopted as the main deposition process. Any variations of this method are clarified in the corresponding experimental sections.

3.2.3 GOx enzyme in electrolyte preparation



Figure 22: Commercial glucose oxidase (GOx) used in this study.

GOx is a flavoprotein enzyme (EC 1.1.3.4) commonly derived from *Aspergillus niger* or *Penicillium* species (refer to Figure 22). It contains a flavin adenine dinucleotide (FAD) cofactor at its active site. GOx specifically catalyzes the oxidation of β -D-glucose to gluconolactone, while the FAD cofactor is reduced to FADH₂. The enzyme is highly selective for glucose, which makes it the gold standard in glucose biosensing. The enzymatic reaction pathway of GOx with glucose has already been described in Section 2.2.1.1. This section provides details about electrolyte preparation using GOx enzyme. Working enzyme solutions were prepared at a constant concentration of 500 U/mL.

A commercially available Phosphate Buffered Saline (PBS) tablet was dissolved in 200 millilitres of deionised water to create a 1× buffer solution with a pH of roughly between 7.2-7.6 that was used to make the PBS. In order to accommodate the various ideal conditions needed by each enzyme, PBS was utilised as the buffer for GOx (refer to Figure 23).

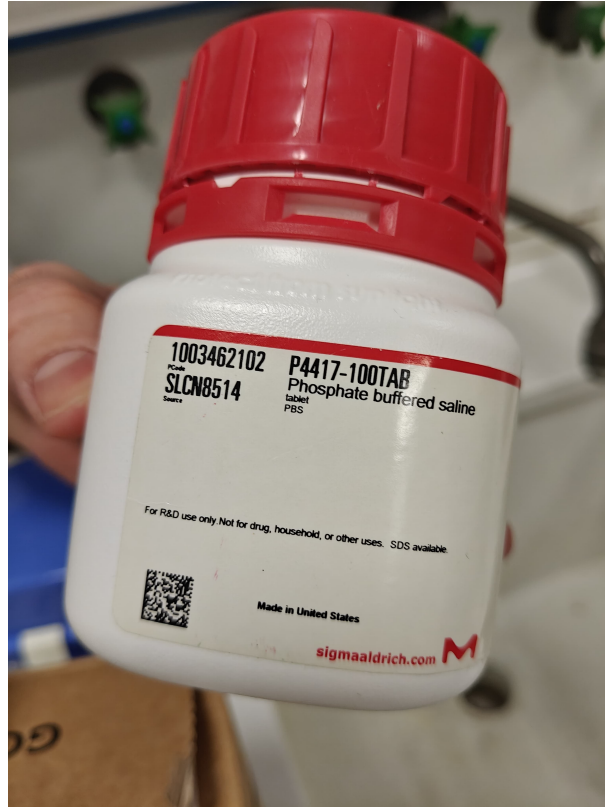


Figure 23: Commercially available PBS tablet used to prepare buffer solution.

3.2.4 GDH enzyme in electrolyte preparation

The detection of glucose using OECTs combined with the enzyme GDH offers a promising alternative to traditional sensing methods. Unlike GOx, GDH operates independently of oxygen and avoids the formation of hydrogen peroxide, which makes it particularly attractive for biosensing applications requiring long-term stability and reduced interference. [31] The GDH enzyme used for testing with OECT is shown in Figure 25.

The electrolyte preparation was optimized to support GDH activity and ensure efficient electron transfer. The formulation contained GDH at a concentration of 1 mg/mL, combined with pyrroloquinoline quinone (PQQ) from a 4 mM stock solution, and buffered in MES solution (5 mM MES + 0.5 mM CaCl₂, pH 7.0). After thorough mixing, the final PQQ concentration was adjusted to approximately 20 μ M, providing sufficient cofactor availability to sustain the enzymatic cycle (refer to Figure 24).

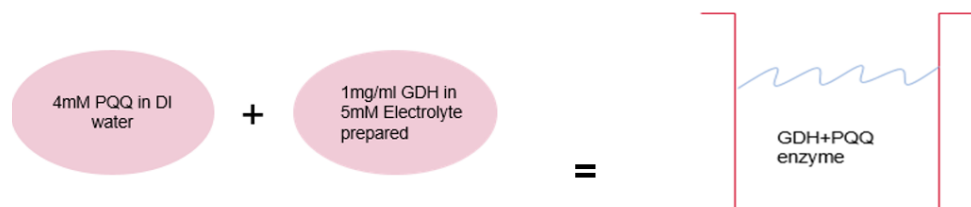


Figure 24: Preparation of the GDH:PQQ enzyme solution in MES buffer for OECT measurements.

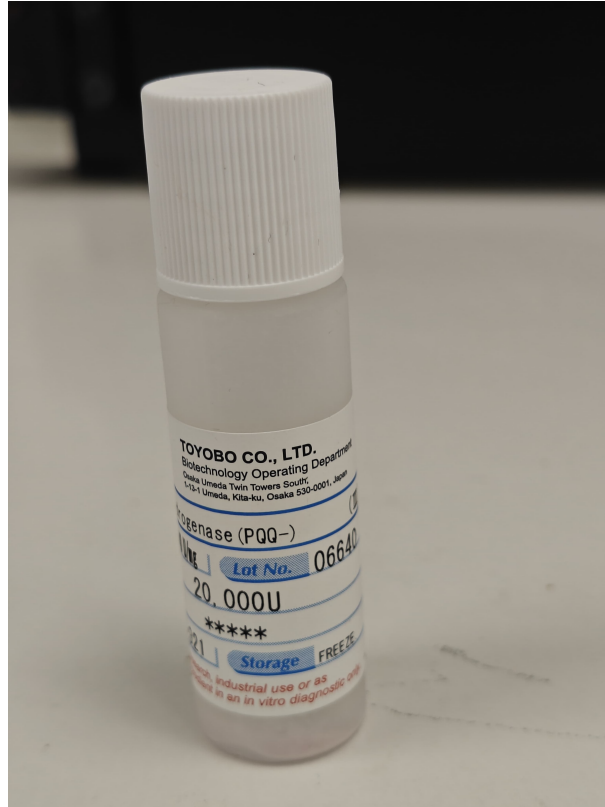


Figure 25: Commercially available GDH enzyme used for OECT experiments.

3.2.5 GDH enzyme immobilization on channel for transfer measurements

The GDH-PQQ enzyme immobilization procedure began with preparation of an enzyme solution containing 1 mg/mL GDH and 20 μ M PQQ cofactor in 5 mM MES buffer (pH 7.0), followed by 30 min incubation to allow PQQ binding. For immobilization, 50 μ L of the GDH-PQQ solution was drop-cast onto OECTs featuring the dual-layer PEDOT:PSS channel architecture (mPEDOT:PSS base layer containing 5% EG + 0.1% DBSA + 1% GOPTS, topped with pure PH1000 PEDOT:PSS). The chips were refrigerated overnight at 4°C to facilitate physical adsorption of the enzyme complex onto the channel surface prior to electrochemical characterization.

3.2.6 Glucose sample preparation for steady state and real time measurements

To prepare glucose solutions for steady-state measurements of OECT devices incorporating GDH immobilized on the channel, 1.80 g of $\geq 99.5\%$ pure D-glucose was dissolved in 10 mL of buffer solution to obtain a 1 M glucose stock solution. From this concentrated stock, a series of intermediate glucose solutions with concentrations of 250 mM, 200 mM, 150 mM, 100 mM, and 50 mM were prepared through systematic volumetric dilution using the same buffer. Each intermediate stock served as the source for the corresponding glucose concentration used during device testing.

For each measurement, a fixed-volume dilution approach was employed in which 1 μ L of the selected intermediate glucose stock was mixed with 49 μ L of electrolyte solution on the OECT channel, resulting in a total volume of 50 μ L. This procedure produced final glucose concentrations of 1 mM, 2 mM, 3 mM, 4 mM, and 5 mM on the device surface. Maintaining a constant total volume ensured consistent pH and ionic strength across all samples, enabling accurate comparison of the device's steady-state response to different glucose levels. The complete preparation details are provided in Table 1.

Table 1: Preparation of glucose concentrations for steady state measurement of OECT with GDH enzyme immobilized on channel

Intermediate Stock Concentration	Dilution Ratio (1:50)	Final Concentration on Device
250 mM	1 μ L + 49 μ L	5 mM
200 mM	1 μ L + 49 μ L	4 mM
150 mM	1 μ L + 49 μ L	3 mM
100 mM	1 μ L + 49 μ L	2 mM
50 mM	1 μ L + 49 μ L	1 mM

In order to prepare glucose solutions for steady state and real time measurements of OECT with GDH/GOx enzyme floating on electrolyte, a set of intermediate glucose stock solutions was prepared by serial dilution from an initial 1 M glucose stock. Using buffer solution, this concentrated stock was used as the starting point for a series of serial dilutions that produced intermediate glucose stocks at concentrations of 580 mM, 57 mM, 5.6 mM, 550 μ M, 54 μ M, 5.3 μ M, 520 nM, and 51 nM. In order to maintain a constant pH and ionic strength appropriate for enzymatic activity and device operation, precise volumes of the more concentrated glucose stock were mixed with buffer to create each intermediate stock.

In order to obtain a final 1:50 dilution for OECT measurements, 1 μ L of the corresponding glucose stock was combined with 49 μ L of the electrolyte solution that was applied to the device. Eight different glucose concentrations, ranging from 1 nM to 10 mM, were created on the device surface by this process (more precisely, 1 nM, 10 nM, 100 nM, 1 μ M, 10 μ M, 100 μ M, 1 mM, and 10 mM). For example, a final glucose concentration of 10 mM was obtained by mixing 1 μ L of the 580 mM glucose stock with 49 μ L of electrolyte. Refer table 2 for the complete glucose preparation.

Table 2: Preparation of glucose concentrations for steady state and real time measurement of OECT with enzyme floating on electrolyte

Intermediate Stock Concentration	Dilution Ratio (1:50)	Final Concentration on Device
580 mM	1 μ L + 49 μ L	10 mM
57 mM	1 μ L + 49 μ L	1 mM
5.6 mM	1 μ L + 49 μ L	100 μ M
550 μ M	1 μ L + 49 μ L	10 μ M
54 μ M	1 μ L + 49 μ L	1 μ M
5.3 μ M	1 μ L + 49 μ L	100 nM
520 nM	1 μ L + 49 μ L	10 nM
51 nM	1 μ L + 49 μ L	1 nM

3.2.7 Electrical characterization and measurement procedures

3.2.7.1 Instrumentation

The Keithley 4200A-SCS is the basis for OECT characterisation, offering a resilient and adaptable platform for electrical measurements (refer to Figure 26). This parameter analyser possesses a modular architecture capable of integrating various source-measure units (SMUs) and specialised instrumentation cards. Its primary functions encompass accurate DC measurements (ranging from femtoamps to amps) and voltage sourcing (up to ± 200 V), rendering it suitable for a diverse array of semiconductor devices. The 4200A-SCS offers essential infrastructure for voltage application, current measurement, and timing control during OECT testing, while integrating effortlessly with specialised add-on modules such as the MBox. [48]

The MBOX is a custom upgrade to the Keithley system that was made just for characterising organic and

electrochemical transistors (refer to Figure 27). This specialised module has precision SMUs that are perfect for OECT measurements, especially the need for ultra-low-current sensitivity. The MBox's design includes a number of important features, such as guarded triaxial connections to reduce parasitic leakage currents, low-noise signal paths for stable nanoampere-level measurements, and custom firmware that optimises measurement sequences for devices with electrolyte gates. Researchers can use the MBOX application software to make and run test protocols that are specifically designed for OECT characterisation. The module connects directly to the 4200A-SCS mainframe, combining its unique features with the Keithley system's strong measurement infrastructure.

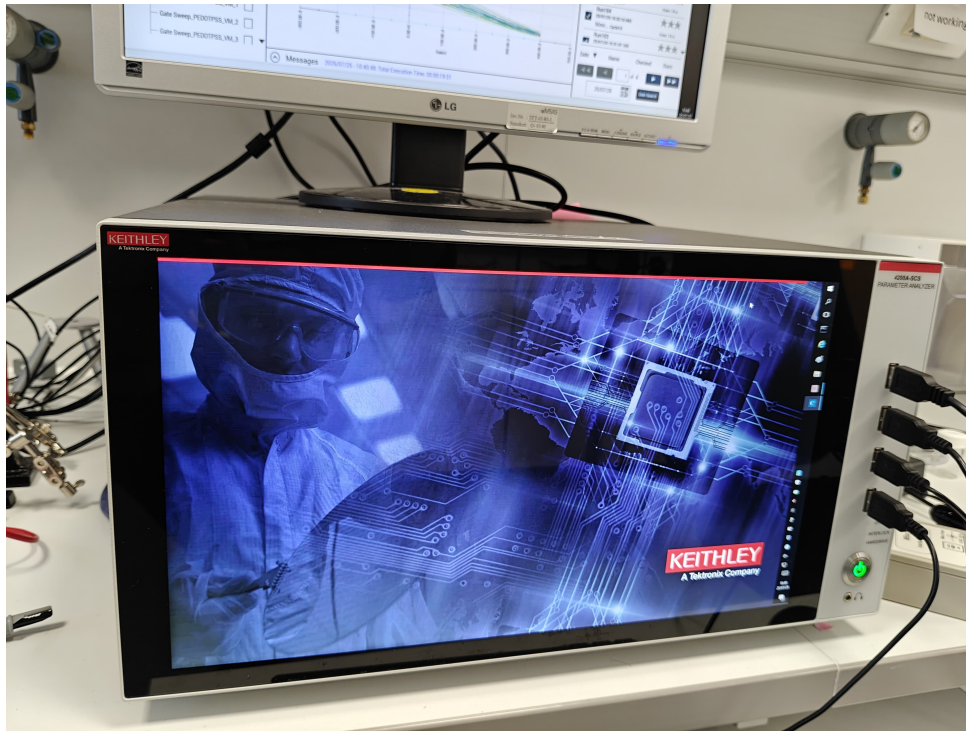


Figure 26: Keithley 4200A-SCS.

The Keithley 4200A-SCS mainframe and MBox module are both fully controlled by the Clarius software platform, which acts as the measurement system's single interface (refer to Figure 28). Multiple workflow options are provided by this advanced software environment, including customisable script editing (supporting Python and TCL), pre-configured test modules for common characterisation tasks, and sophisticated data visualisation tools. Clarius makes it possible to create specific measurement protocols for OECT research, like the output and transfer characterisation files previously mentioned, which specify all test parameters, such as voltage ranges, sweep rates, and measurement intervals. Depending on the sensitivity requirements, the software automatically routes measurements to the relevant hardware (either the MBox module or the mainframe SMUs). Real-time plotting during measurements, automated parameter extraction (transconductance, threshold voltage, etc.), and reliable data management for comparing multiple devices are some of the key features. From measurement execution to data analysis, this integrated software environment greatly simplifies the OECT characterization process.

The electrical characterization was carried out using SMUs to apply the required voltages and record the resulting currents. The drain current (I_d) was monitored while varying the gate voltage (V_g), providing information on the modulation of channel conductivity by ionic gating. The MBox module ensured accurate current measurement in the nA– μ A range, which is particularly important for detecting low-level signals. This straightforward measurement setup provided the essential data needed to evaluate channel performance, contact quality, and the

electrochemical response of the OEET devices.



Figure 27: MBox Module.

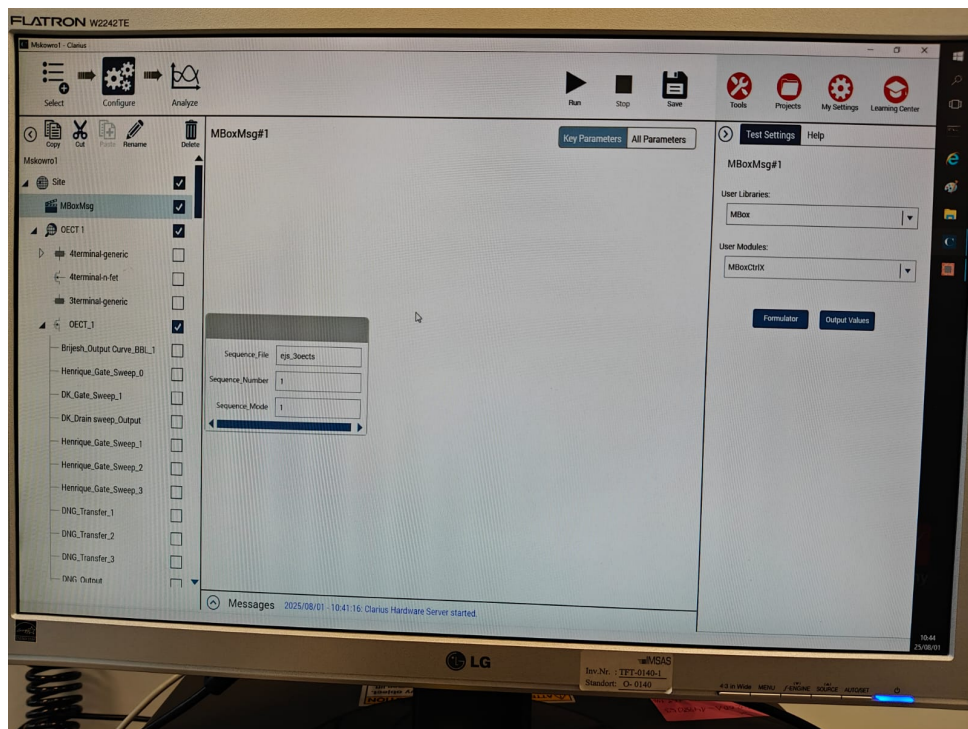


Figure 28: Clarius software platform.

3.2.7.2 Steady-state measurement procedure

The OEET device was set up on the MBox and secured to the SMU1 unit, which was connected to the source and drain electrodes. An external, overhanging platinum gate electrode was used for every OEET measurement.

A total of 49 μL of electrolyte containing the floating enzyme was pipetted onto the channel area at room temperature, and a baseline transfer measurement was taken without glucose to serve as a control. A final 1:50 dilution was achieved on the device by directly adding 1 μL of the selected glucose stock solution to the electrolyte droplet on the channel for the glucose test (refer to Section 3.2.6). Without any incubation time, the transfer measurement was carried out immediately after glucose addition. Using the same addition and measurement sequence, this process was repeated for each of the eight prepared glucose concentrations. Depending on the specific test run, the electrolyte was either retained for overall exposure or replaced to reset the device between concentrations. To ensure uniformity across the entire concentration range, all measurements were conducted following the same procedure. The reactions occurring during this experiment are discussed in Section 2.2.1.1.

The transfer characteristics of the OECTs were measured by sweeping the gate voltage (V_g) from 0 V to 0.6 V at a scan rate of 0.01 V/s, while keeping the drain voltage constant at $V_d = -0.2$ V. From the transfer characteristics, the on current (I_{on}), off current (I_{off}), on/off current ratio ($I_{\text{on}}/I_{\text{off}}$), maximum transconductance (g_m), and average current modulation $(I_{\text{off}} - I_{\text{on}})/I_{\text{on}}$ were also plotted to evaluate device behavior under different glucose concentrations.

3.2.7.3 Real time measurement procedure

As part of a comparative study with enzyme-based OECTs, the study is done with real-time measurements in this section. The same chip configuration, PBS buffer for GOx and MES buffer for GDH as the electrolyte, and an overhanging external platinum gate electrode were all used in the same experimental setup as previously discussed. Because of this consistency, any changes in current that were noticed could be mainly described to the interaction between the enzyme and glucose rather than setup variations.

Glucose solutions were prepared following the same initial stock preparation method as in steady state transfer measurements (refer to Section 3.2.6). A 1 M D-glucose stock solution was obtained by dissolving 1.80 g of $\geq 99.5\%$ pure D-glucose in 10 mL of buffer solution. Serial dilutions were then performed in the respective buffer solutions (PBS for GOx and MES for GDH) to obtain intermediate glucose stocks. For this experiment, only five concentrations were selected: 1 μM , 10 μM , 100 μM , 1 mM, and 10 mM.

The real-time measurements were performed by placing the OECT chip onto the MBox and securing it to the SMU1 unit. An initial mixture of 50 μL of enzyme solution in buffer was pipetted over the channel region. After allowing approximately 10 minutes for the system to stabilize, 1 μL of glucose was incrementally added at defined intervals to yield final concentrations of 1 μM , 10 μM , 100 μM , 1 mM, and 10 mM. The total runtime of each experiment was 1 hour, with 10-minute intervals between each concentration step.

In real-time measurements, two types of plots were generated. The first type displays the drain current I_d (A) as a function of time during sequential additions of glucose, allowing direct observation of the transient response for each concentration step. The second type of plot shows the absolute change in drain current (ΔI_d), as a function of glucose concentration.

To generate the ΔI_d vs. glucose concentration plot, the drain current change for each concentration step was calculated by comparing the current value 5 s before the droplet addition of the next concentration with the current value 5 s before the addition of the first concentration. For the 1 μM glucose point, ΔI_d was obtained by subtracting the current 5 s before the 1 μM addition (denoted as I_{d0}) from the current 5 s before the 10 μM addition. The same procedure was repeated for subsequent concentrations, as follows in table 3:

For real-time measurements, the drain voltage was fixed at $V_d = -0.2$ V, while the gate voltage was held at

Glucose Concentration	Calculation of ΔI_d
1 μM	$I_d(1195 \text{ s}) - I_{d0}$
10 μM	$I_d(1795 \text{ s}) - I_{d0}$
100 μM	$I_d(2395 \text{ s}) - I_{d0}$
1 mM	$I_d(2995 \text{ s}) - I_{d0}$
10 mM	$I_d(3595 \text{ s}) - I_{d0}$

Table 3: Calculation of ΔI_d values for each glucose concentration in real-time measurements. I_{d0} is the drain current measured 5 s before the addition of the 1 μM glucose droplet.

$V_g = 0.6 \text{ V}$, corresponding to the point of maximum transconductance obtained from the transfer characteristics. Under these biasing conditions, the dynamic response of the device to sequential additions of glucose was recorded. A drain current (I_d) versus time graph, which directly shows the transient changes in I_d upon stepwise additions of glucose at defined time intervals and a ΔI_d versus glucose concentration graph, where ΔI_d was calculated as the absolute change in drain current between consecutive concentration steps.

4 Results and Discussion

4.1 Wildau's Experiment with GDH Enzyme in Polymer Systems

The motivation for this study stems from preliminary tests conducted on PEDOT:PSS films deposited on gold electrodes at TH Wildau, where the enzyme PQQ-dependent glucose dehydrogenase (PQQ-GDH) was immobilized on polymer-modified electrode surfaces (refer to Figure 29). The aim of those experiments was to investigate the possibility of achieving direct electron transfer (DET) between the enzyme and the conducting polymer matrix, as DET is considered a key strategy for developing mediator-free biosensors. [12] Their results demonstrated that polymer-modified gold electrodes could successfully interact with PQQ-GDH, thereby providing a foundation for exploring enzyme integration in OECT platforms. Two types of PEDOT:PSS-based formulations were studied: Polymer 1 (P1), consisting of pure PEDOT:PSS (Clevios PH 1000), and Polymer 2 (P2), a modified PEDOT:PSS formulation (mPEDOT:PSS) (refer to Section 3.2.2 for mPEDOT:PSS ingredients).

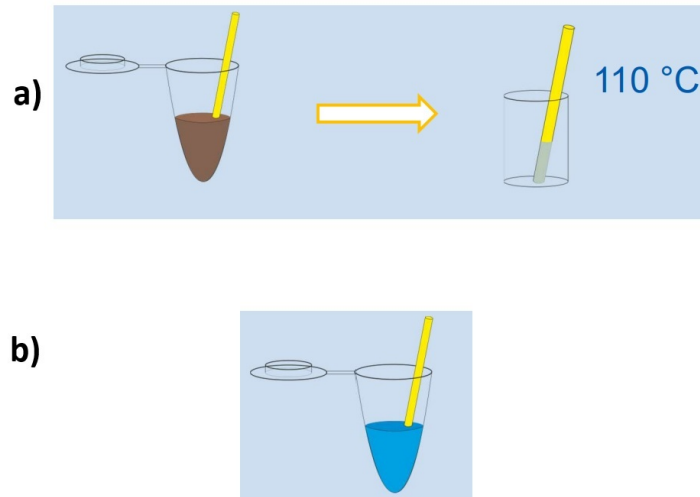


Figure 29: Wildau's method: (a) Polymer stabilisation by heating, (b) Adding PQQ-GDH enzyme.

They examined various combinations, comprising single-layer coatings, successive polymer/enzyme layers, and enzyme-polymer blends (refer to Figure 30) and evaluated their redox activity and glucose sensitivity by linear sweep voltammetry (LSV). Despite being DET focused, the work underscored critical elements including the enzyme compatibility with PEDOT:PSS-based polymers, the electrochemical behavior of enzyme-polymer combinations, and the variability in glucose sensitivity upon enzyme immobilization.

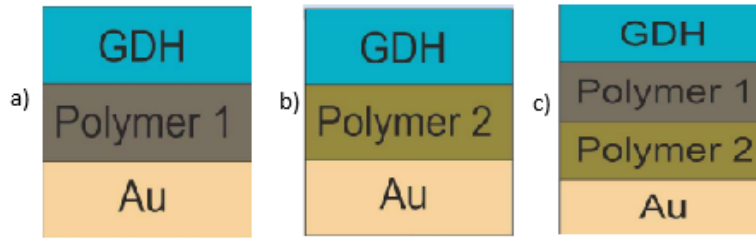


Figure 30: Schematic representation of different polymer-enzyme configurations used in the Wildau study for gold electrode modification. [49] (a) GDH enzyme deposited on pure PEDOT:PSS (Polymer 1) — Au-P1-GDH. b) GDH enzyme deposited on mPEDOT:PSS formulation (Polymer 2) — Au-P2-GDH. c) GDH enzyme deposited on a bilayer of Polymer 1 over Polymer 2 — Au-P2-P1-GDH.

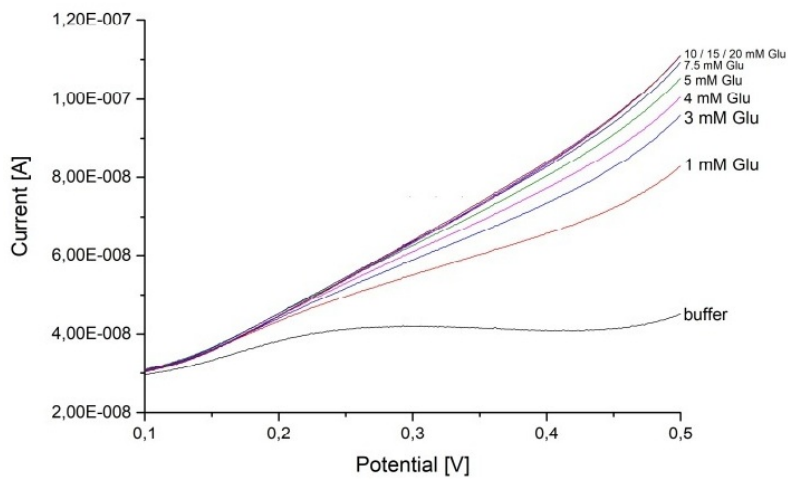


Figure 31: Current-potential curves for GDH immobilized on Au electrodes coated with P1 (pure PEDOT:PSS).

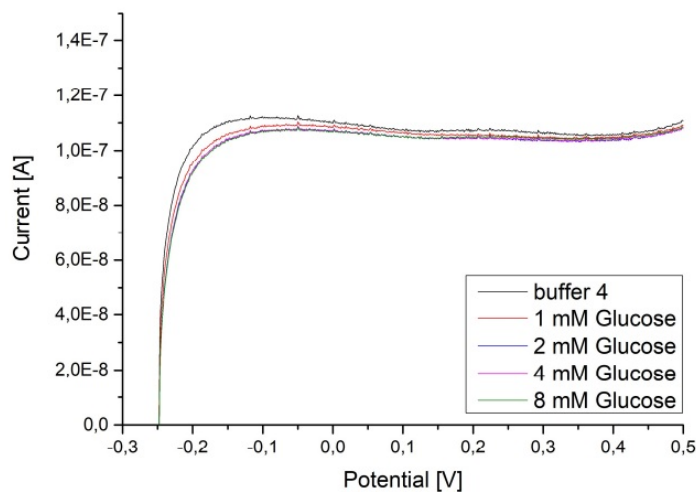


Figure 32: Current-potential curves for GDH immobilized on Au electrodes coated with P2 (mPEDOT:PSS).

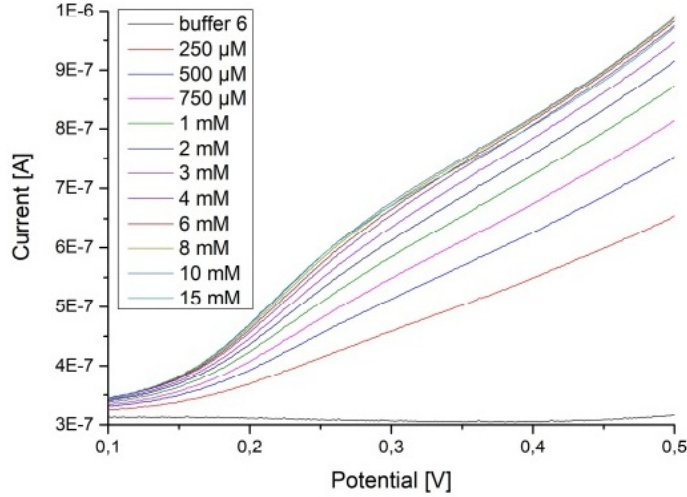


Figure 33: Current–potential curves for GDH immobilized on Au electrodes coated with combined P1 and P2 layers.

Figures 31, 32, and 33 shows representative results from the Wildau experiments, [49] illustrating the response of GDH immobilized on the different configurations of polymer-coated electrodes. The current–potential curves show that polymer composition strongly influenced the catalytic activity and electron transfer efficiency of GDH. In particular, the combination of P1 and P2 enabled the most pronounced and concentration-dependent current response to glucose, suggesting that optimized PEDOT:PSS matrices can indeed promote DET with the enzyme.

In the OECT platform, however, this DET process is coupled with the intrinsic volumetric ion-electronic interaction of PEDOT:PSS. As a result, glucose oxidation at the enzyme interface can influence the channel conductivity in two ways: a) Through DET, where electrons from the reduced GDH:PQQ complex are transferred directly to the PEDOT:PSS channel, and b) through ionic modulation of the polymer bulk, which is subsequently amplified by the OECT architecture.

Electropolymerised PEDOT (eP1) was also reported by Wildau as an alternative immobilisation route. Although this method can promote direct electron transfer in LSV, their results indicated that it did not provide a linear or reproducible glucose response, most likely due to enzyme orientation or partial deactivation during the electrochemical process. This approach was not pursued further in the present work

4.1.1 Control Measurements on OECT using Wildau’s strategy

As part of the control experiments on OECT using GDH enzyme [49], the first step was to transfer the idea of TH Wildau’s experiment from electrode configuration to OECT. Thus, the strategies previously demonstrated on Au electrodes with PEDOT:PSS films (Au–P1–GDH and Au–P1–P2–GDH) were adapted into OECT channels, where the dual contributions of DET and ionic coupling were investigated for their potential to enable sensitive and integrable glucose sensing.

To evaluate the stability of channel materials, four chips were fabricated and measured in MES buffer as controls. Before polymer deposition, the OECT substrates showed clean and well-defined electrode structures (refer to Figure 34). Chips 1 and 2 were coated with pure PH1000 PEDOT:PSS on the channel using spin-coating (equivalent to Polymer 1 from the Wildau study, Au–P1–GDH) and baked at 110 °C. Chips 3 and 4 channels were first coated with mPEDOT:PSS, baked and then topped with a layer of pure PH1000 PEDOT:PSS (equivalent to Au–P1–P2–GDH).

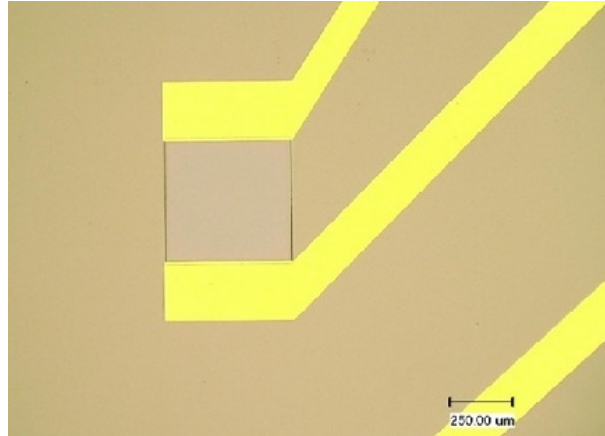


Figure 34: OEET chip before channel deposition. Electrode patterns are intact and clean. Scale shown is 250μm.

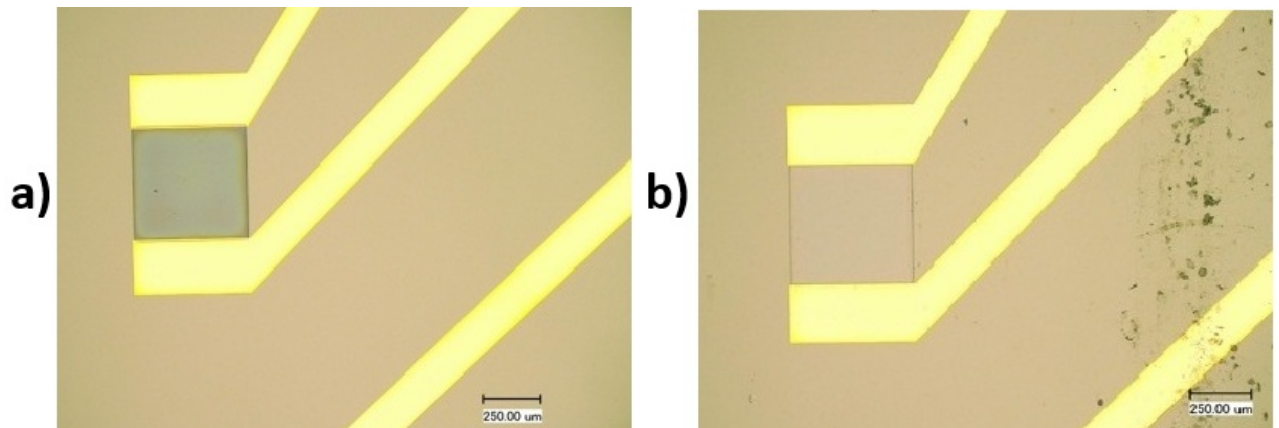


Figure 35: Microscopic image of a pure PH1000 PEDOT:PSS channel before and after testing. Significant delamination occurred after electrolyte exposure. Scale shown is 250μm.

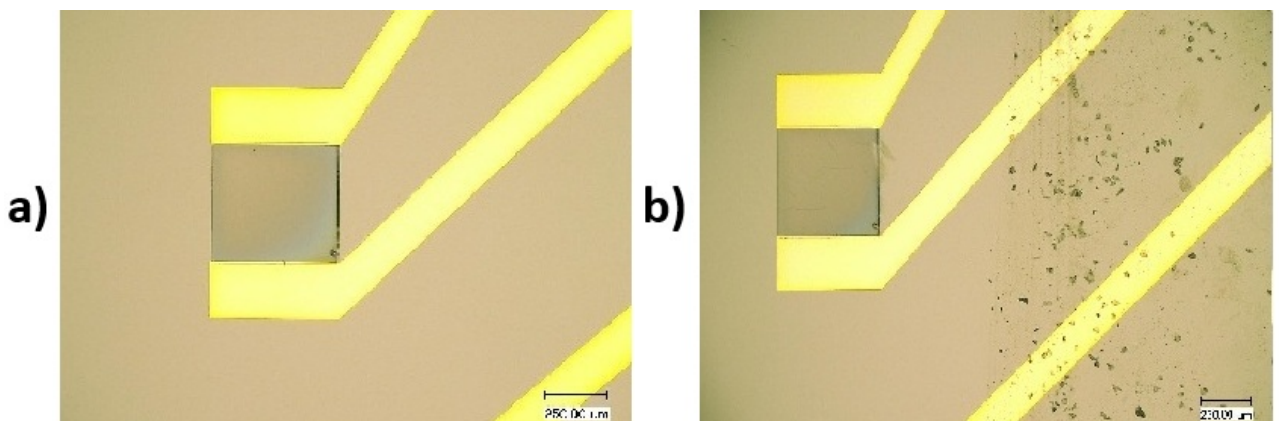


Figure 36: Microscopic image of an mPEDOT:PSS + PH1000 PEDOT:PSS bilayer channel before and after testing. The structure remained stable. Scale shown is 250μm.

After deposition of pure PH1000 PEDOT:PSS, the channels initially appeared smooth and uniform. The control measurement was done by conducting the steady state measurement on the chips without GDH deposition on the PEDOT:PSS channel. MES buffer used with GDH enzyme was used as electrolyte and an overhanging Pt gate electrode was employed in a standard gate-electrolyte-channel setup. The electrical characterization is

explained in Section 3.2.7.2. The transfer characteristics were plotted thereafter. However, after electrical testing and rinsing, the polymer film showed damage and partial delamination (refer to Figure 35). This confirmed that pure PEDOT:PSS is not stable in aqueous environments without crosslinkers.

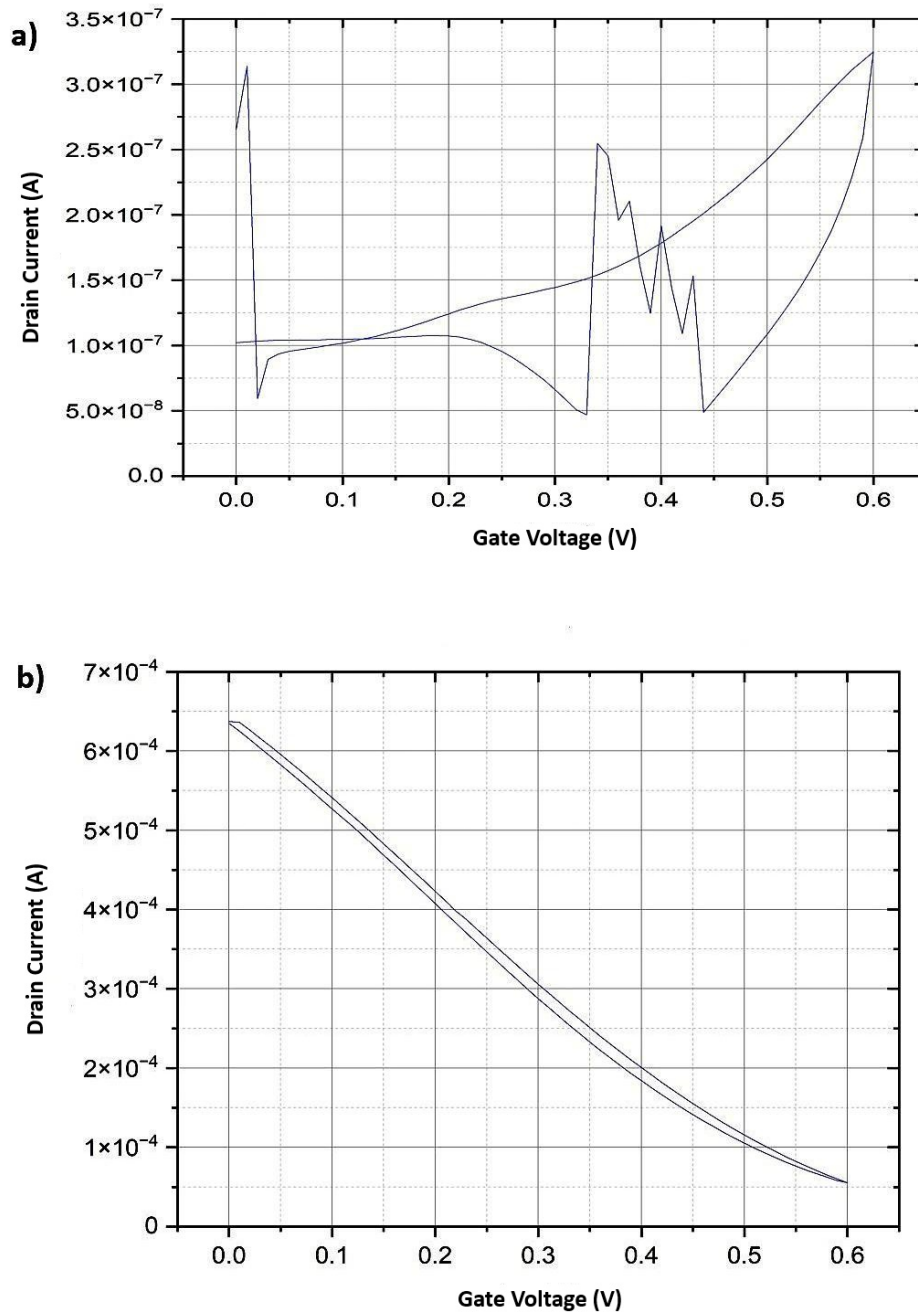


Figure 37: (a) Transfer characteristics of a device with a pure PH1000 PEDOT:PSS channel, showing low drain current and uneven curves. (b) Transfer characteristics of a device with an mPEDOT:PSS + PH1000 PEDOT:PSS bilayer channel, showing stable and reproducible response.

In contrast, channels prepared with a base layer of mPEDOT:PSS and a top layer of PH1000 retained their morphology after repeated testing (refer to Figure 36). The crosslinked formulation provided better adhesion and mechanical robustness.

These physical observations were supported by electrical measurements. Devices with only pure PEDOT:PSS showed degraded transfer characteristics, including low drain currents and poorly reproducible curves (refer to Figure 37a). In contrast, the mPEDOT:PSS + PH1000 bilayer devices exhibited stable and repeatable transfer curves typical of well-functioning OECTs (refer to Figure 37b).

Together, these results confirm that while pure PEDOT:PSS films are unsuitable for stable measurements, the mPEDOT:PSS + PH1000 combination provides reliable electrical behaviour and mechanical robustness. This configuration was therefore selected as the channel material for subsequent biosensing experiments.

4.2 Steady state measurement of OECT with GDH enzyme immobilized on channel

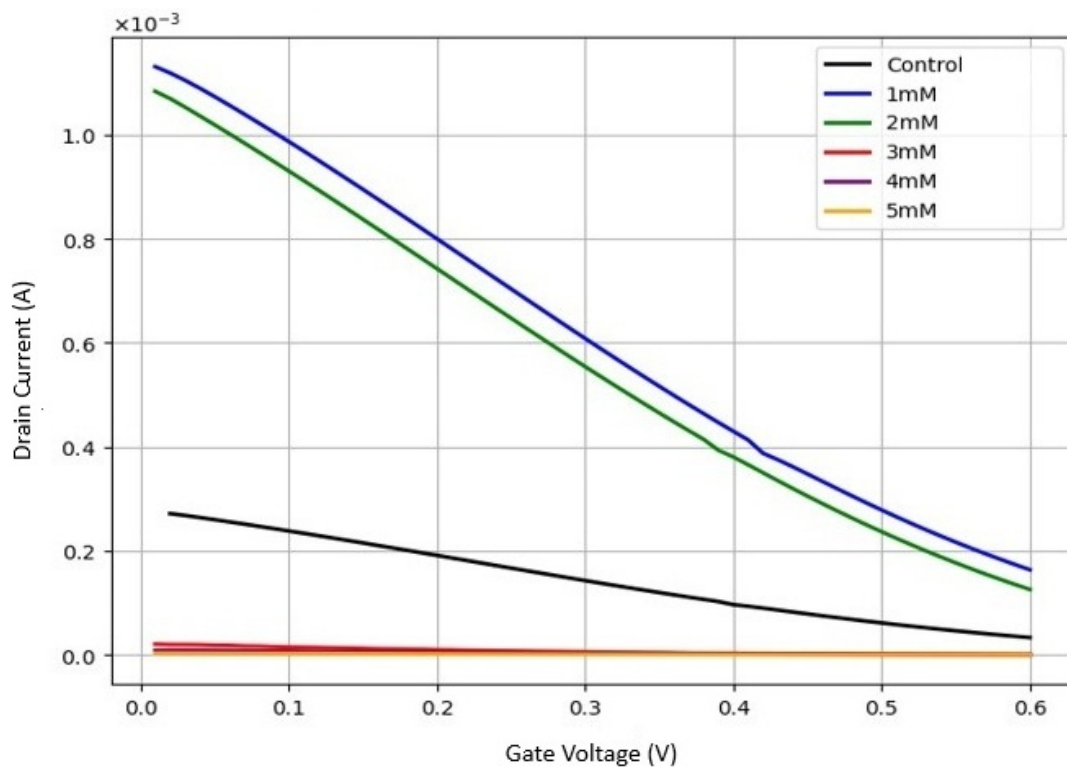


Figure 38: Transfer characteristics of GDH-PQQ immobilized OECT (OECT 1) showing drain current (I_d) response to different glucose concentrations (1-5 mM).

In this work, the OECT device was functionalized by immobilizing GDH directly onto the channel surface. The detailed procedure for enzyme immobilization is described in Section 3.2.5. The steady-state measurement procedure followed the same protocol described in Section 3.2.7.2, with the only difference being that, in this case, the GDH enzyme was immobilized on the channel instead of being mixed with the electrolyte. The transfer characteristics were recorded for three OECT devices at different glucose concentrations (1-5 mM), along with a control measurement using MES buffer only (refer to Figure 38 for OECT 1 and Figures A18 and A19 for OECT 2 and 3 in the Appendix). The results from the three devices were averaged to minimize device-to-device variability. From these measurements, five key parameters were extracted for each glucose concentration (0-5 mM): the on current (I_{on}), off current (I_{off}), on/off ratio (I_{on}/I_{off}), maximum transconductance (g_m), and average current modulation ($|I_{off} - I_{on}|/I_{on}$). The averaged values of these parameters (mean $\pm$ SD) were plotted as a function of glucose concentration to evaluate the steady-state behavior of the GDH-PQQ immobilized OECTs.

In a properly functionalized OECT, the control should exhibit the highest I_d , because in the absence of glucose the PEDOT:PSS channel remains in its fully doped and highly conductive state. [5] With increasing glucose concentration, the enzymatic reaction (GDH:PQQ-mediated glucose oxidation) produces reduced PQQH₂ in theory, this should transfer electrons to the platinum gate. This redox activity effectively shifts the gate potential and drives ionic redistribution in the channel, progressively dedoping PEDOT:PSS and lowering the number of mobile holes. As a result, I_d should decrease systematically with increasing glucose, that reflect change in the channel's electrochemical potential.

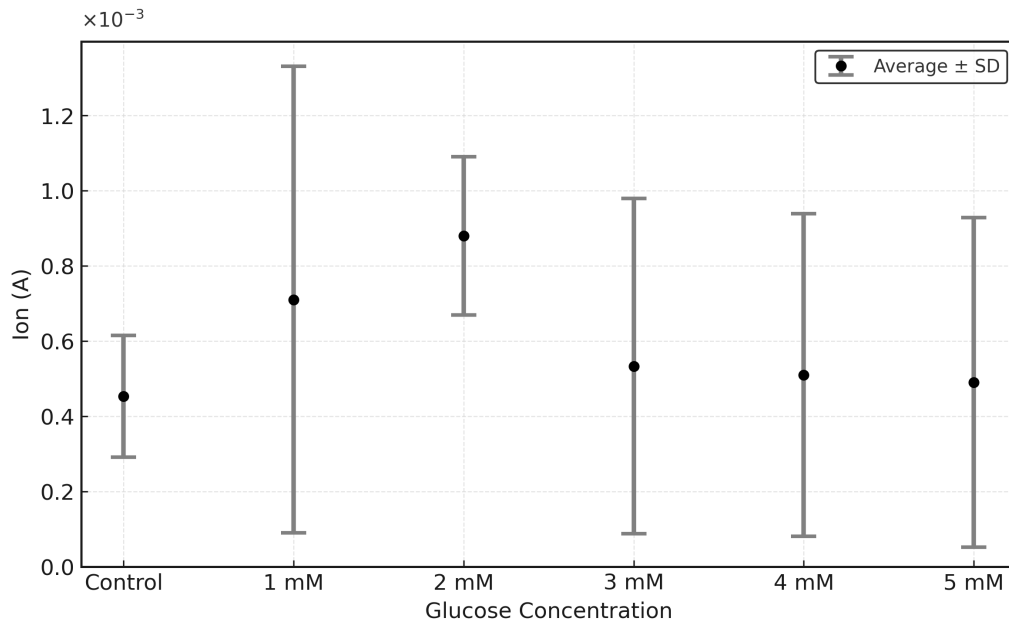


Figure 39: Average on current (I_{on}) vs glucose concentration for GDH-PQQ immobilized OECT.

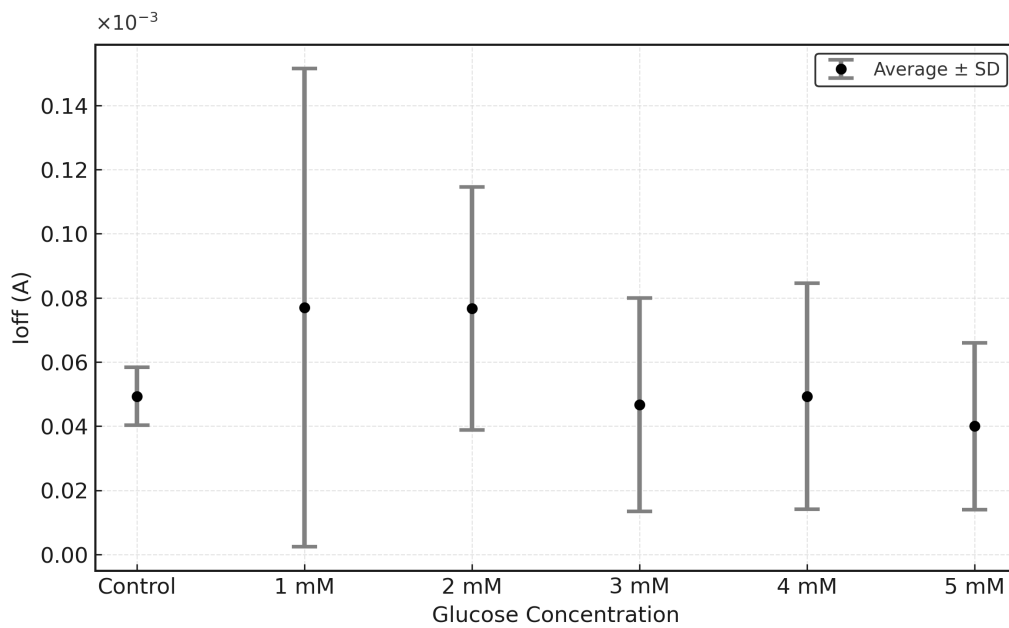


Figure 40: Average off current (I_{off}) vs glucose concentration for GDH-PQQ immobilized OECT.

The on current was highest at 1–2 mM glucose and decreased sharply at higher concentrations (refer to Figure 39). This non-monotonic behavior may be attributed to redox-mediated interactions between the PQQ/PQQH₂

couple and the PEDOT:PSS channel that initially enhance doping at low glucose concentrations but lead to dedoping at higher levels due to the accumulation of oxidized species [5,16].

The off current (I_{off}) displayed a similar decreasing trend with increasing glucose concentration and stabilized beyond 3 mM (refer to Figure 40). The reduction in I_{off} suggest stronger dedoping of the PEDOT:PSS channel driven by enzymatic oxidation of glucose, which depletes mobile charge carriers and diminishes the channel's residual conductivity. [25].

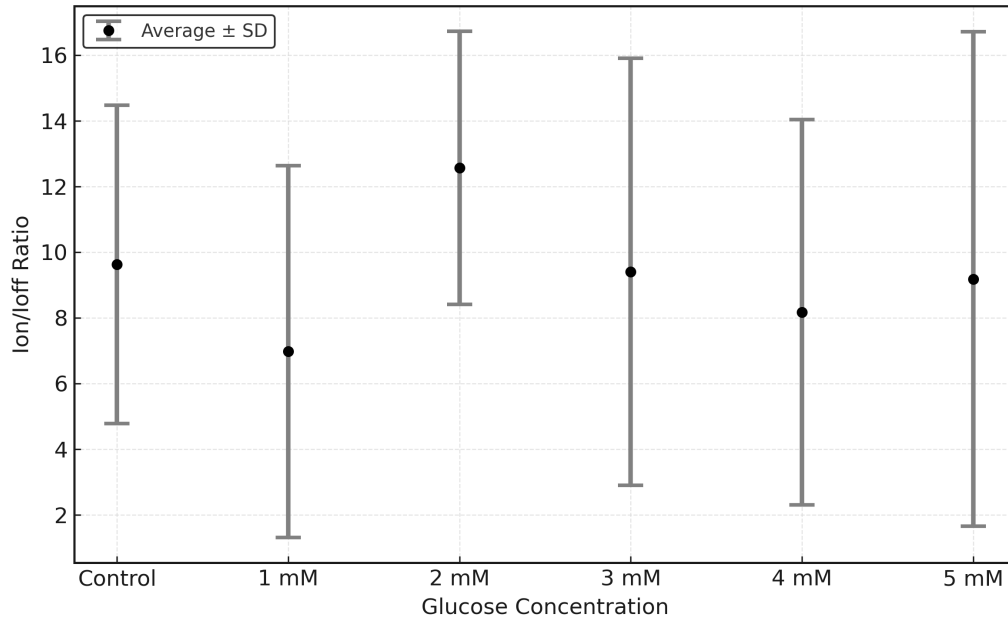


Figure 41: Average on/off ratio vs glucose concentration for GDH-PQQ immobilized OECT.

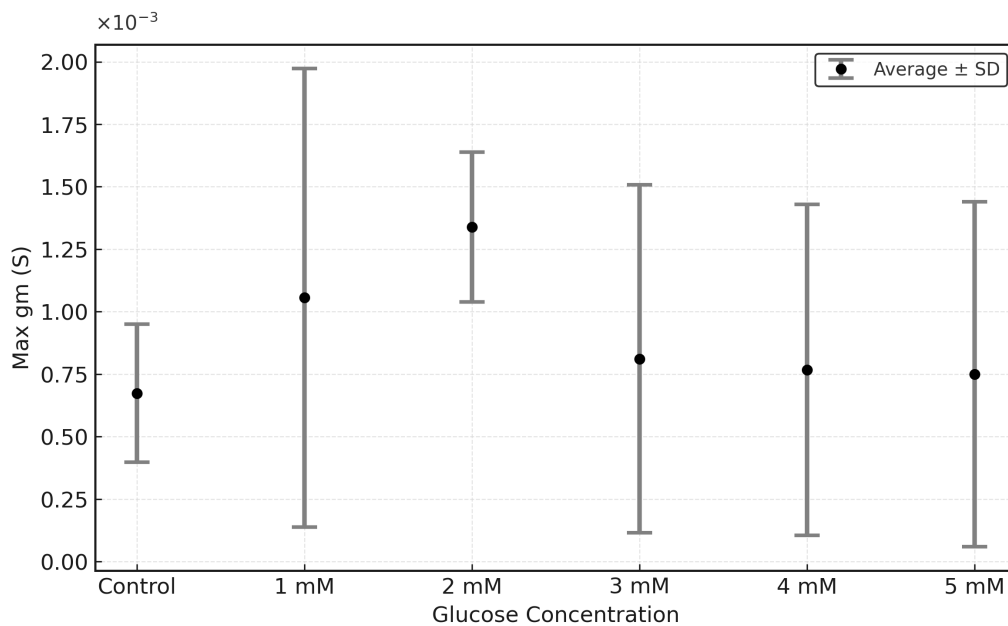


Figure 42: Average maximum transconductance vs glucose concentration for GDH-PQQ immobilized OECT.

The on/off ratio exhibited fluctuations with glucose concentration rather than a smooth increase (refer to Figure

41). Such irregular variations often arise from non uniform enzyme immobilization or partial detachment during operation, which affects the gate potential and charge transport reproducibility [5].

The maximum transconductance peaked near 2 mM glucose and decreased at higher concentrations (refer to Figure 42). This reduction suggests lower gate efficiency and mobility caused by channel dedoping and electrochemical degradation under repeated redox cycling [5, 25].

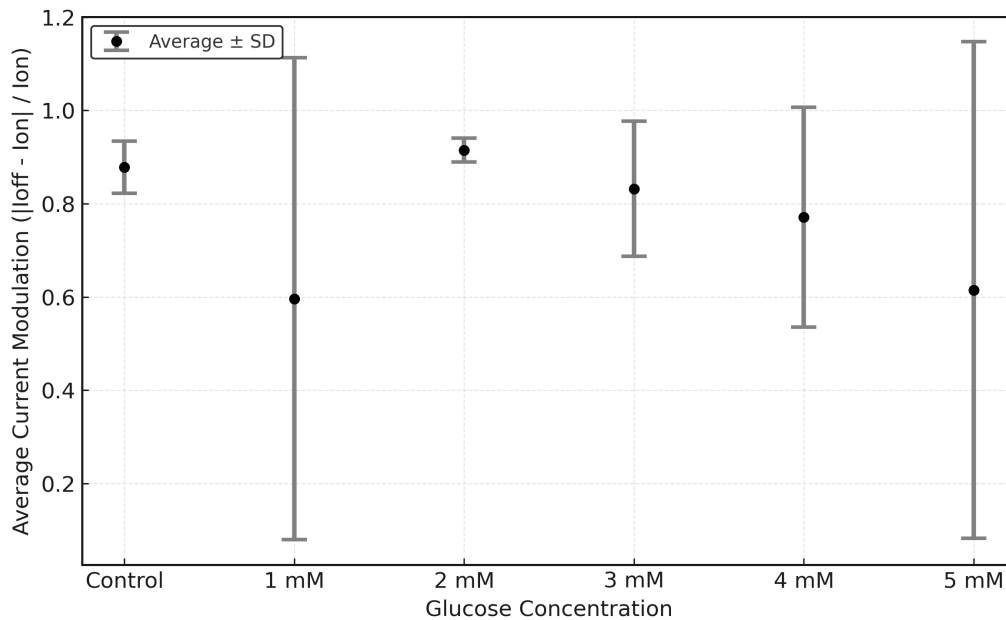


Figure 43: Average current modulation vs glucose concentration for GDH-PQQ immobilized OECT.

The average current modulation displayed a non-monotonic trend with glucose concentration (refer to Figure 43). Ideally, modulation should increase consistently with glucose levels; however, fluctuations are common when enzyme immobilization is unstable or when local enzyme distribution is nonuniform, leading to inconsistent redox kinetics across the channel surface [9, 43].

The GDH-PQQ immobilized OECT exhibited partial glucose sensitivity but lacked reproducibility across devices. The irregular trends and deviations indicate weak enzyme adhesion and unstable electrochemical coupling with the PEDOT:PSS channel. Therefore, in later experiments, GDH was dissolved directly in the electrolyte to improve device stability and eliminate adhesion-related inconsistencies [5, 25].

4.3 Comparative study between GOx and GDH enzyme based OECT using steady state measurements

4.3.1 GOx steady state measurements

The steady-state measurement procedure is explained in Section 3.2.7.2. As the glucose concentration increased from 1 nM to 10 mM, the transfer characteristics derived from OECT 1 demonstrated a steady and concentration-dependent decrease in drain current. The enzymatic reaction that GOx catalyses, which oxidises glucose to gluconolactone while generating protons and hydrogen peroxide, is responsible for this behaviour. The PEDOT:PSS channel becomes dedoped as a result of the electrolyte's proton buildup changing the local pH. By lowering the concentration of hole carriers in the channel, this dedoping effect lowers the conductivity of the channel and consequently the observed drain current (refer to Figure 44).

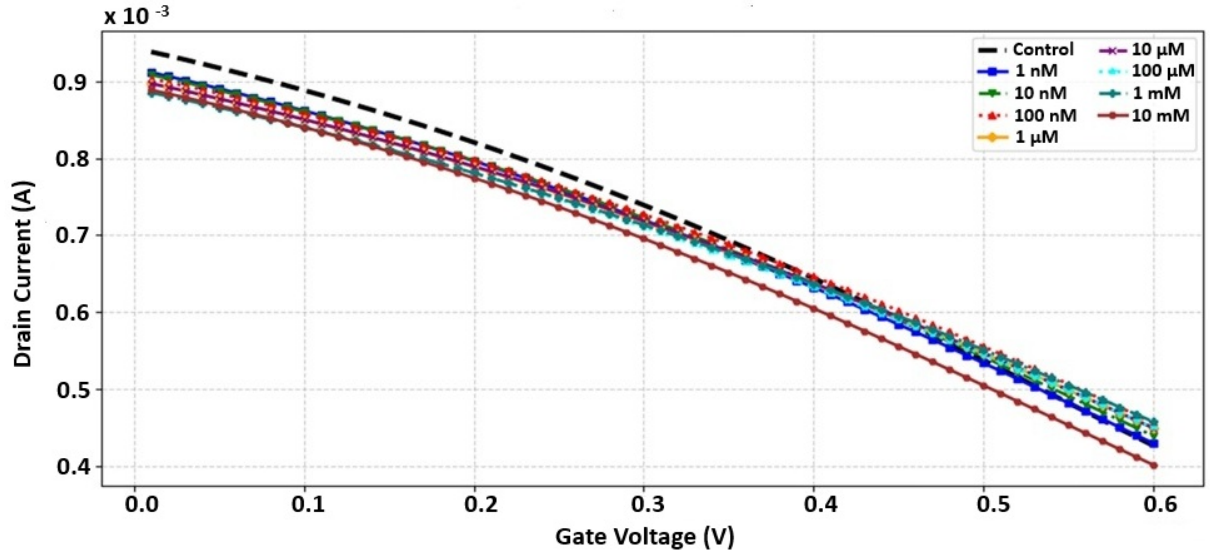


Figure 44: Transfer characteristics of a GOx-based OECT (OECT1) sensor under different glucose concentrations

The increased modulation effect with higher glucose levels is further supported by the transfer curves downward shift, especially at higher gate voltages. The curves form and progression point to a distinct and consistent correlation between OECT response and glucose concentration. Additional context is provided by the Appendix, which also includes transfer characteristics from GOx based OECT 2 sensor (refer to Figure A1) and GOx based OECT 3 (refer to Figure A2). These additional findings demonstrate consistency and variability in sensor responses and enable cross-comparison across various devices.

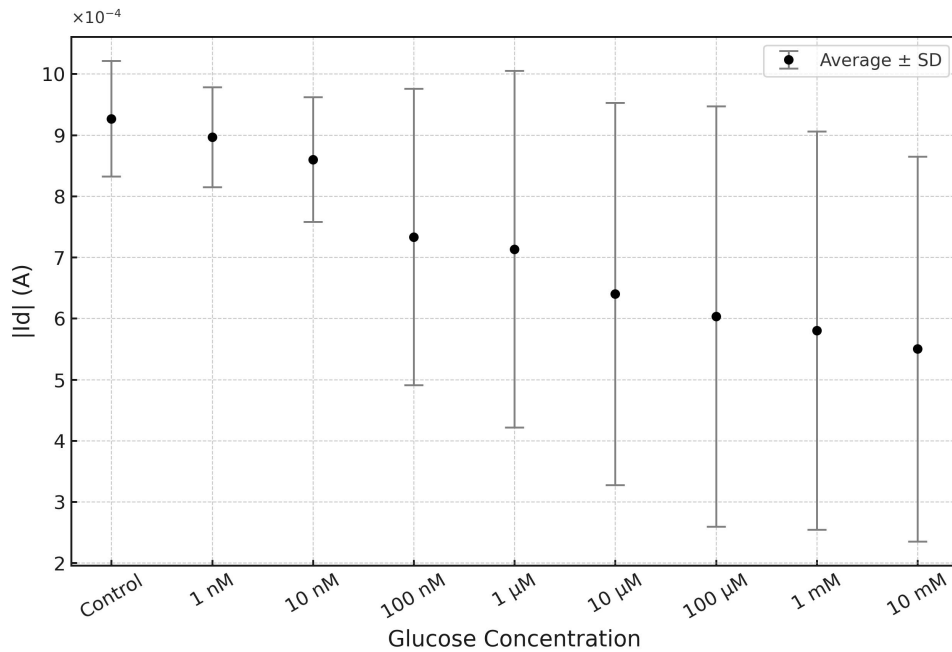


Figure 45: Average on current vs glucose concentrations comparison of OECTs 1-3 with GOx enzyme

Similarly, the on current, measured at $V_g = 0$ V, also showed a decreasing trend with rising glucose levels. This decline reflects reduced channel conductivity, as enzymatic dedoping lowers the hole density in PEDOT:PSS (refer to Figure 45) and (refer to Figure A11) for scatter plots. Since the enzyme reacts immediately after glucose

addition, the nominal $V_g = 0$ V may effectively shift (e.g., $\sim +0.2$ V), explaining why modulation appears even before external gate bias changes.

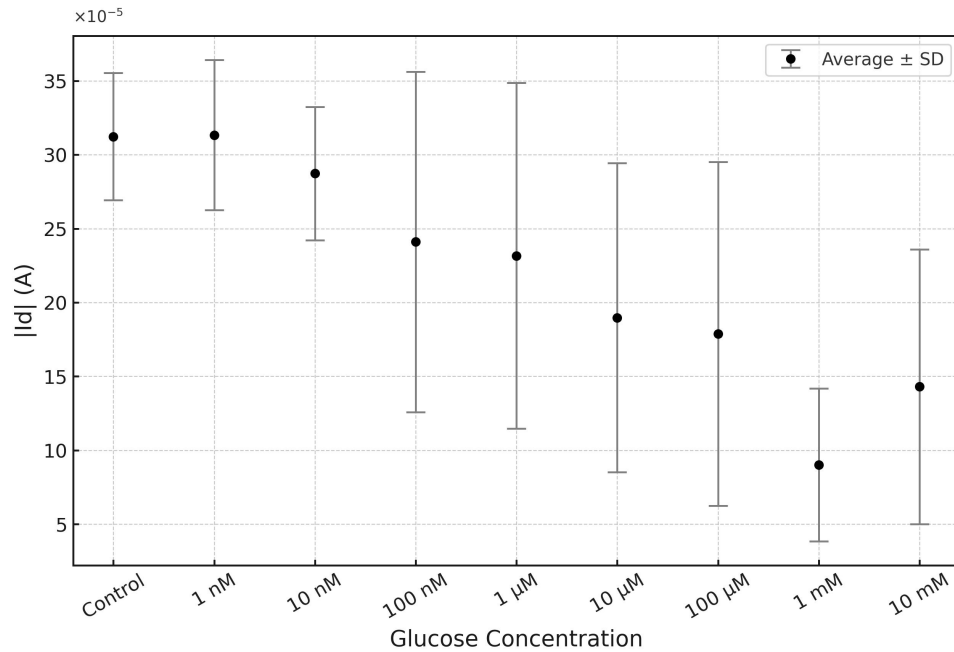


Figure 46: Average off current vs glucose concentrations comparison of OECTs 1-3 with GOx enzyme

The off current, measured at $V_g = 0.6$ V, showed a gradual decline as glucose concentration increased. This trend can be attributed to increased resistance in the PEDOT:PSS channel due to progressive dedoping caused by proton accumulation in the electrolyte. As the channel becomes less conductive, the current that flows in the off state is reduced (refer to Figure 46) and (refer to Figure A10) for the scatter plot of the three chips tested.

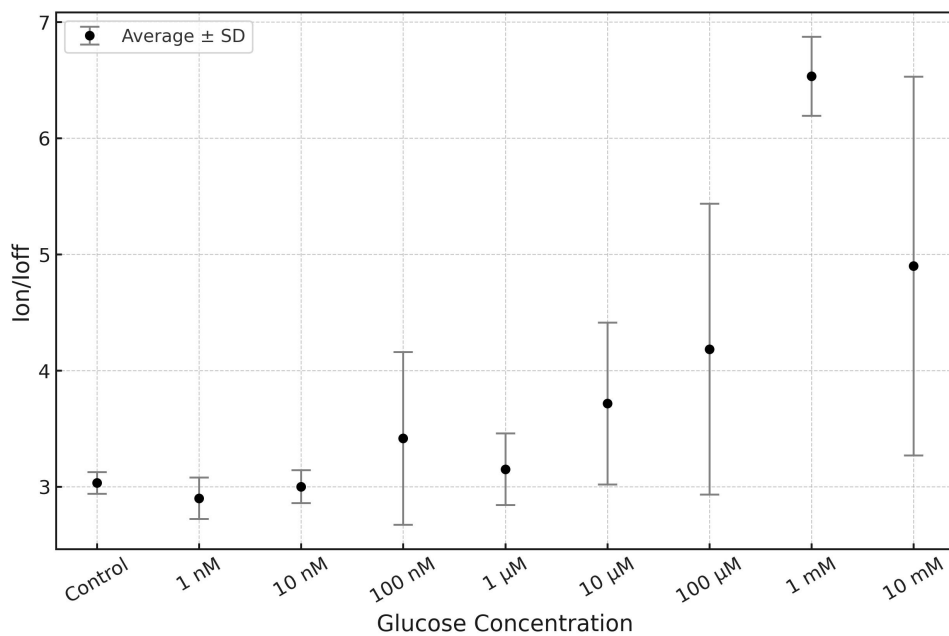


Figure 47: Average on/off ratio vs glucose concentrations comparison of OECTs 1-3 with GOx enzyme

An analysis of the on/off current ratio revealed a significant increase with increasing glucose concentration, peaking

at 1 mM and thereafter reducing. This behaviour results from the growing disparity between the doped (on) and dedoped (off) states of the channel. As dedoping becomes more pronounced, the contrast between minimum and maximum current values widens, leading to improved distinguishability between different glucose concentrations (refer to Figure 47) and (refer to Figure A12) for the scatter plots.

The decrease in maximum transconductance can be attributed to proton accumulation in the electrolyte, which modifies the effective gate bias and reduces the coupling capacitance between the gate and the channel. Both effects diminish the channel's responsiveness to gate modulation, consistent with earlier reports on OECD behavior under strong ionic flux and biochemical reactions (refer to Figure 48) and (refer to Figure A13) for the scatter plots. [5]

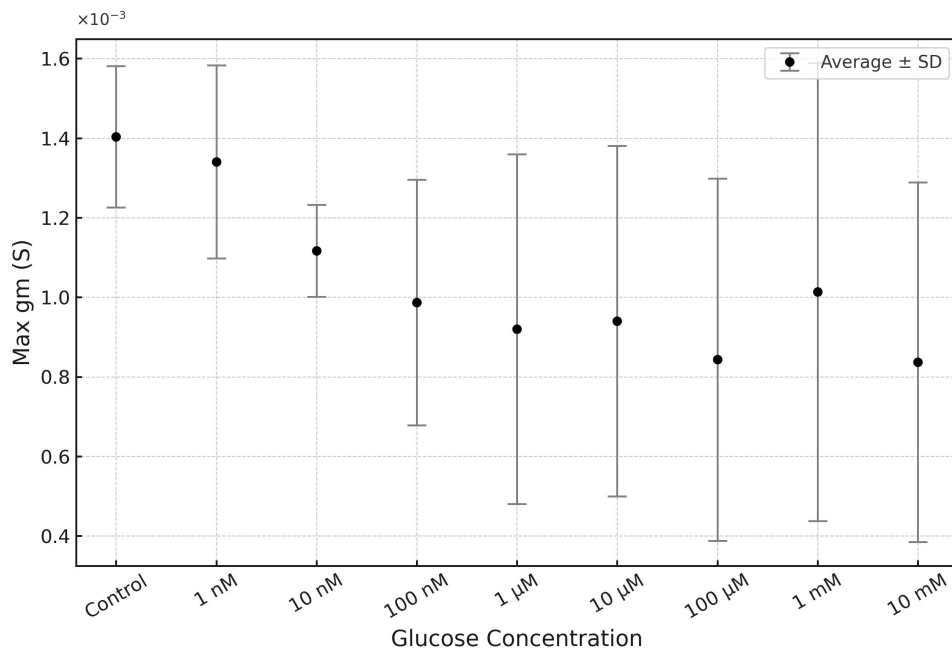


Figure 48: Average maximum transconductance vs glucose concentrations comparison of OECDs 1-3 with GOx enzyme

The average current modulation, calculated as $(I_{\text{off}} - I_{\text{on}})/I_{\text{on}}$, was also analyzed as a function of glucose concentration. The modulation remained relatively constant from the control up to 100 nM, indicating limited enzymatic impact on the channel at very low glucose levels. However, a distinct increase in modulation was observed from 1 µM to 100 µM, suggesting this range as the most responsive window where enzymatic reaction begins to significantly influence channel conductivity. At higher concentrations (1 mM and 10 mM), modulation continued to increase but with a reduced slope, possibly indicating a saturation in enzymatic activity or dedoping efficiency. [9, 43] The rise in modulation correlates with increased proton generation by GOx, enhancing the dedoping effect of PEDOT:PSS and thereby amplifying the device response. This trend confirms the effective sensitivity of the device in the mid-micromolar glucose range (refer to Figure 49).

The behaviour seen across all electrical parameters validates that the GOx-functionalized OECD is a good fit for glucose sensing. The enzymatic reaction in the electrolyte initiates electrochemical processes that are directly correlated with changes in drain current, transconductance, and current modulation. The reproducibility of these effects across various devices is further supported by the data from other chips that are included.

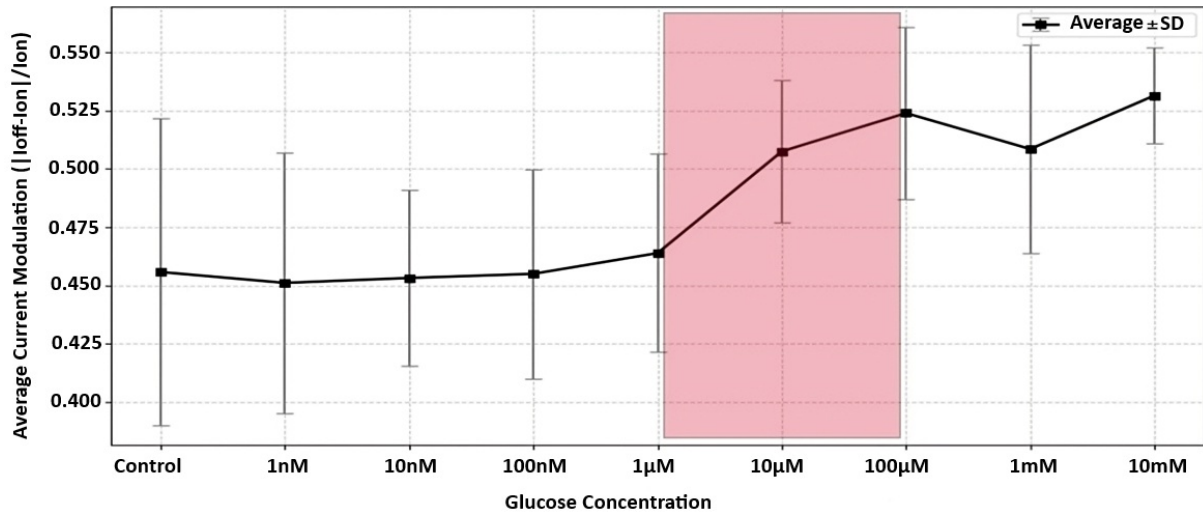


Figure 49: Average current modulation vs glucose concentrations of a GOx based OECT sensor

4.3.2 GDH steady state measurements

In the conducted experiments according to Section 3.2.7.2, transfer characteristics of GDH-based OECTs were first measured across a series of glucose concentrations ranging from nanomolar to millimolar levels. Figure 50 shows the results for OECT 1, while the corresponding data for OECTs 2, 3, and 4 are provided in the Appendix (refer to Figures A3, A4 and A5). OECT 1 demonstrates a clear, concentration-dependent reduction in drain current, with the control (no glucose) consistently yielding the highest current. The strongest modulation is observed between 1 nM and 100 μ M, suggesting effective sensing within this range.

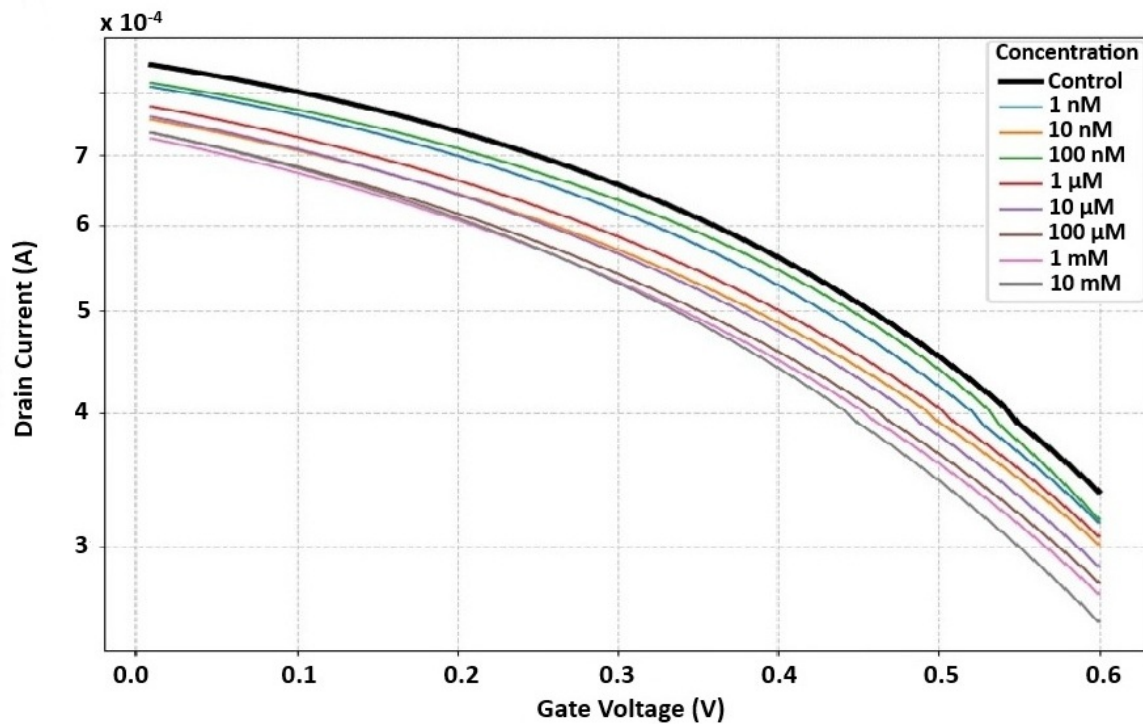


Figure 50: Transfer characteristics of a GDH-based OECT (OECT 1) under different glucose concentrations.

Across the four tested chips, notable variability was observed. OECT 1 and 2 displayed smooth, reproducible,

concentration-dependent responses, while OECT 3 produced irregular, noisy, and non-monotonic trends. OECT 4 exhibited a sharp non-linear drop in current beginning around 1 μM , along with significantly reduced current levels overall. These differences suggest that OECTs 1 and 2 functioned reliably, whereas OECT 3 likely suffered from fabrication defects or unstable gate behavior, and OECT 4 may have experienced electrical failure or enzyme degradation.

The on-current (I_{on}) exhibited a more pronounced decrease at higher glucose concentrations, particularly above 1 μM (refer to Figure 51). This trend of modulation of channel current may arise due to enzymatic glucose oxidation, though the absolute values were varied strongly across devices. The scatter plot of on-current data for each chip is provided in the Appendix (refer to Figure A15).

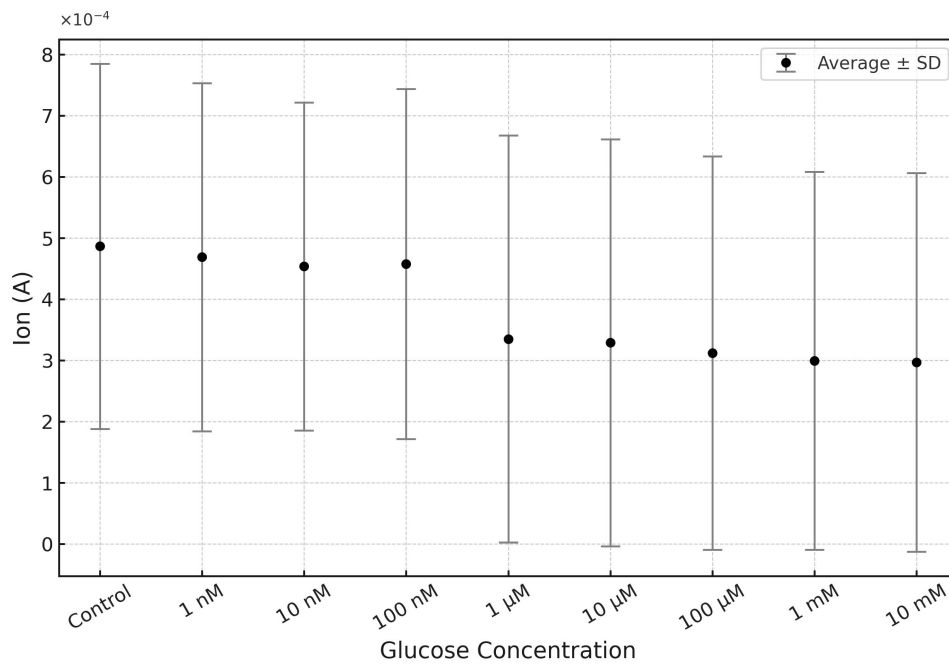


Figure 51: Average on current vs glucose concentrations across OECTs 1–4 with GDH enzyme.

The off-current (I_{off}) decreased slightly with increasing glucose concentration but showed high variability across chips (refer to Figure 52). While some devices displayed a modest concentration dependence, others showed nearly constant or noisy responses, limiting the reliability of this parameter for glucose quantification. The detailed scatter plot of individual chip responses is provided in the Appendix (refer to Figure A14).

The on/off ratio ($I_{\text{on}}/I_{\text{off}}$) provided an additional metric for assessing device performance. A moderate increase in the ratio was observed as glucose concentration increased, particularly in the micromolar-to-millimolar range (refer to Figure 53). This suggests that even when absolute current levels varied across chips, their relative modulation still captured the enzymatic activity. The corresponding scatter plot of on/off ratios from all devices is shown in the Appendix (refer to Figure A16).

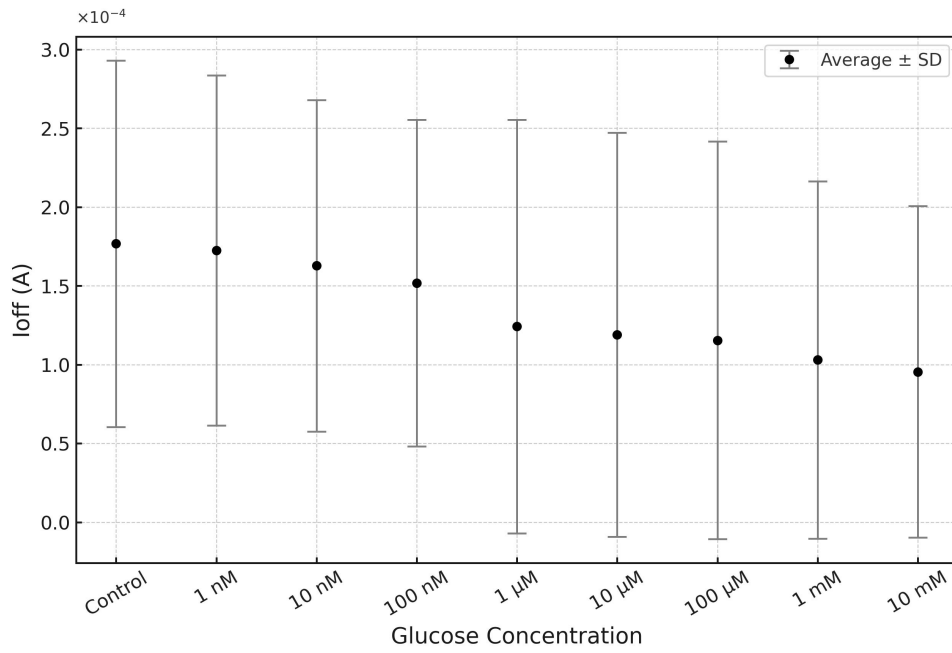


Figure 52: Average off current vs glucose concentrations across OECTs 1–4 with GDH enzyme.

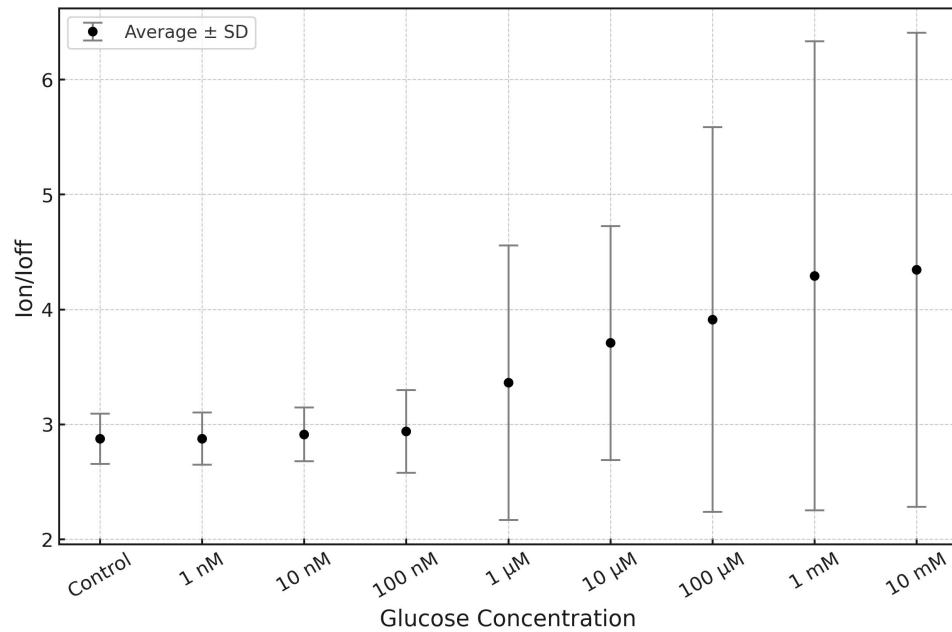


Figure 53: Average on/off ratio vs glucose concentrations across OECTs 1–4 with GDH enzyme.

The maximum transconductance (g_m) did not exhibit a consistent concentration-dependent trend across devices. While some OECTs showed a weak decrease with glucose, others displayed noisy or flat responses, likely due to variability in enzyme stability (refer to Figure 54). A detailed view of chip-to-chip scatter in maximum transconductance is included in the Appendix (refer to Figure A17).

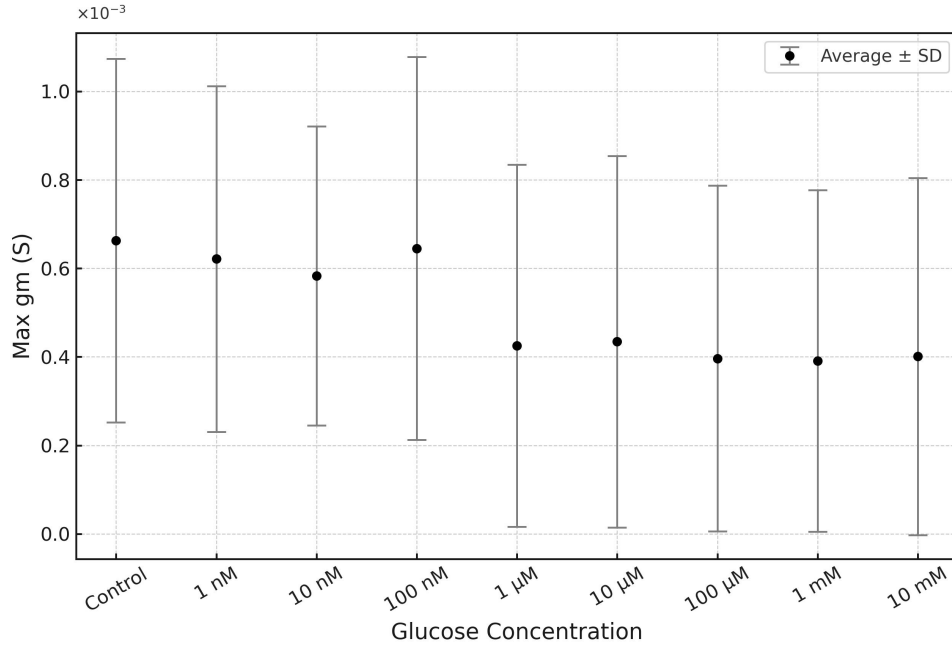


Figure 54: Average maximum transconductance vs glucose concentrations across OECTs 1–4 with GDH enzyme.

Despite these limitations, the results confirm that glucose detection is feasible in GDH-based OECTs, particularly in the sub-micromolar to low-micromolar concentration range. Variability in gate-electrolyte contact area, repeated freeze-thaw cycles of the enzyme, and fabrication inconsistencies likely contributed to these noisy responses.

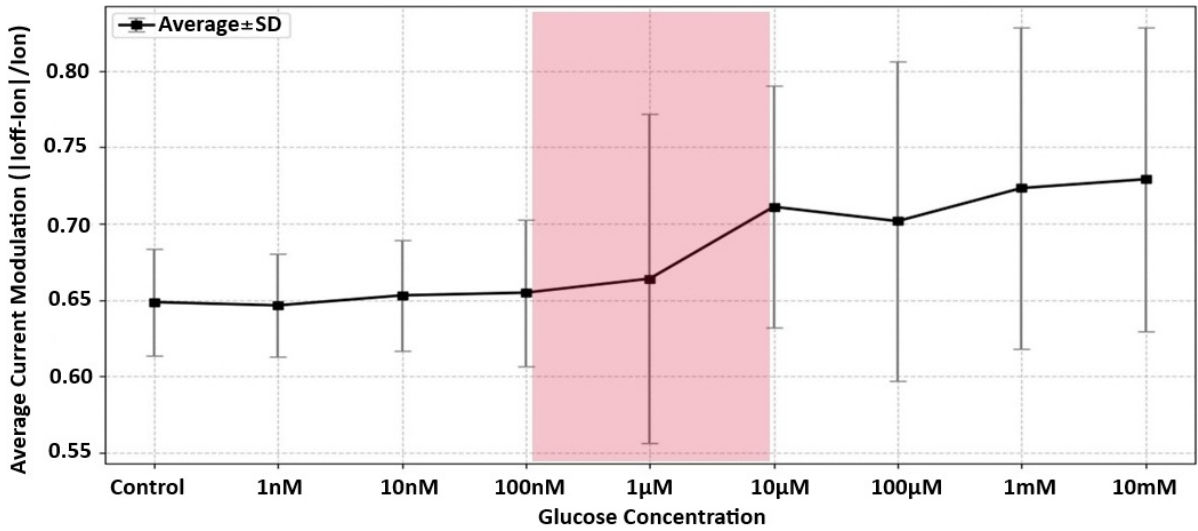


Figure 55: Average current modulation (ΔI_d) vs glucose concentrations for GDH-based OECTs (Average $\pm$ SD across devices).

To summarize device performance, Figure 55 presents the average current modulation (ΔI_d , mean $\pm$ SD) across functional devices. A weak but detectable increase is observed between 100 nM and 10 μ M, while the response becomes approximately linear between 10 μ M and 10 mM. This range overlaps with physiological blood glucose levels (4–10 mM) and also includes concentrations typically reported in sweat and saliva, confirming the potential applicability of GDH-based OECTs for biofluid sensing. [3, 26, 60]

Early attempts to measure glucose using only the traditional transfer characteristic method did not consistently show the anticipated responses. Standardizing experimental techniques, such as enzyme handling and gate electrode immersion depth and expanding the number of data points in the most sensitive concentration window may improve reproducibility. To better capture the direct impact of glucose on enzyme activity, the following section employs real-time drain current monitoring as an alternative measurement approach.

4.4 Comparative study between GOx and GDH enzyme based OECT using real time transfer measurements

4.4.1 GOx real time measurements

The testing procedure for real-time measurement is explained in Section 3.2.7.3. The drain current I_d vs. time response of the GOx-based OECT upon sequential addition of each glucose concentrations (1 μ M to 10 mM) at a 10-minute interval shows a clear stepwise change in I_d after each droplet addition, indicating the device's ability to respond to varying glucose levels (refer to Figure 56). The current changes are concentration-dependent, with more pronounced shifts at higher glucose concentrations. Similar I_d vs. time plots for the measurements (refer to Figures A8 and A9) are shown in the Appendix for reference.

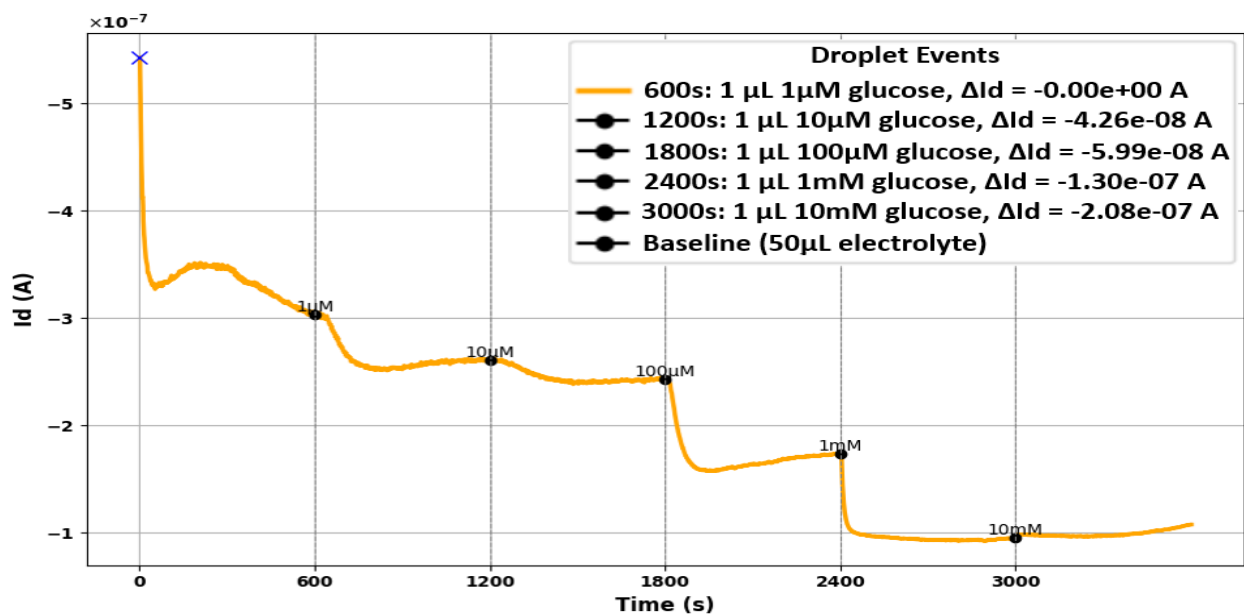


Figure 56: Transfer graph showing drain current I_d vs time graph for GOx based OECT

The ΔI_d vs glucose concentration plot for the GOx device shows an overall increase in ΔI_d with increasing glucose concentration, indicating that the device's sensitivity improves at higher analyte levels (refer to Figure 57). The $\Delta|I_d|$ values were calculated as described in Table 3. The relatively small changes at lower concentrations (1 μ M and 10 μ M) suggest a lower signal-to-noise ratio in this range, while the larger changes at 1 mM and 10 mM reflect a stronger enzymatic and electrochemical interaction between GOx and glucose. Notably, the error bars indicate higher standard deviations at the higher concentration intervals, which may arise from increased reaction kinetics, [35] enhanced ion migration effects, [5] or variability in droplet mixing at elevated analyte levels. [46]

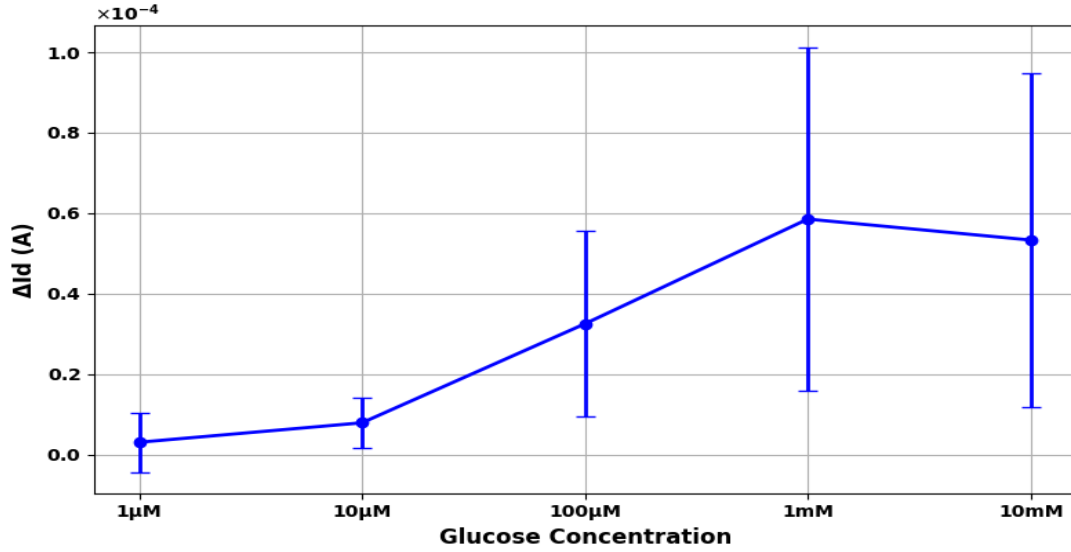


Figure 57: Drain current change ΔI_d vs glucose concentrations for a GOx-based OECT ($V_g = 0.6$ V, $V_d = -0.2$ V). The response is approximately linear up to ~ 1 mM, matching the range of sweat and saliva glucose. [24]

4.4.2 GDH real time measurements

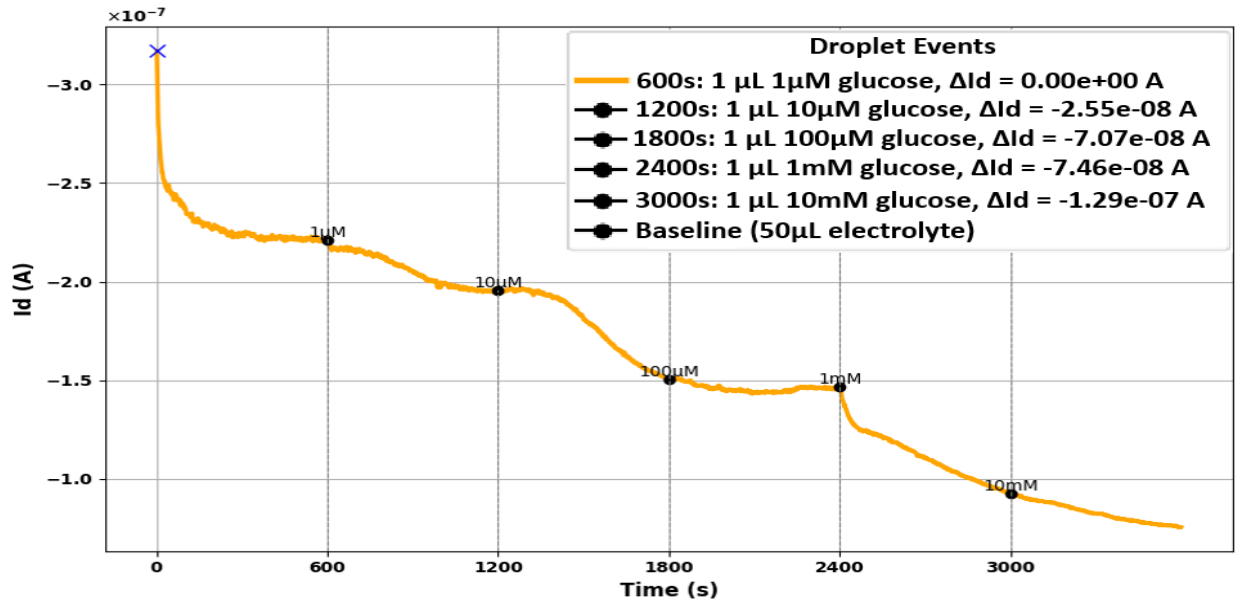


Figure 58: Transfer graph showing drain current I_d vs time graph for GDH based OECT

The real-time measurement procedure of a GDH-based OECT when glucose was added sequentially at concentrations of 1 μM, 10 μM, 100 μM, 1 mM, and 10 mM is explained in Section 3.2.7.3. Each droplet addition was done at a 10-minute interval (refer to Figure 58). After each injection, a distinct step-like change in I_d was observed, with the magnitude of the change gradually increasing as the glucose concentration increased. In contrast to the positive changes commonly seen with GOx-based devices, the overall behaviour shows a consistent downward drift in drain current as glucose concentration increases. This signal inversion may be linked to GDH's oxygen-independent redox cycle, in which the cofactor-mediated electron transfer process alters the local ionic distribution, thereby lowering channel conductivity. The electrolyte composition (MES buffer) may also

contribute to this effect by changing the ion-semiconductor interactions at the interface.

The Appendix includes two I_d vs. time plots from replicate measurements (refer to Figures A6 and A7), demonstrating the GDH response's repeatability across different devices and runs. Although the baseline current values and absolute magnitudes of ΔI_d vary slightly between devices, the downward stepwise trend remains consistent.

The plot (refer to Figure 59) shows the $\Delta|I_d|$ vs. glucose concentration for the GDH-based OEET. The $\Delta|I_d|$ values are calculated using the method outlined in Table 3. $\Delta|I_d|$ increases with higher glucose concentrations, indicating that the GDH response is concentration-dependent. However, at 1 mM and 10 mM, the error bars show significantly larger standard deviations than at lower concentrations. This suggests that, while the sensor maintains sensitivity at high analyte levels, measurement variability increases. Possible explanations for this increased variability include increased ionic flux and screening effects at high glucose concentrations, [5] variability in droplet spreading and mixing at the device surface [31] and nonlinearities in the enzymatic reaction rate at high substrate levels. [55] Despite this, the overall trend is consistent, confirming GDH's suitability for oxygen-independent glucose sensing applications, though further optimisation may be required to improve stability at higher concentration ranges.

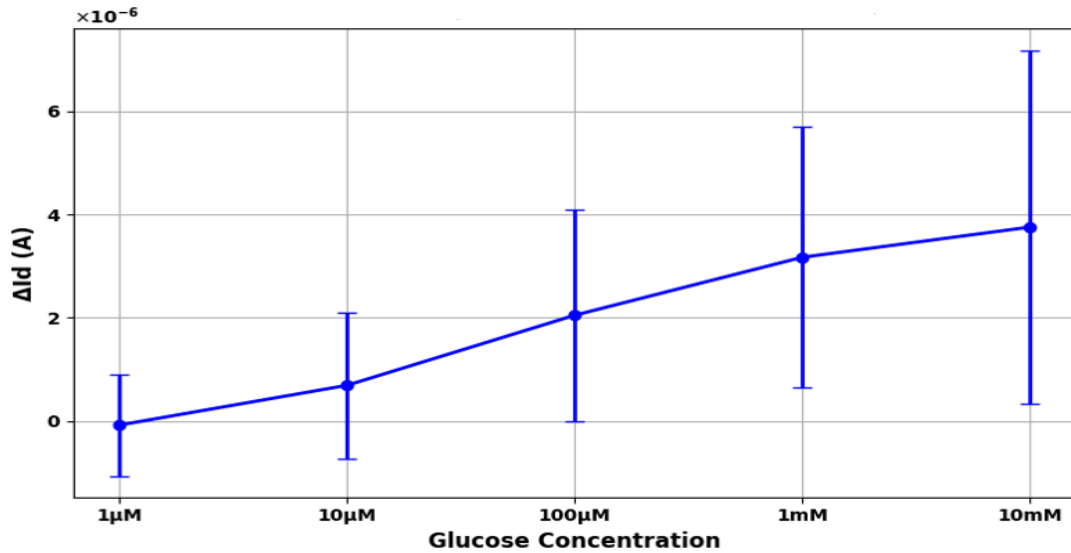
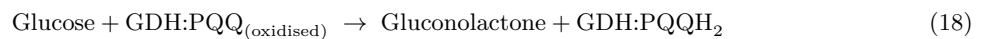
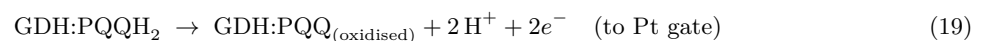


Figure 59: Drain current change ΔI_d vs glucose concentrations for a GDH-based OEET ($V_g = 0.6$ V, $V_d = -0.2$ V). The device exhibits an approximately linear response from 10 μ M to 10 mM, covering the physiological blood-glucose range (4–10 mM) and extending into sweat and saliva levels. [24]

Based on the chemistry of PQQ-dependent GDH and the electrochemical gate coupling in OEETs, it is hypothesized that glucose is transduced as follows. First, the enzyme–cofactor complex catalyzes glucose oxidation while reducing the bound PQQ to PQQH₂, a reaction established for PQQ-GDH by Xie *et al.* [55]:



At the electrolyte–platinum gate interface, the reduced cofactor is reoxidized, delivering electrons to the gate and regenerating the active cofactor (a net two-electron process). Importantly, this mechanism proceeds without molecular oxygen, consistent with the oxygen-independent operation of PQQ-GDH described by Milton *et al.* [31]:



The two steps yield the overall sensing reaction:



Because the gate/electrolyte redox couple sets the interfacial potential, the gate offset follows a Nernstian dependence on the PQQ/PQQH₂ ratio:

$$E_g = E^0 + \frac{RT}{2F} \ln \left(\frac{[\text{PQQ}]}{[\text{PQQH}_2]} \right), \quad (21)$$

so that increasing glucose drives the reduction $\text{PQQ} \rightarrow \text{PQQH}_2$, shifts E_g , and changes the effective gate bias seen by the channel. Under steady-state conditions, the OECT drain current therefore varies with the redox-induced gate shift; in the common depletion-mode model this appears as:

$$I_d = \frac{1}{2} \mu C^* \frac{WT}{L} (V_{g,\text{eff}} - V_{\text{th}})^2, \quad (22)$$

where μ is the electronic mobility, C^* the volumetric capacitance, and W, T, L are the channel dimensions. This expression follows the steady-state OECT model formulated by Bernardis and Malliaras [5]. As equations. (18)–(19) proceed faster at higher glucose concentrations, $V_{g,\text{eff}}$ shifts according to equation. (21), leading to predictable modulation of I_d in equation. (22). In a p-type PEDOT:PSS device, the net effect is progressive dedoping of the channel and a systematic decrease of I_d with increasing glucose.

5 Conclusion

Diabetes mellitus is a prevalent metabolic disorder characterized by the inability of the body to regulate blood glucose levels effectively. This imbalance can lead to severe health complications, including cardiovascular disease, neuropathy, and kidney failure. Therefore, accurate and continuous glucose monitoring is essential for managing the disease and preventing both hyperglycemia and hypoglycemia. Organic Electrochemical Transistors (OECTs) have gained attention as promising candidates for glucose sensing due to their high sensitivity, low operating voltage, and excellent compatibility with biological environments.

In this study, OECTs were designed and fabricated for enzymatic glucose detection using glucose oxidase (GOx) and glucose dehydrogenase (GDH). Preliminary tests using cyclic voltammetry (CV) were conducted on channels with two architectures: a dual-layer design with a modified PEDOT:PSS base and a pure PEDOT:PSS top layer, and a single-layer made of pure PEDOT:PSS. The dual-layer configuration showed improved stability and reproducibility, while the single-layer devices performed poorly with unstable responses in OECTs. Hence, the dual-layer channel was adopted for all subsequent OECT measurements.

Two enzyme integration strategies were examined. In the first method, GDH was immobilized on the mPEDOT:PSS channel by drop-casting a GDH-PQQ solution and refrigerating it overnight to promote adsorption. However, this approach was unsuccessful due to enzyme wash-off during measurements, leading to unstable and inconsistent responses.

The second and more effective approach involved the use of a floating-enzyme electrolyte, in which the enzyme was mixed directly into the buffer solution. This method enabled enzymatic reactions to occur freely at the electrolyte-channel interface. Under this configuration, both GOx and GDH were tested in steady-state and real-time conditions, and measurable drain current modulation was observed following glucose addition. The two enzymes exhibited comparable responses, confirming their capability for glucose detection within the tested range.

GOx was simpler to prepare, as it required only PBS buffer, whereas GDH preparation involved the cofactor PQQ and pH adjustment using MES buffer. Preliminary results indicated the possibility of direct electron transfer (DET) between PQQ-GDH and PEDOT:PSS, although this could not be conclusively verified due to the dominant ionic effects inherent to OECT operation. Further optimization of enzyme-electrode coupling and channel modification could help determine this mechanism in future work.

Overall, the OECTs developed in this study demonstrated effective glucose detection within the concentration range typical of blood (1–10 mM). GOx showed advantages in terms of simplicity, stability, and cost, while GDH offered potential for oxygen-independent operation. The findings of this study highlights the influence of enzyme, channel composition, and electrochemical behavior on the performance of OECT-based glucose sensors designed for diabetes monitoring.

6 Next steps

Based on the current results and feedback, two big changes are suggested to make the glucose sensing work even better. For real time measurements, the gate voltage could be lowered from 0.6 V to 0.3 V while measurements were being taken. It is expected that this lower bias will make the saturation period longer in real-time measurements, which will make the signal response to changes in glucose concentration clearer and more distinct.

Second, the process for keeping enzymes in place needs to be improved. This study utilised both channel immobilisation and floating enzyme electrolyte ; however, neither method consistently produced results. A new way to do this would be to directly immobilize the enzyme on the gate electrode. This setup would make it easier to change the gate potential and make the drain current response stronger and more sensitive by letting the electrons released during the enzyme–glucose reaction reach the gate faster.

References

- [1] I. Ahmed and coauthor TitleUnknown. Recent advances in optical sensors for continuous glucose monitoring. *RSC Sensors & Diagnostics*, 1, 2022.
- [2] American Diabetes Association. Hyperglycemia (high blood glucose). <https://www.diabetes.org/healthy-living/medication-treatments/blood-glucose-testing-and-control/hyperglycemia>, 2022. Accessed: July 6, 2022.
- [3] American Diabetes Association. Standards of medical care in diabetes—2023. *Diabetes Care*, 46(Supplement 1):S1–S300, 2023.
- [4] American Diabetes Association. Low blood glucose (hypoglycemia). <https://www.diabetes.org/living-with-diabetes/hypoglycemia-low-blood-glucose>, 2025. Accessed: September 6, 2025.
- [5] David A. Bernards and George G. Malliaras. Steady-state and transient behavior of organic electrochemical transistors. *Advanced Functional Materials*, 17(17):3538–3544, 2007.
- [6] Jessica R. Castle and W. Kenneth Ward. Amperometric glucose sensors: Sources of error and potential benefit of redundancy. *Journal of Diabetes Science and Technology*, 4(1):221–225, 2010.
- [7] Qin Chen, Wenwen Zhang, and Wenzhi Wu. Vis/nir/fir spectroscopy for blood glucose monitoring: A review. *AIP Advances*, 15(3):030703, 2025.
- [8] Yuhua Chen, Yanghan Sun, Yi Li, Zhuo Wen, Xinyu Peng, Yuanke He, Yuanfang Hou, Jingchuan Fan, Guangchao Zang, and Yuchan Zhang. A wearable non-enzymatic sensor for continuous monitoring of glucose in human sweat. *Biosensors and Bioelectronics*, 172:112782, 2021.
- [9] Fabio Cicoira, Michele Sessolo, Omid Yaghmazadeh, John A. DeFranco, Sang Yoon Yang, and George G. Malliaras. Influence of device geometry on sensor characteristics of planar organic electrochemical transistors. *Advanced Materials*, 22(9):1012–1016, 2010.
- [10] Renan Colucci, Henrique Frulani de Paula Barbosa, Florian Günther, Priscila Cavassin, and Gregório Couto Faria. Recent advances in modeling organic electrochemical transistors. *Journal of Physics: Materials*, 3(1):012001, 2020.
- [11] Stefano Ferri, Katsuhiko Kojima, and Koji Sode. Review of glucose oxidases and glucose dehydrogenases: A bird’s eye view of glucose sensing enzymes. *Journal of Diabetes Science and Technology*, 5(5):1068–1076, 2011.
- [12] Victoria Flexer, Fabien Durand, Seiya Tsujimura, and Nicolas Mano. Efficient direct electron transfer of pqq-glucose dehydrogenase on carbon cryogel electrodes at neutral ph. *Analytical Chemistry*, 83(14):5721–5727, 2011.
- [13] F. Franceschini et al. Nickel-based catalysts for non-enzymatic electrochemical glucose sensors. *Current Opinion in Electrochemistry*, 34:100976, 2022.
- [14] Jacob T. Friedlein, Robert R. McLeod, and Jonathan Rivnay. Device physics of organic electrochemical transistors. *Organic Electronics*, 63:398–414, 2018.
- [15] P. Ghosh et al. Development of a conductometric glucose sensor in human blood serum phantom using nanomolar detection. *Biosensors and Bioelectronics*, 180:113009, 2021.
- [16] Alexander Giovannitti, Daria-Tereza Sbircea, Sahika Inal, Christopher B Nielsen, Matthew J Donahue, Muhammad R Niazi, David A Hanifi, Aram Amassian, George G Malliaras, and Jonathan Rivnay. Controlling

-
- the mode of operation of organic transistors through side-chain engineering. *Proceedings of the National Academy of Sciences*, 113(43):12017–12022, 2016.
- [17] K. Gupta, R. Rani, and D.S. Chauhan. The progress of glucose monitoring—a review of invasive to minimally and non-invasive techniques, devices and sensors. *Sensors*, 19(4):800, 2019.
 - [18] Adam Heller and Ben Feldman. Electrochemical glucose sensors and their applications in diabetes management. *Chemical Reviews*, 108(7):2482–2505, 2008.
 - [19] Jin Hu, Jiajia Dai, Caiping Huang, Xierong Zeng, Weiwei Wei, Zhezhe Wang, and Peng Lin. Organic electrochemical transistor with mos₂ nanosheets modified gate electrode for sensitive glucose sensing. *Sensors*, 23(17):7449, 2023. These authors contributed equally. Correspondence to: Weiwei Wei, Peng Lin.
 - [20] Lin Hu, Jiaying Song, Xinxing Yin, Zhen Su, and Zaifang Li. Research progress on polymer solar cells based on pedot:pss electrodes. *Polymers*, 12(1):145, 2020. China-Australia Institute for Advanced Materials and Manufacturing (IAMM), Jiaying University, Jiaying 314001, China.
 - [21] Sahika Inal, George G. Malliaras, and Jonathan Rivnay. Benchmarking organic mixed conductors for transistors. *Nature Communications*, 8:1767, 2017.
 - [22] International Diabetes Federation. Idf diabetes atlas, 11th edition. <https://idf.org/news/idf-diabetes-atlas-11th-edition/>, 2023. Accessed: 15-Aug-2025.
 - [23] Won-Yong Jeon, Han-Sem Kim, Hye-Won Jang, Ye-Sung Lee, Ueon Sang Shin, Hyug-Han Kim, and Young-Bong Choi. A stable glucose sensor with direct electron transfer, based on glucose dehydrogenase and chitosan hydro bonded multi-walled carbon nanotubes. *Biochemical Engineering Journal*, page 108589, 2022.
 - [24] M. Johnston, L. Wang, X. Zhan, J. Wong, and X. Zhang. Non-invasive and wearable sensors for monitoring glucose in body fluids. *Biosensors*, 11(7):246, 2021.
 - [25] D. Khodagholy, J. Rivnay, M. Sessolo, M. Gurfinkel, P. Leleux, L. H. Jimison, and G. G. Malliaras. High transconductance organic electrochemical transistors. *Nature Communications*, 4:2133, 2013.
 - [26] H. Lee, Y. Hong, S. Park, and T. Ha. Wearable and flexible electrochemical sensors for sweat analytes: A review. *Microsystems & Nanoengineering*, 8(1):1–19, 2022.
 - [27] R. Li et al. Nanostructured nickel-based non-enzymatic glucose sensors. *Chemistry – An Asian Journal*, 17(6):e202101111, 2022.
 - [28] Y. Lu, W. Wang, J. Liu, M. Xie, Q. Liu, and S. Li. Vascular complications of diabetes: A narrative review. *Medicine*, 102(40):e35285, 2023.
 - [29] S. M. Lundsgaard-Nielsen, A. Pors, S. O. Banke, J. E. Henriksen, D. K. Hepp, and A. Weber. Critical-depth raman spectroscopy enables home-use non-invasive glucose monitoring. *PLoS ONE*, 13(5):e0197134, 2018.
 - [30] Ana J. L. Martins, Reinaldo J. Velásquez, Denis B. Gaillac, Vanessa N. Santos, Diego C. Tami, Rodrigo N. P. Souza, Fernan C. Osorio, Gabriel A. Fogli, Beatriz S. Soares, Cassio G. do Rego, Gilberto Medeiros-Ribeiro, Juliana B. Drummond, Clara M. Mosquera-Lopez, and Jhonattan C. Ramirez. A comprehensive review of non-invasive optical and microwave biosensors for glucose monitoring. *Biosensors and Bioelectronics*, 271:117081, 2025.
 - [31] Richard D. Milton, Fabio Giroud, Sahar Thumser, L. Jared Minter, and Erwin Reisner. Hydrogen peroxide produced by glucose oxidase affects the performance of laccase cathodes in glucose/oxygen fuel cells: Fad-dependent glucose dehydrogenase as a replacement. *Physical Chemistry Chemical Physics*, 15(44):19371–19379, 2013.

-
- [32] Jamal S. Mohamed, Sasi Karan Ramanujam Rajakumar, Murugasenapathi N. K., and Tamilarasan Palanisamy. Fabrication of cylindrical diffusion-facilitated organic electrochemical transistor and its sweat glucose sensing application. *Journal of Materials Science: Materials in Engineering*, 20(51), 2025.
 - [33] David L. Nelson and Michael M. Cox. *Lehninger Principles of Biochemistry*. W. H. Freeman and Company, 7 edition, 2017.
 - [34] Josefin Nissa, Per Janson, and Magnus Berggren. The role of relative capacitances in impedance sensing with organic electrochemical transistors. *Advanced Electronic Materials*, 7(4):2001173, 2021.
 - [35] Ezekiel Odebunmi and Samson Owolude. Kinetic and thermodynamic studies of glucose oxidase catalysed oxidation reaction of glucose. *The Pacific Journal of Science and Technology*, 11(4):123–131, 2007. Cited 33 times, accessed Jan 2007.
 - [36] David Ohayon, Victor Druet, and Sahika Inal. A guide for the characterization of organic electrochemical transistors and channel materials. *Chemical Society Reviews*, 2022. Received 7th November 2022.
 - [37] World Health Organization. Diabetes. <https://www.who.int/news-room/fact-sheets/detail/diabetes>, 2023. Accessed: 2024-06-05.
 - [38] Anna-Maria Pappa, Vincenzo F. Curto, Markus Braendlein, Xenofon Strakosas, Matthew J. Donahue, Matteo Fiocchi, Jonathan Rivnay, Fabio Benfenati, Roisin M. Owens, and George G. Malliaras. Organic transistor arrays integrated with finger-powered microfluidics for multianalyte saliva testing. *Advanced Healthcare Materials*, 5(17):2295–2302, 2016.
 - [39] J.C. Pickup, F. Hussain, N.D. Evans, O.J. Rolinski, and D.J.S. Birch. Fluorescence-based glucose sensors. *Biosensors and Bioelectronics*, 20(12):2555–2565, 2005.
 - [40] J. Rivnay and et al. Organic electrochemical transistors. *Nature Reviews Materials*, 3:17086, 2018.
 - [41] Jonathan Rivnay, Pierre Leleux, Marco Ferro, Michele Sessolo, Ashley Williamson, Dimitrios A Koutsouras, Dario Khodagholy, Manuella Ramuz, Xenofon Strakosas, and Richard M Owens. High-performance transistors for bioelectronics through tuning of channel thickness. *Science Advances*, 1(4):e1400251, 2015.
 - [42] Matteo Sensi, Marcello Berto, Andrea Candini, Andrea Liscio, Andrea Cossarizza, Valerio Beni, Fabio Biscarini, and Carlo Augusto Bortolotti. Modulating the faradic operation of all-printed organic electrochemical transistors by facile *in situ* modification of the gate electrode. *Advanced Electronic Materials*, 5(3):1800566, March 2019.
 - [43] M. Singh, U. Z. A. Ungku Faiz, S. Gravelsins, Y. Suganuma, N. K. Kotoulas, M. Croxall, A. Khan-Trottier, C. Goh, and A. Dhirani. Glucose oxidase kinetics using mno_2 nanosheets: confirming michaelis-menten kinetics and quantifying decreasing enzyme performance with increasing buffer concentration. *Nanoscale Advances*, 3:3816–3823, 2021.
 - [44] Takao Someya, Zhenan Bao, and George G Malliaras. The rise of plastic bioelectronics. *Nature*, 540(7633):379–385, 2016.
 - [45] C. Song et al. Electrochemical glucose sensors: Classification, catalyst innovation, and sampling mode evolution. *Biotechnology Journal*, 19(e2200319), 2024.
 - [46] V. Srinivasan. Droplet-based microfluidic lab-on-a-chip for glucose detection. *Lab-on-a-Chip Platform for In Vitro Measurement of Glucose*, 2004.
 - [47] Author(s) TBD. Recent advances in optical sensors for continuous glucose monitoring. *Sensors & Diagnostics*, 2022.

-
- [48] Inc. Tektronix. Keithley 4200a-scs parameter analyzer. <https://www.tek.com/en/products/keithley/4200a-scs-parameter-analyzer>, 2025. Accessed: 2025-07-19; "The highest performance parameter analyzer, it delivers synchronizing current-voltage (I-V), capacitance-voltage (C-V) and ultra-fast pulsed I-V measurements." :contentReference[oaicite:1]index=1.
- [49] TH Wildau Project Team. Pedot-project: Direct electron transfer gdh. **Internalprojectpresentation, notpubliclyavailable**, 2023. Status Meeting Presentation, June 2023.
- [50] Deyu Tu and Simone Fabiano. Mixed ion-electron transport in organic electrochemical transistors. *Applied Physics Letters*, 117(8):080501, 2020.
- [51] J. Wang. Electrochemical glucose biosensors. *Chemical Reviews*, 108(2):814–825, 2008.
- [52] Henry S. White, Gregg P. Kittlesen, and Mark S. Wrighton. Chemical derivatization of an array of three gold microelectrodes with polypyrrole: Fabrication of a molecule-based transistor. *Journal of the American Chemical Society*, 106(18):5375–5377, 1984.
- [53] World Health Organization. Definition and diagnosis of diabetes mellitus and intermediate hyperglycemia: report of a who/idf consultation. *World Health Organization*, 2006.
- [54] Yingjie Wu, Qihan Zhang, He Wang, and Mingfeng Wang. Multiscale engineering of functional organic polymer interfaces for neuronal stimulation and recording. *Materials Chemistry Frontiers*, 4(11):3444–3471, 2020.
- [55] Weihong Xie, Leif Bülow, and Bin Xie. Pyrroloquinoline quinone glucose dehydrogenase adopted in thermometric analysis for enhancement of glucose determination. *Analytical and Bioanalytical Chemistry*, 410(15):3655–3664, 2018.
- [56] Feng Yan. Noninvasive glucose detection in saliva with organic electrochemical transistors. Technical report, The Hong Kong Polytechnic University, Research and Innovation Office, 2021. Technology showcase report. Principal Investigator: Prof. Feng Yan.
- [57] Eun-Hyung Yoo and Soo-Young Lee. Glucose biosensors: An overview of use in clinical practice. *Sensors*, 10(5):4558–4576, 2010.
- [58] Guangye Zhang, Chen Xie, Peng You, and Shunpu Li. Organic electrochemical transistor. In *Introduction to Organic Electronic Devices*, pages 207–220. Springer, Singapore, 2022. First Online: 14 October 2022.
- [59] Y. Zhang, Y. Bai, and J. Jia. The application of wearable glucose sensors in point-of-care testing. *Frontiers in Bioengineering and Biotechnology*, 9:774210, 2021.
- [60] Yu Zhang, Jun Guo, Hong Lin, Hong Ma, and Bo Lu. Noninvasive glucose monitoring using saliva nano-biosensor. *Nano Communication Networks*, 6(3):96–101, 2015.
- [61] C. Zhao, J. Yang, and W. Ma. Transient response and ionic dynamics in organic electrochemical transistors. *Nano-Micro Letters*, 16:233, 2024.

A Appendix-Figures

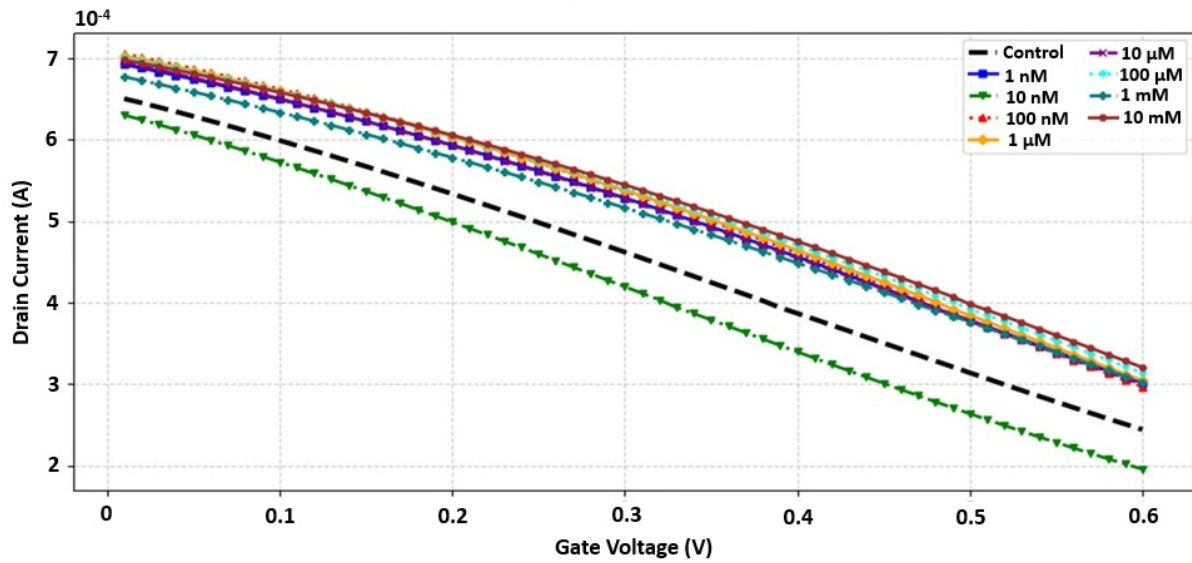


Figure A1: Transfer characteristics of GOx-based OECT 2 sensor under different glucose concentrations.

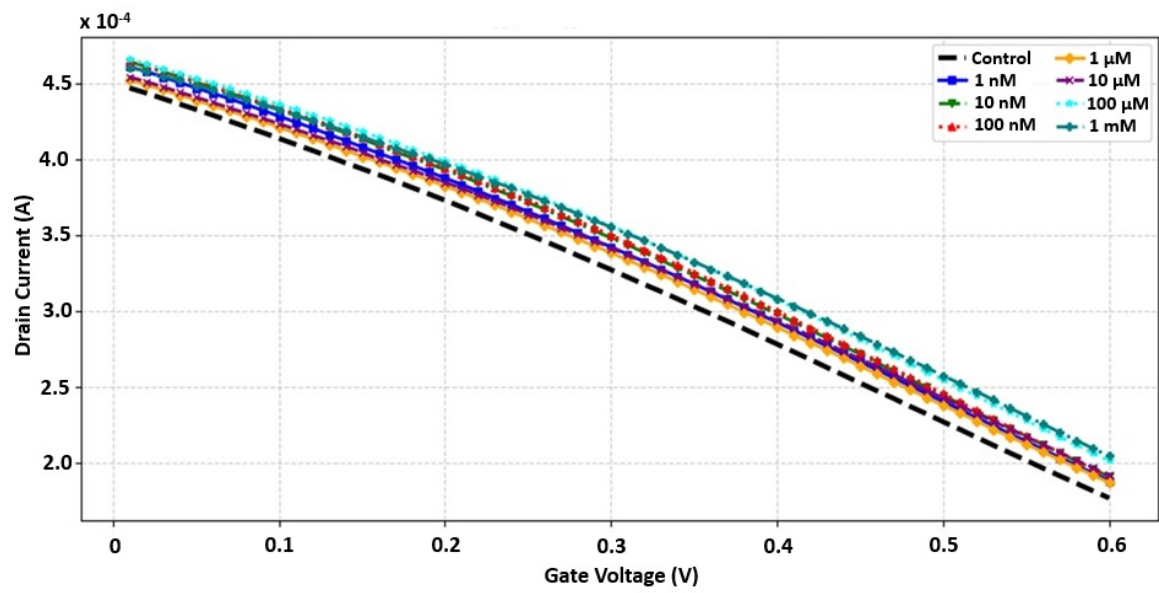


Figure A2: Transfer characteristics of GOx-based OECT 3 sensor under different glucose concentrations.

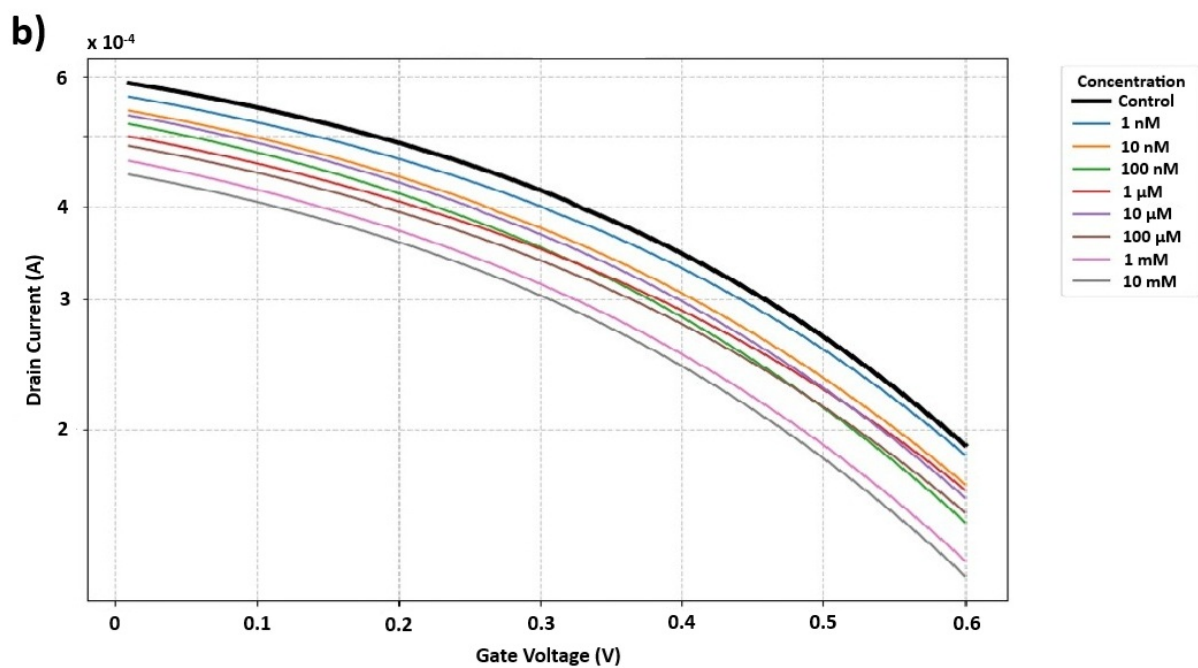


Figure A3: Transfer graphs showing drain current vs gate voltage under different glucose concentrations for OECT 2.

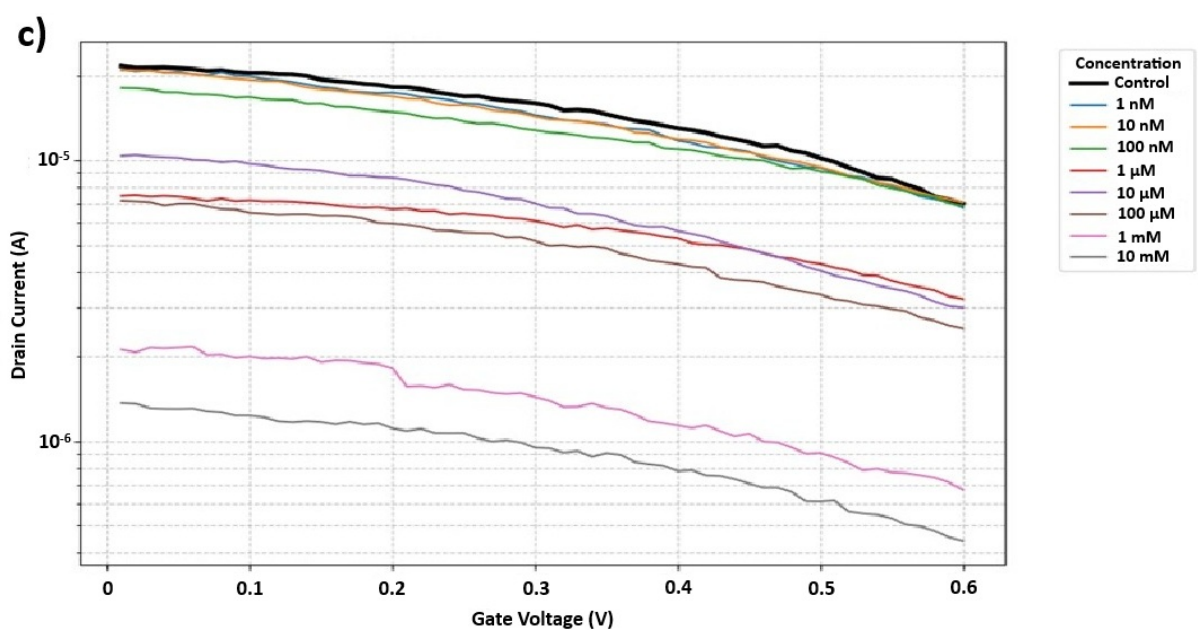


Figure A4: Transfer graph showing drain current vs gate voltage under different glucose concentrations for OECT 3.

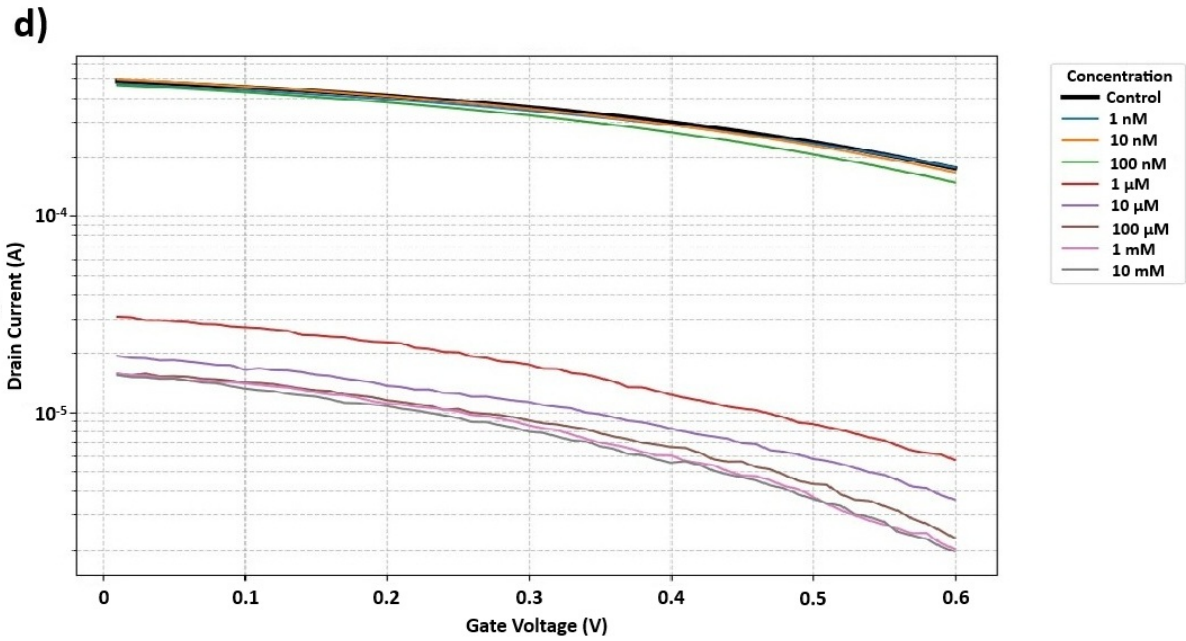


Figure A5: Transfer graph showing drain current vs gate voltage under different glucose concentrations for OECT 4.

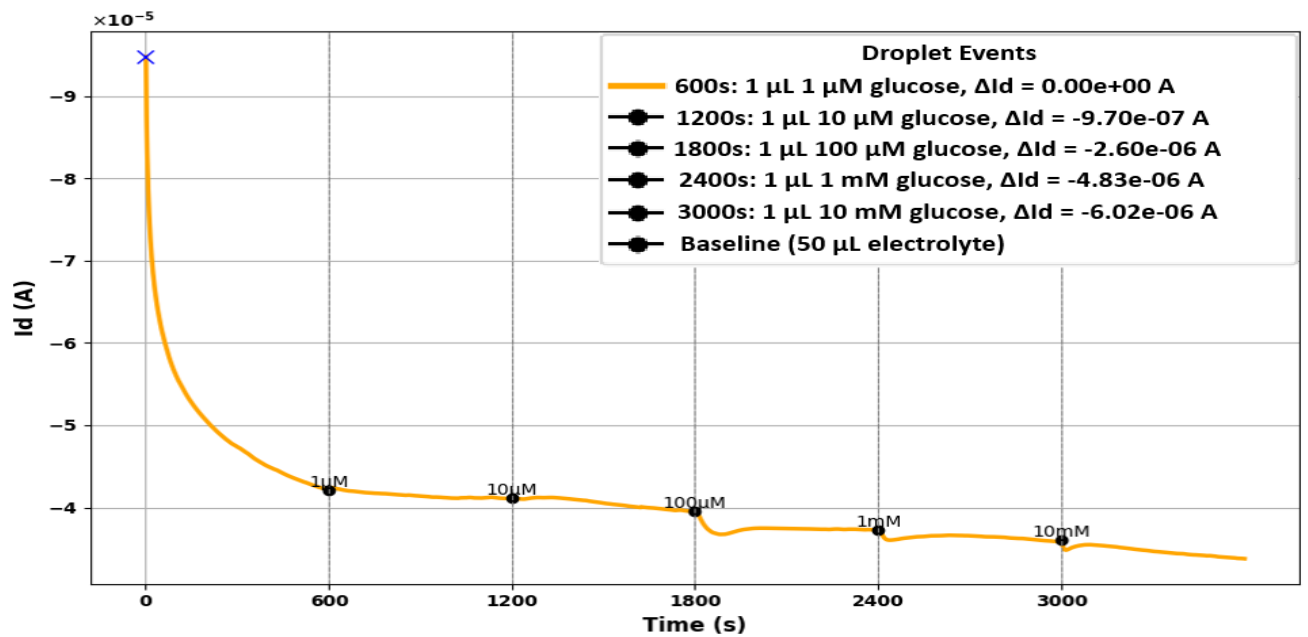


Figure A6: Transfer graph showing drain current I_d vs time graph for GDH based OECT 2.

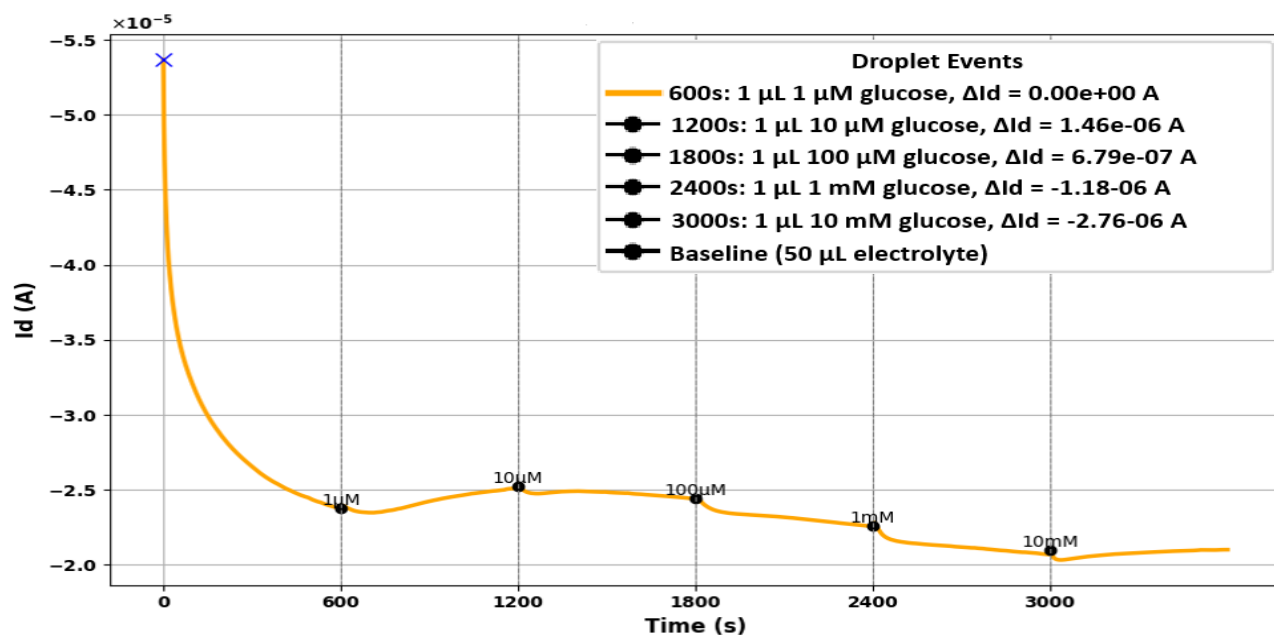


Figure A7: Transfer graph showing drain current I_d vs time graph for GDH based OECT 3.

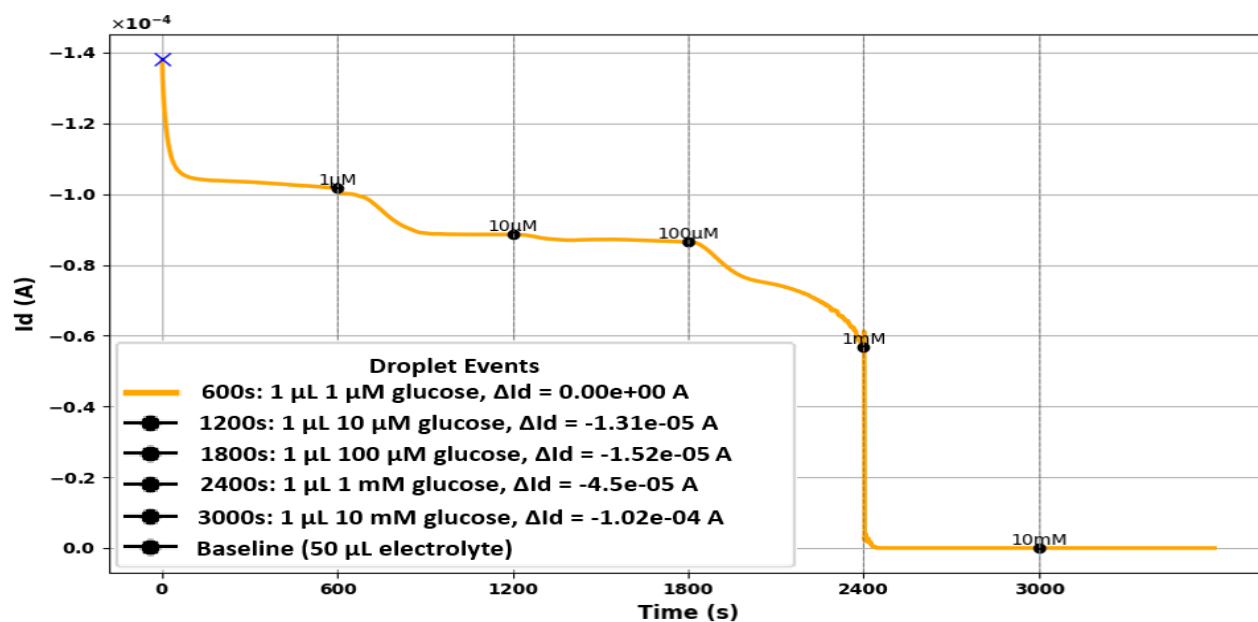


Figure A8: Transfer graph showing drain current I_d vs time graph for GOx based OECT 2.

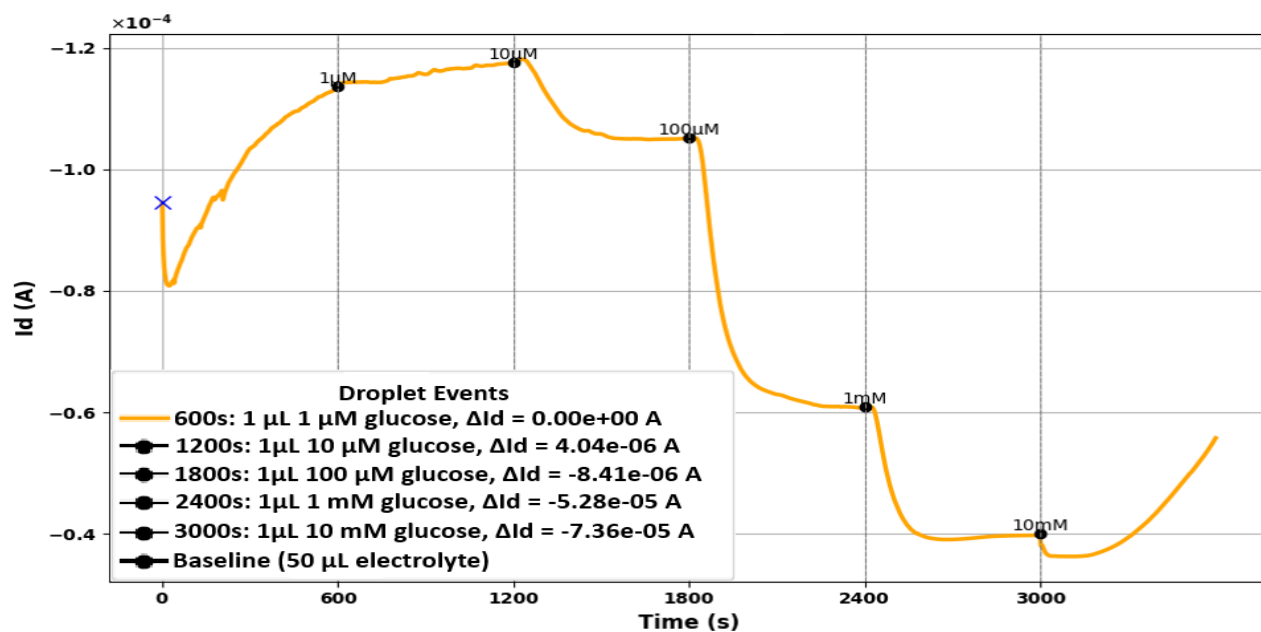


Figure A9: Transfer graph showing drain current I_d vs time graph for GOx based OECT 3.

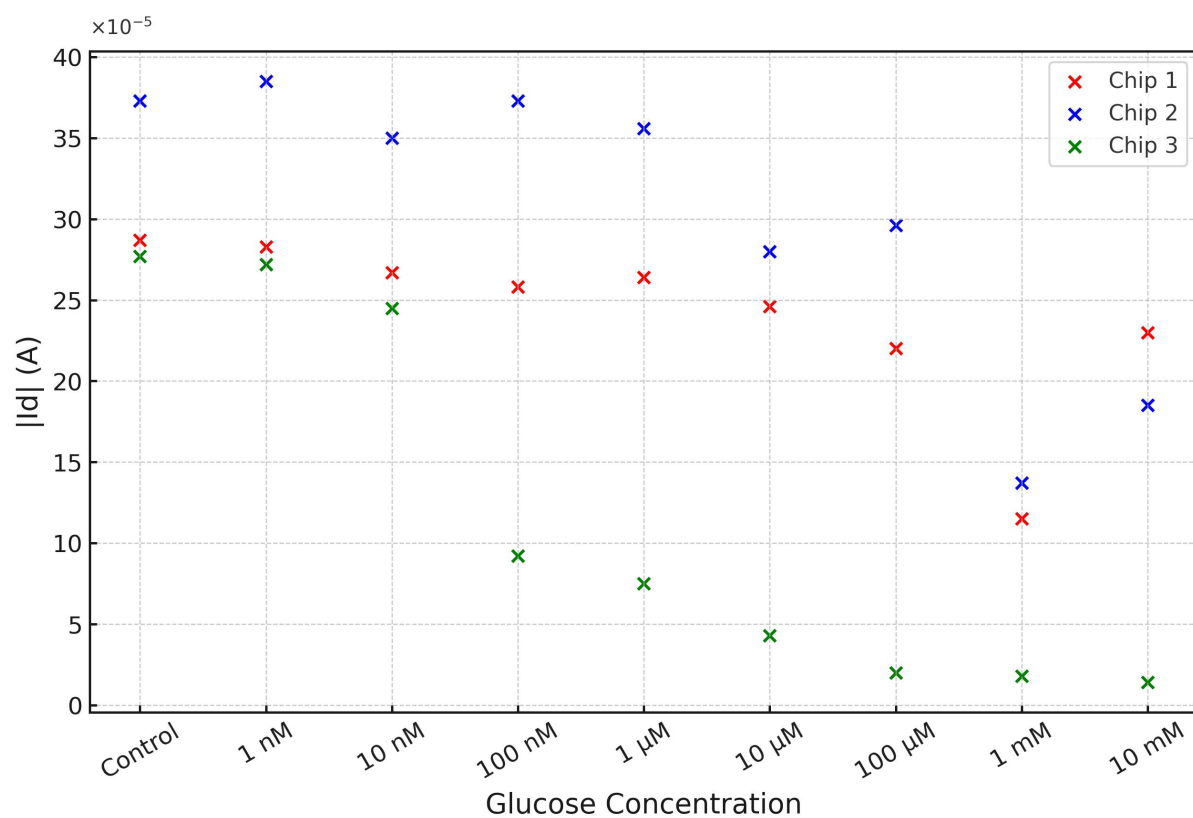


Figure A10: Off current vs glucose concentrations comparison of OECT 1, OECT 2, OECT 3 with GOx enzyme.

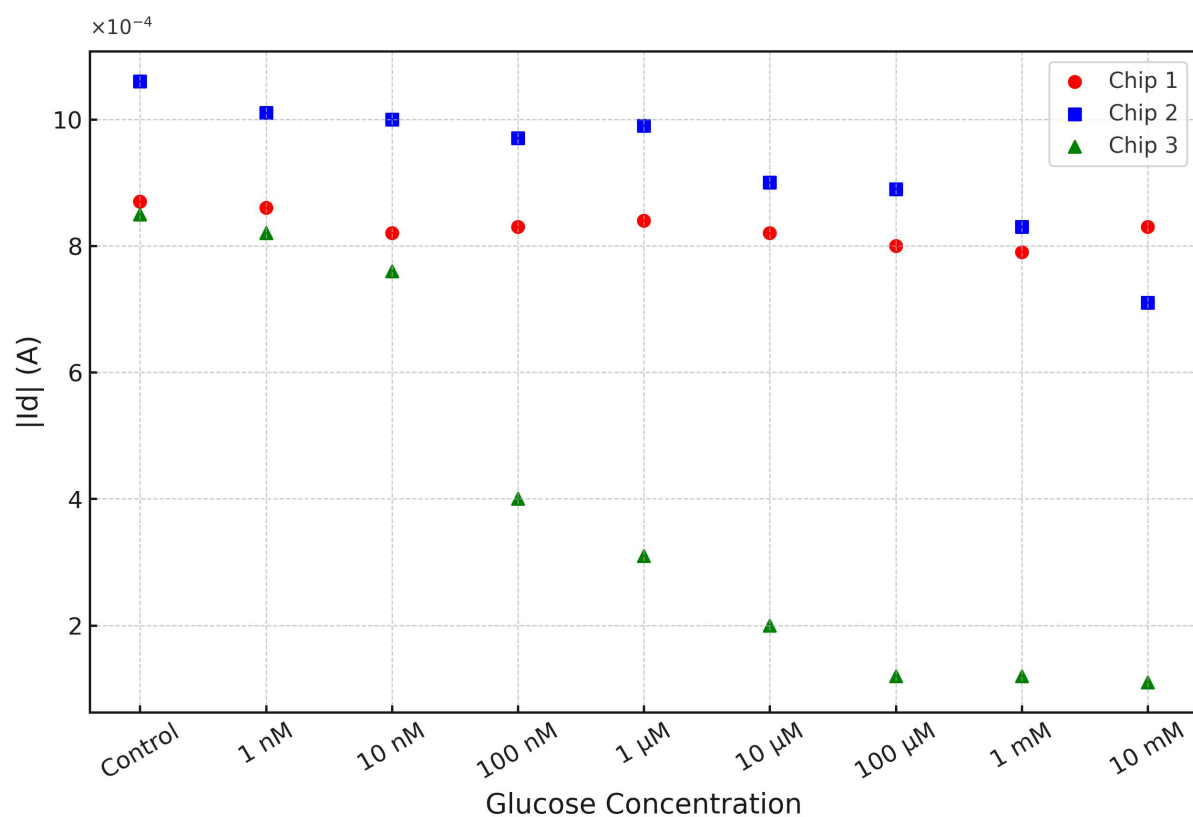


Figure A11: On current vs glucose concentrations comparison of OECT 1, OECT 2, OECT 3 with GOx enzyme.

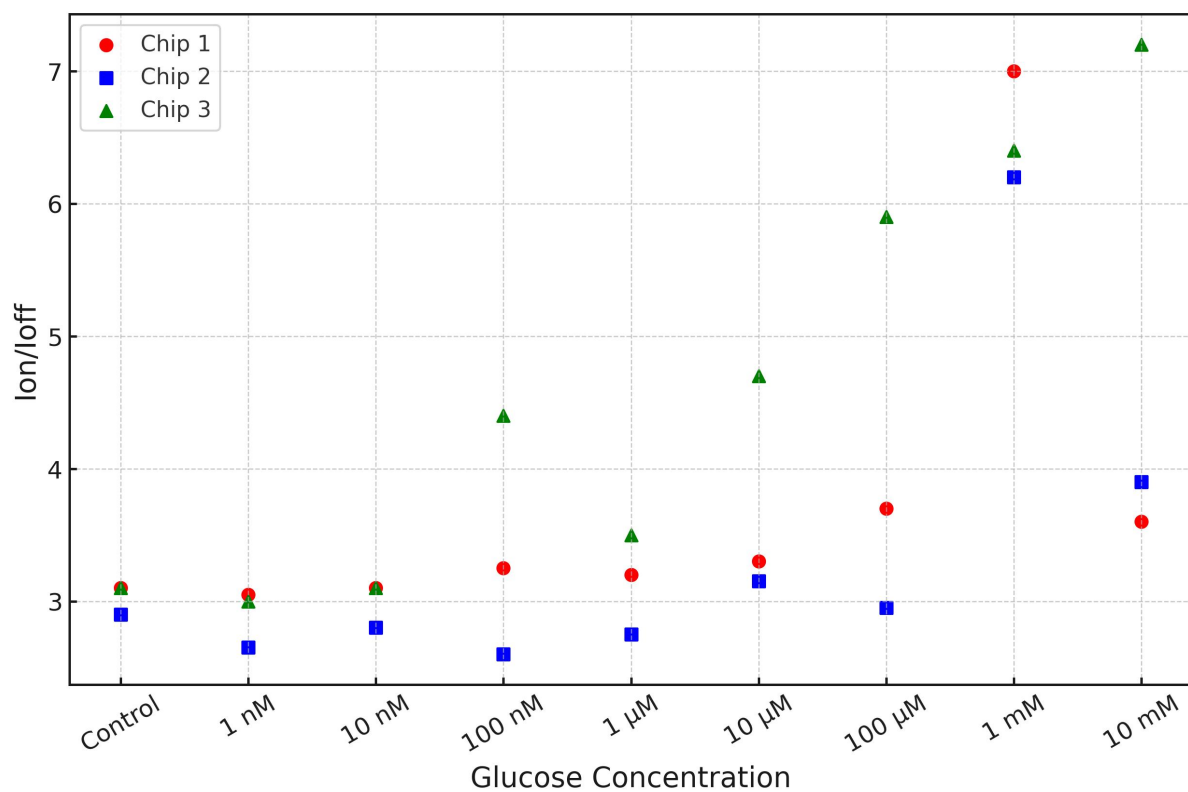


Figure A12: On/off ratio vs glucose concentrations comparison of OECT 1, OECT 2, OECT 3 with GOx enzyme.

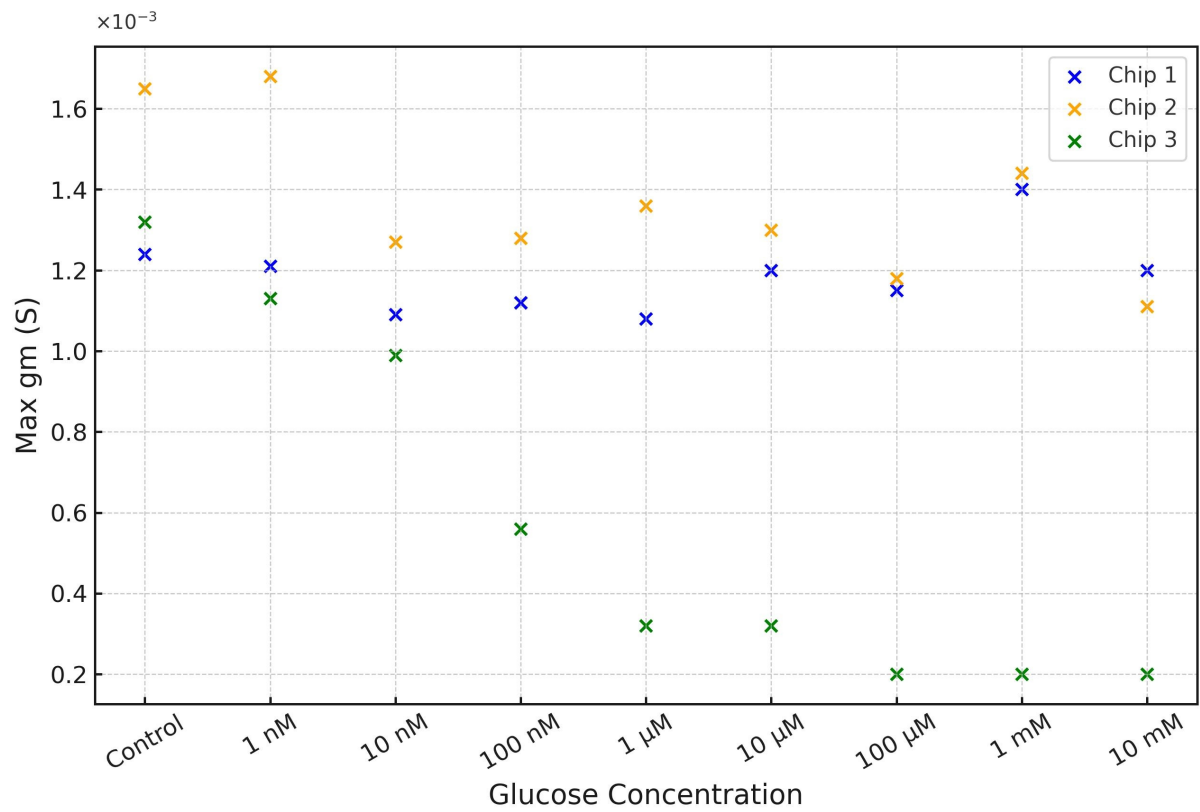


Figure A13: Maxtransconductance vs Glucose Concentrations comparison of OECT 1, OECT 2, OECT 3 with GOx enzyme.

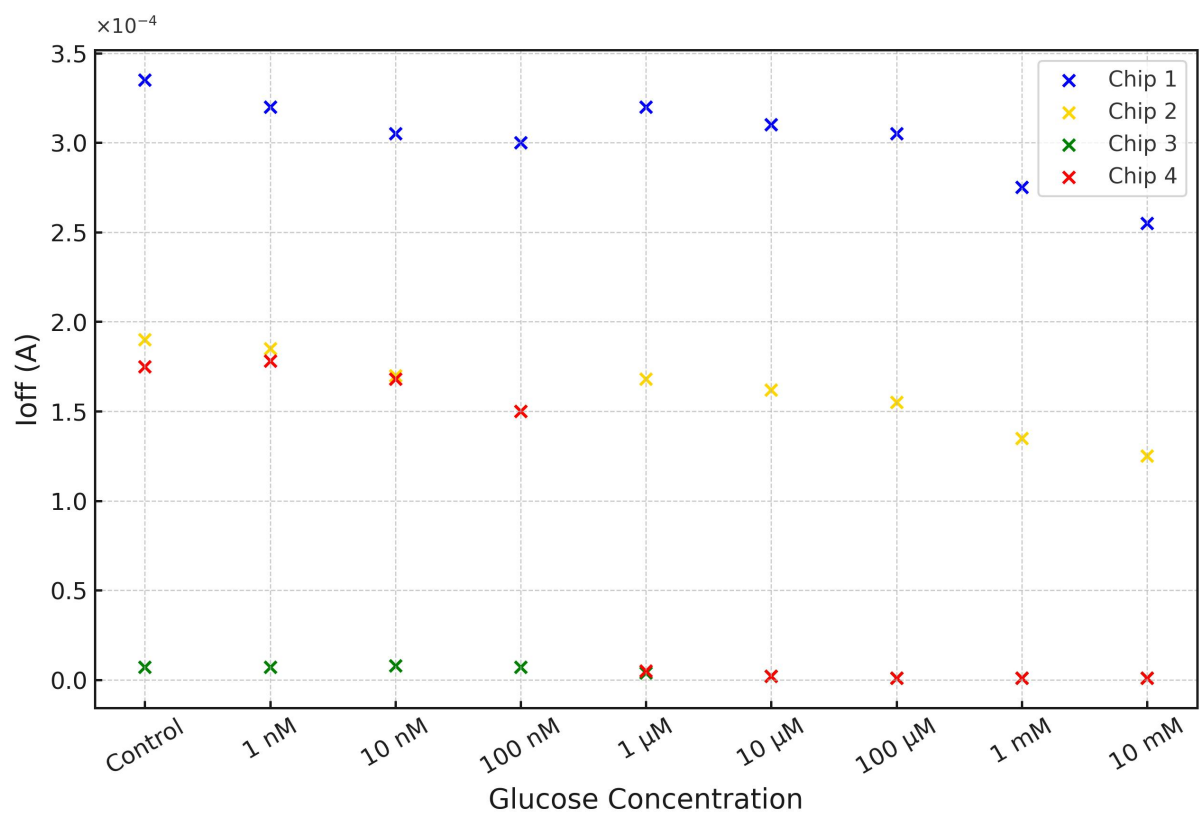


Figure A14: Off current vs glucose concentrations across OECTs 1–4 with GDH enzyme.

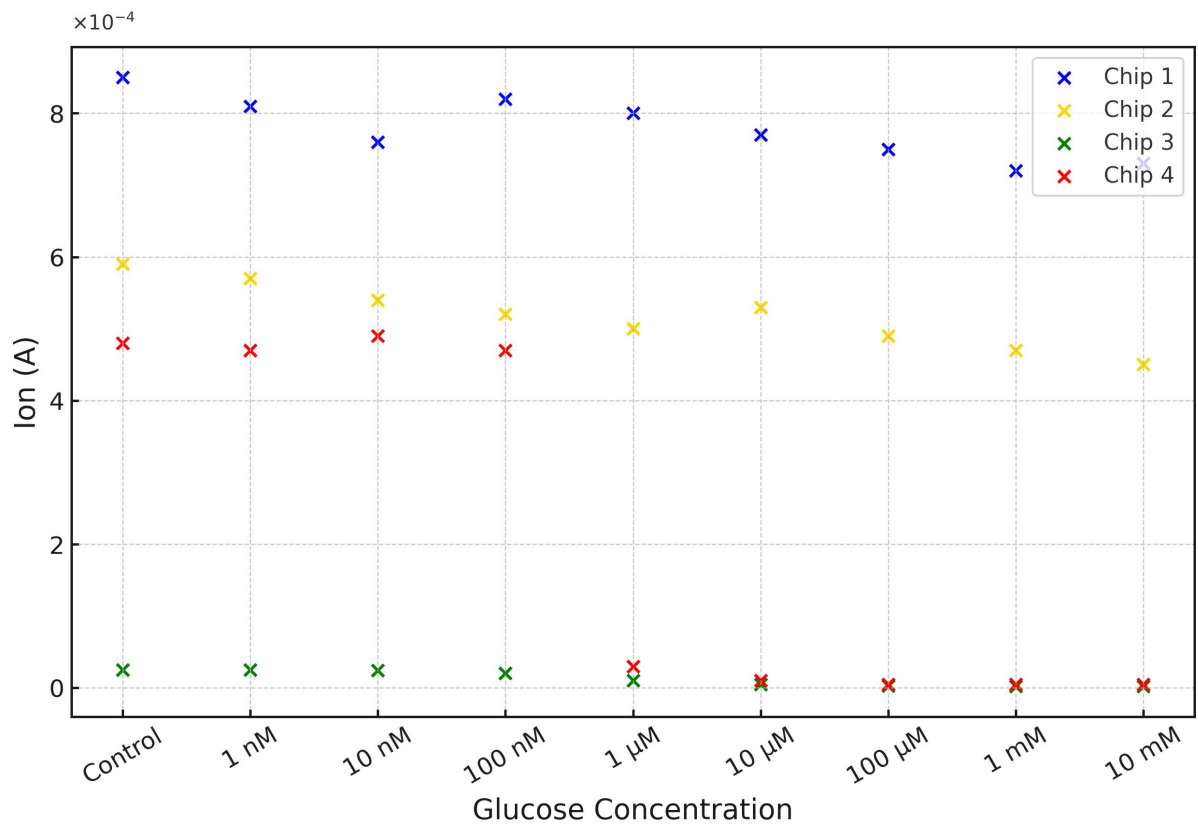


Figure A15: On current vs glucose concentrations across OECTs 1–4 with GDH enzyme.

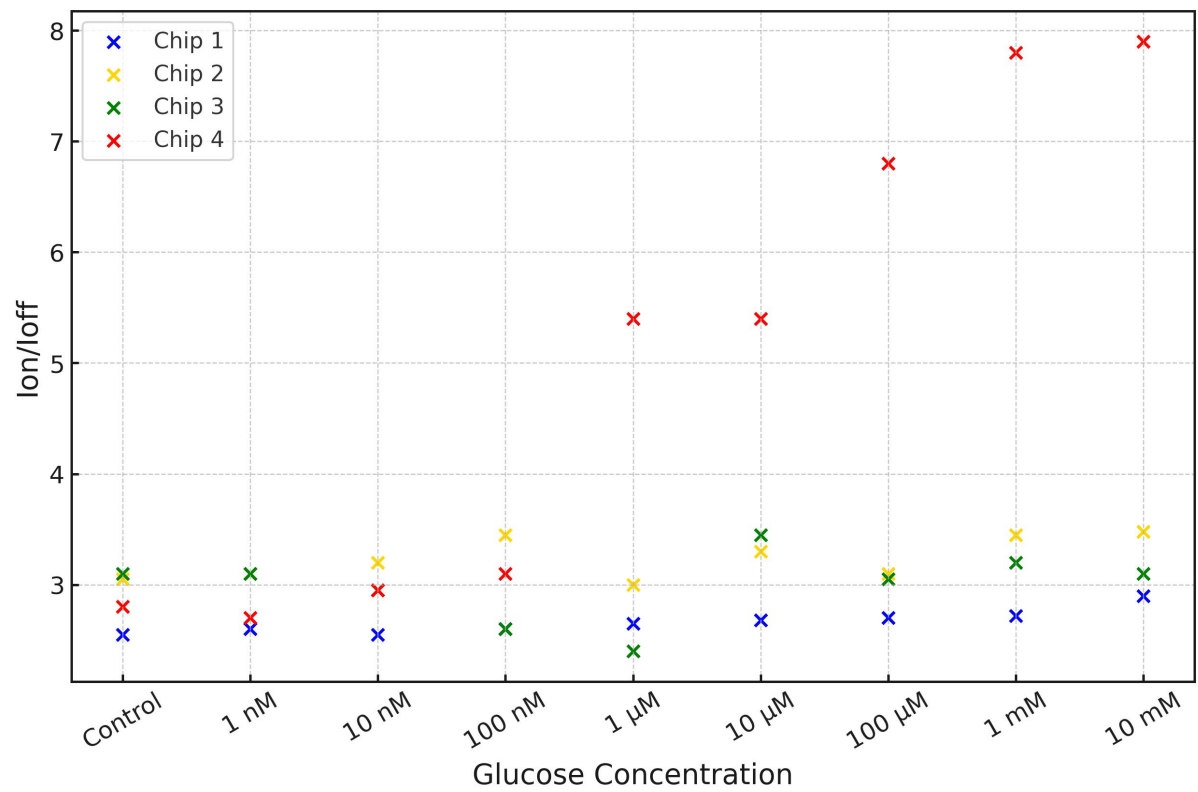


Figure A16: On/off ratio vs glucose concentrations across OECTs 1–4 with GDH enzyme.

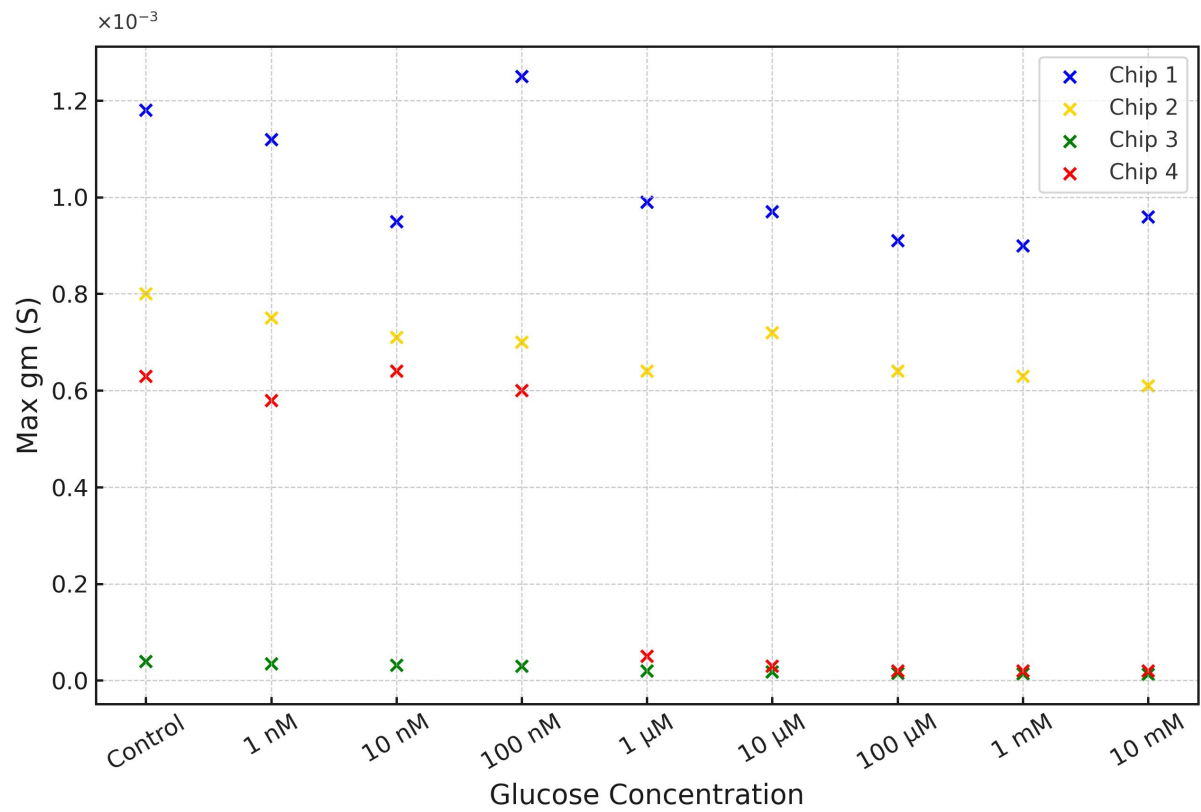


Figure A17: Maximum transconductance vs glucose concentrations across OECTs 1–4 with GDH enzyme.

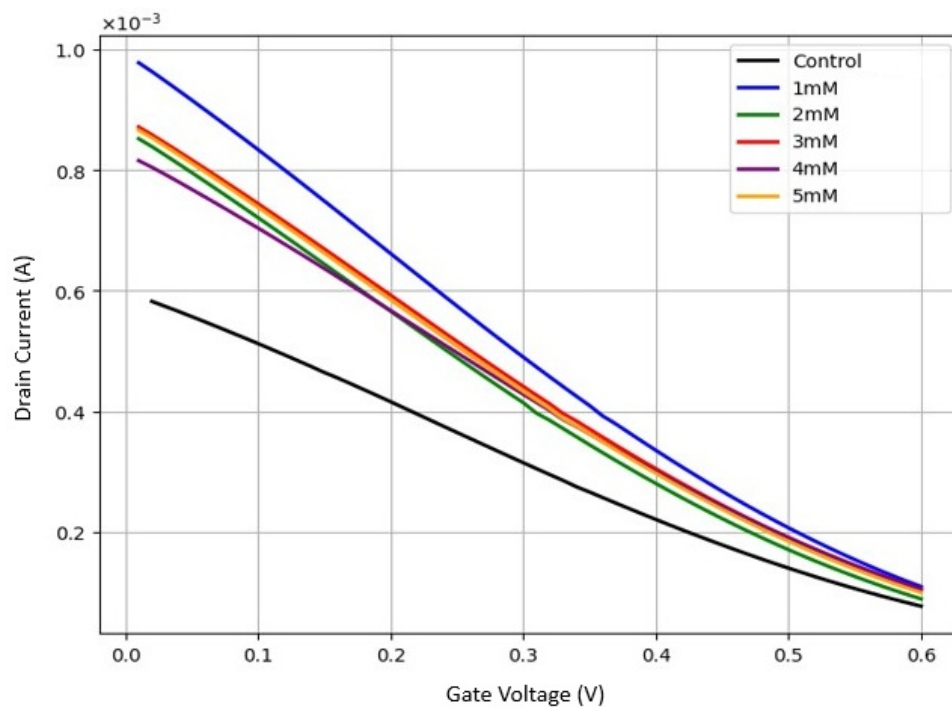


Figure A18: Transfer characteristics of GDH-PQQ immobilized OECT (OECT 2) showing drain current (I_d) response to different glucose concentrations (1-5 mM).

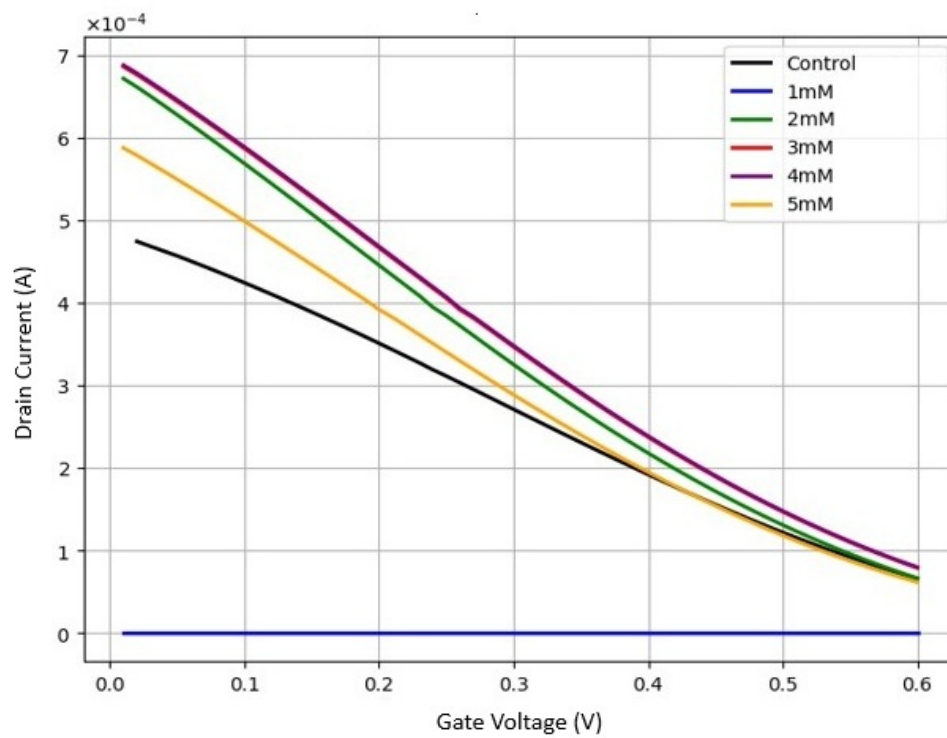


Figure A19: Transfer characteristics of GDH-PQQ immobilized OECT (OECT 3) showing drain current (I_d) response to different glucose concentrations (1-5 mM).

B Appendix-AI usage in this work

Artificial intelligence tools, including Quillbot and ChatGPT were utilized during the preparation of this report to support certain non-technical aspects of the work. Specifically, these tools assisted in grammar correction and language refinement to improve the readability and coherence of the document.

It is important to note that all technical content, data analysis, experimental design, results interpretation, and conclusions presented in this report are entirely the author's own work. The AI tools were used solely as writing aids and did not contribute to the generation or validation of scientific results or original ideas. Here are the main prompts that has been used:-

Abstract

- Please make this abstract sound smoother and fix small grammar or flow issues.
- Rephrase a bit so the sentences connect naturally from problem to result.
- Keep it short, clear, and professional without changing the meaning.

Introduction 1

- Please make this section flow naturally from one idea to the next.
- Fix grammar and simplify long sentences, but keep the scientific meaning.
- Make the connection between diabetes, glucose monitoring, and OECTs sound smooth.
- Rephrase the part about GOx and GDH so it reads clearly and naturally.
- Make sure the last paragraph clearly shows the goal and importance of the work.

2.1 Glucose and Its Importance in Biological Systems 2.1

- Please fix grammar and make this section read smoothly.
- Make the sentences flow better when explaining glucose levels in blood and other fluids.
- Simplify the long sentences but keep all technical details correct.

2.2 Currently Employed Glucose Testing Techniques 2.2

- Please rephrase this introduction so it sounds clear and formal, but still easy to read.
- Fix grammar and make the section flow naturally between different sensing methods.

2.2.1 Electrochemical Glucose Sensors

- Can you polish this paragraph so it's easier to follow and technically correct?
- Rephrase to clearly show the difference between amperometric, potentiometric, and conductometric sensors.

2.2.1.1 Enzymatic Electrochemical Glucose Sensors

- Please make this section about GOx and GDH enzymes sound smooth and logical.
- Fix grammar and reword long sentences for better flow.
- Simplify the reaction explanations while keeping them accurate.

2.2.1.2 Non-Enzymatic Electrochemical Glucose Sensors

- Rephrase this section to make it sound more natural and clear.
- Make the comparison with enzymatic sensors smoother and better connected.

2.2.2 Optical Glucose Detection Methods

- Please rewrite this so it sounds more fluent and easy to read.
- Fix grammar and make the explanation of optical methods clearer.

2.2.3 Other Glucose Detection Methods

- Polish this section — make it simple, formal, and well connected.
- Fix awkward phrasing and make the ideas flow more naturally.

2.3 Organic Electrochemical Transistor (OECT) 2.3

- Please reword this introduction to OECTs so it sounds clear and confident.
- Make the comparison with FETs flow better and fix small grammar issues.

2.3.1 p-Type Depletion and Accumulation Mode in OECTs

- Rephrase this part to make the explanation of accumulation and depletion modes easier to follow.
- Fix grammar and connect the sentences better.

2.3.2 Electronic and Ionic Circuit

- Make this section sound smoother when explaining how the two circuits work.
- Simplify the sentences and fix grammar without losing technical meaning.

2.3.3 Transient-State Behaviour in OECTs

- Please polish this section — make the time constant explanation clear and natural.
- Reword long sentences for better readability.

2.3.4 Steady-State Behaviour in OECTs

- Rephrase this part so the explanation of current–voltage relationships sounds smoother.
- Fix small grammar errors and make the section more fluent.

2.3.5 OECT in Enzymatic Sensing

- Please rewrite this to sound natural and connected to the earlier OECT discussion.
- Fix grammar and make the glucose sensing explanation flow better.
- Simplify sentences where possible, but keep all key points.

3.2.6 Glucose Sample Preparation for Steady-State and Real-Time Measurements 3.2.6

- Make this section sound smoother and fix any grammar mistakes.
- Rephrase this part so it's easy to follow when reading about how glucose solutions were prepared.
- Simplify long sentences and make the explanation sound clear and professional.
- Make sure the concentrations and dilution steps are described neatly and in correct scientific English.
- Reword this to make it flow better between steady-state and real-time preparation.

3.2.7 Electrical Characterization and Measurement Procedures 3.2.7

- Rephrase this section so it sounds clear and professional..
- Fix grammar and improve sentence flow without changing the meaning.

3.2.7.1 Instrumentation

- Make the description of the Keithley 4200A-SCS and MBox sound smooth and well-written.
- Fix grammar, especially in the longer technical explanations.
- Simplify wording where possible so it's easier to follow.

3.2.7.2 Steady-State Measurement Procedure

- Make this section sound clear and consistent while keeping the technical tone.
- Rephrase the steps so they flow properly and sound natural.
- Fix grammar and make the explanation of measurements cleaner.

3.2.7.3 Real-Time Measurement Procedure

- Rewrite this section so it sounds smooth and clear.
- Make the explanation of how glucose was added step-by-step easier to follow.
- Fix grammar and remove repeated or unwanted phrasing.
- Rephrase to make the ΔI_d calculation part flow better.
- Make the part about voltage settings and conditions sound clean and professional.

Conclusion 5

- Please make this section sound smooth and professional while keeping my writing style.
- Make the explanation of both enzyme methods (immobilized and floating) read clearly and naturally.
- Rephrase a bit so the ending highlights the main findings and importance more strongly.
- Keep the language simple, confident, and easy to read — not too formal or too casual.